TRIAL BIOSTATISTICS TOOLKIT · PLANNING REFERENCE

Protocol to CSR: the early-phase biostatistics deliverables pipeline

Every statistics deliverable of an early-phase clinical-pharmacology study, from protocol development to the clinical study report, results posting and archiving: owner, inputs, what it feeds, governing standard, quality control and common pitfalls.

180	5 · 6	9	62
deliverables	stages · gates	study designs	references checked 2026-11-08

1. Study design & protocol
closes at Protocol final

2. Planning & specifications
closes at SAP and specifications final

3. Data & CDISC
closes at Database lock

4. Analysis & QC
closes at Outputs QC-passed

5. Reporting, disclosure & archiving
closes at Archived

A planning reference for a biostatistics consultant or lead biostatistician, not a record of client work, an SOP or a validated system. Adapt it to the protocol, the client's SOPs and the regulations that apply. The interactive explorer, the poster map and the Excel register hold the same content: https://patrickiben.github.io/biometrics-ai-portfolio/biostat_pipeline/. Generated 2026-11-08 from pipeline.json.

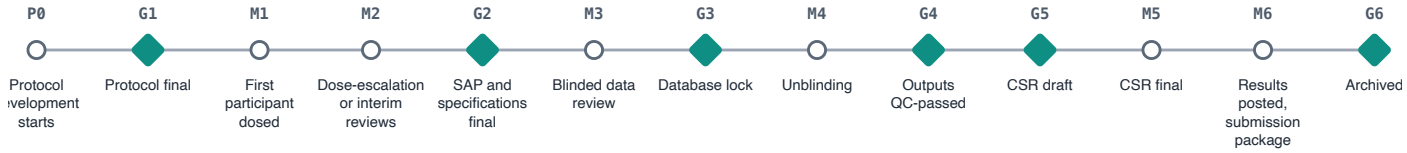
BEFORE YOU START

How to read this handbook

Each deliverable has one card. The card says what the deliverable is and why it matters, what it needs and which deliverables it depends on, what it produces and which deliverables it feeds, how it is checked, the mistakes that cost time later, and the standards that govern it.

Milestones and gates

Every deliverable is placed between two milestones: the point where work on it can start and the point by which it must be final. Diamonds are gates: a stage is complete when its gate is passed.



Tracks

Biostatistics	51 deliverables
Programming & CDISC	50 deliverables
PK & pharmacometrics	23 deliverables
Data management interface	11 deliverables
Safety & committees	13 deliverables
Quality & compliance	18 deliverables
Reporting & disclosure	14 deliverables

Study designs

Most deliverables apply to every early-phase study. Design-specific deliverables are marked with the designs they belong to; the design chapters list them together.

SAD	Single ascending dose
MAD	Multiple ascending dose
BE	Bioequivalence
BA	Relative bioavailability
FE	Food effect
DDI	Drug-drug interaction
QT	Concentration-QTc or thorough QT
RI	Renal impairment
HI	Hepatic impairment

Field guide

Depends on	Direct predecessors: deliverables that must exist, at least in draft, before this one can be finished.
Feeds	Deliverables that list this one as a predecessor. A change here should trigger a check of each of them.
CONDITIONAL	Applies only in some studies; the card says when.
WORKED STUDY	The toolkit's synthetic bioequivalence study produced this deliverable; its card links to the real output.

OVERVIEW

The pipeline at a glance

Five stages, each closed by a gate. Rows are stages, columns are tracks; each cell counts the deliverables in that stage and track.

STAGE	BIOSTATISTICS	PROGRAMMING & CDISC	PK & PHARMACOMETRICS	DATA MANAGEMENT INTERFACE	SAFETY & COMMITTEES	QUALITY & COMPLIANCE	REPORTING & DISCLOSURE	TOTAL	GATE
1. Study design & protocol	18		1		2	1		**22**	Protocol final
2. Planning & specifications	13	13	4	5	2	4		**41**	SAP and specifications final
3. Data & CDISC	3	33	3	6	5	2		**52**	Database lock
4. Analysis & QC	16	2	15		4	5		**42**	Outputs QC-passed
5. Reporting, disclosure & archiving	1	2				6	14	**23**	Archived

Longest dependency chains

The chains below are the longest paths through the "depends on" links that end at a stage-5 deliverable. They are the order in which delays propagate.

S1–01 → S1–02 → S1–05 → S1–06 → S1–12 → S1–07 → S1–11 → S2–01 → S2–03 → S2–21 → S3–02 → S3–30 → S3–40 → S3–32 → S3–41 → S3–33 → S3–04 → S3–18 → S3–19 → S4–07 → S3–06 → S3–20 → BE–03 → S4–10 → BA–03 → S4–20 → S4–21 → S5–01 → S5–03 → S5–12 → S5–11 → S5–13 → S5–20 → S5–21

S1–01 → S1–02 → S1–05 → S1–06 → S1–12 → S1–07 → S1–11 → S2–01 → S2–03 → S2–21 → S3–02 → S3–30 → S3–40 → S3–32 → S3–41 → S3–33 → S3–04 → S3–18 → S3–19 → S4–07 → S3–06 → S3–20 → BE–03 → S4–10 → BA–03 → S4–20 → S4–21 → S5–01 → S5–03 → S5–12 → S5–11 → S5–13 → S5–14

S1–01 → S1–02 → S1–05 → S1–06 → S1–12 → S1–07 → S1–11 → S2–01 → S2–03 → S2–21 → S3–02 → S3–30 → S3–40 → S3–32 → S3–41 → S3–33 → S3–04 → S3–18 → S3–19 → S4–07 → S3–06 → S3–20 → BE–03 → S4–10 → BA–03 → S4–20 → S4–21 → S5–01 → S5–03 → S5–12 → S5–11 → S5–18 → S5–19

S1–01 → S1–02 → S1–05 → S1–06 → S1–12 → S1–07 → S1–11 → S2–01 → S2–03 → S2–21 → S3–02 → S3–30 → S3–40 → S3–32 → S3–41 → S3–33 → S3–04 → S3–18 → S3–19 → S4–07 → S3–06 → S3–20 → BE–03 → S4–10 → BA–03 → S4–20 → S4–21 → S5–01 → S5–03 → S5–12 → S5–11 → S5–13 → S5–20

STAGE 1 · CLOSES AT PROTOCOL FINAL

Study design & protocol

Biostatistics shapes the design: objectives and estimands, endpoints, the sample-size basis, randomization, decision rules and the statistical sections of the protocol. 22 deliverables.

ID	DELIVERABLE	OWNER	FROM → TO	DESIGNS
S1–01	Biostatistics input to the protocol: design rationale and design selection	Lead biostatistician	P0 → G1	All
S1–02	Objectives, estimands and endpoints alignment (ICH E9(R1) estimand framework)	Lead biostatistician	P0 → G1	All
S1–03	Protocol statistical considerations section (analysis overview)	Lead biostatistician	P0 → G1	All
S1–04	Analysis-set definitions (protocol level)	Lead biostatistician	P0 → G1	All
S1–05	Sample-size or precision justification (or documented rationale when there is no formal sizing, as in first-in-human cohorts)	Lead biostatistician	P0 → G1	All
S1–06	Randomization design and specification (including sentinel dosing and active-to-placebo allocation) CONDITIONAL	Lead biostatistician	P0 → G1	All
S1–07	Safety review committee or DMC charter: statistical decision rules and dose-escalation criteria	Lead biostatistician	P0 → M1	SAD, MAD
S1–08	Dose-escalation operating-characteristics simulation (design stage) CONDITIONAL	Lead biostatistician	P0 → G1	SAD, MAD
S1–09	Interim and adaptive design elements specification (if any)	Lead biostatistician	P0 → G1	All
S1–10	PK precision and variability assumptions memo (CV, BLQ and LLOQ, sampling-schedule adequacy)	Lead biostatistician	P0 → G1	All
S1–11	Biostatistics protocol review and sign-off	Lead biostatistician	G1 → G3	All
S1–12	Emergency unblinding and code-break plan (design-stage definition) CONDITIONAL	Lead biostatistician	P0 → G1	All
S1–13	Blinded sample-size re-estimation (BSSR) plan: design-stage provision CONDITIONAL	Lead biostatistician	P0 → G1	All
S1–14	Estimand-to-design traceability and assumptions log (intercurrent-event strategy register)	Lead biostatistician	P0 → G1	All
S1–15	Statistics input to critical-to-quality factors and quality tolerance limits	Lead biostatistician	P0 → G1	All
BA–01	Relative bioavailability or formulation-bridging design and precision basis	Lead biostatistician	P0 → G1	BA
BE–01	BE design and sample-size basis (2x2, replicate or parallel)	Lead biostatistician	P0 → G1	BE
DDI–01	Drug interaction study design and sample-size basis	Lead biostatistician	P0 → G1	DDI
FE–01	Food-effect design and sample-size basis	Lead biostatistician	P0 → G1	FE
HI–01	Hepatic impairment design and sample-size basis	Lead biostatistician	P0 → G1	HI
QT–01	Cardiac safety (QT) design and sample-size basis CONDITIONAL	Lead biostatistician	P0 → G1	QT, SAD, MAD
RI–01	Renal impairment design and sample-size basis	Lead biostatistician	P0 → G1	RI

S1–01 Biostatistics input to the protocol: design rationale and design selection

Document · Biostatistics · Lead biostatistician with Clinical pharmacologist, PK scientist, Medical monitor, Regulatory affairs · Protocol development starts → Protocol final · All designs

The biostatistician's written justification of the overall design: randomization, blinding and placebo control, sequential cohorts and sentinel dosing for SAD and MAD, crossover versus parallel groups, and randomized versus fixed sequences for BE, BA, food-effect and DDI studies. It explains how the design answers the objectives and controls bias, including washout length relative to the expected half-life.

Why it matters. Without a documented rationale, choices such as cohort size, washout or sequence structure cannot be defended, and the sample-size, randomization and escalation documents have no agreed basis.

NEEDS

Development plan or target product profile; nonclinical PK and toxicology (NOAEL, exposure margins, predicted human PK and half-life); prior data on the compound or class; clinical pharmacology objectives; applicable guidance on first-in-human starting dose, early-trial risk mitigation and design-specific studies.

PRODUCES

Design section of the protocol; basis for the estimands, sample-size rationale, randomization design, dose-escalation rules and SAP design assumptions.

Feeds: S1–02, S1–08, S1–09, S1–10

QUALITY CONTROL

Independent biostatistician peer review; check of the design against objectives and estimands; medical, clinical pharmacology and regulatory review; protocol team agreement.

COMMON PITFALLS

- Washout chosen without checking the expected terminal half-life, so pre-dose concentrations in later periods force exclusions.
- Fixed-sequence DDI design chosen without stating that period effects cannot be separated from the interaction.
- Design rationale not updated after amendments change cohorts, dose levels or sampling.

STANDARDS AND GUIDANCE

- ICH E9: design techniques to avoid bias (blinding, randomization) and crossover designs, carryover and washout
- ICH E8(R1): quality by design and critical-to-quality factors in study planning
- Clinical Electronic Structured Harmonised Protocol (CeSHarP): protocol template trial design section
- EMEA/CHMP/SWP/28367/07 Rev. 1: first-in-human design, including sentinel dosing and dose escalation
- ICH M13A: randomized single-dose crossover design and washout for BE studies
- ICH M12: design of clinical drug interaction studies
- Assessing the Effects of Food on Drugs in INDs and NDAs:…: design of food-effect studies

IN THIS TOOLKIT

Worked study: protocol trial design section (2x2 crossover) (study/01_design/protocol_statistics.html) WORKED STUDY

S1–02 Objectives, estimands and endpoints alignment (ICH E9(R1) estimand framework)

Specification · Biostatistics · Lead biostatistician with Clinical pharmacologist, Medical monitor, PK scientist · Protocol development starts → Protocol final · All designs

Maps each objective to a target estimand and its endpoints, defining the five attributes: treatment condition, population, variable (endpoint), intercurrent events with the strategy for each, and population-level summary. Many early-phase objectives are descriptive (PK parameters, safety), while comparative objectives (BE, food effect, DDI) estimate a geometric mean ratio with a confidence interval (90% for BE under ICH M13A); emesis, dosing errors, missed doses, prohibited medications and discontinuation each need an explicit strategy.

Why it matters. Without explicit estimands, analysis sets, PK exclusions and intercurrent-event handling are decided ad hoc later, and the analysis may not answer the question the protocol asked.

NEEDS

Protocol objectives; endpoint definitions; PK and PD variable list; anticipated intercurrent events (emesis, dosing errors, missed doses, rescue or concomitant medications, discontinuation); clinical input on the questions of interest.

Depends on: S1–01

PRODUCES

Objectives and estimands section of the protocol; basis for analysis-set definitions, intercurrent-event handling rules in the SAP, and TLF planning.

Feeds: S1–04, S1–05, S1–15

QUALITY CONTROL

Independent biostatistician review of estimand completeness (all five attributes, every anticipated intercurrent event); clinical and clinical pharmacology review; consistency check of objectives, estimands, endpoints and analysis sets.

COMMON PITFALLS

- PK exclusions such as emesis applied in the SAP without a matching intercurrent-event strategy in the protocol.
- Estimand populations described in words that do not map to any analysis-set flag.
- Withdrawal before the second crossover period not addressed, leaving the BE set definition unsupported.

STANDARDS AND GUIDANCE

- ICH E9(R1): estimand attributes and intercurrent-event strategies
- Clinical Electronic Structured Harmonised Protocol (CeSHarP): protocol template section on objectives and associated estimands
- ICH M13A: BE assessed by the 90% confidence interval of the geometric mean ratio

IN THIS TOOLKIT

Worked study: objectives and estimands in the protocol statistics page (study/01_design/protocol_statistics.html) · Worked study: estimand tables in the SAP excerpt (study/02_sap/sap_excerpt.html) WORKED STUDY

S1–03 Protocol statistical considerations section (analysis overview)

Document · Biostatistics · Lead biostatistician with PK scientist, Medical writer · Protocol development starts → Protocol final · All designs

The protocol's statistical overview: analysis sets, planned analyses by domain (PK, PD, safety and comparative PK where applicable), descriptive-statistics conventions, summary-level handling of missing and BLQ data, interim or escalation decision references, and sample size. It states that full detail is in the SAP without pre-empting it.

Why it matters. A thin or inconsistent statistical section leaves the SAP without a protocol anchor, so later analysis choices can be judged post hoc.

NEEDS

Design rationale; estimand framework; analysis-set definitions; sample-size rationale; interim or adaptive elements; PK and PD analysis intent; safety review structure; the study's SOPs and protocol template.

Depends on: BA–01, BE–01, DDI–01, FE–01, HI–01, QT–01, RI–01, S1–04, S1–05, S1–09

PRODUCES

Statistical considerations section of the protocol; the frame the SAP must operationalize; reference for the safety committee charter.

Feeds: S1–11, S2–33, BA–02, BE–02, DDI–02, FE–02, HI–02, QT–02, RI–02

QUALITY CONTROL

Independent biostatistician peer review; alignment check against the estimands and the planned SAP scope; cross-functional protocol review.

COMMON PITFALLS

- Protocol fixes a model detail (factor, covariate, transformation) that the SAP later changes, forcing an amendment.
- BLQ and missing-data rules stated differently in the protocol and the SAP.

STANDARDS AND GUIDANCE

- Clinical Electronic Structured Harmonised Protocol (CeSHarP): protocol template statistical considerations section
- ICH E9: principal features of the analysis described in the protocol
- ICH E9(R1): estimators aligned with the estimands
- ICH E3: CSR Section 9.7 reports the statistical methods planned in the protocol

IN THIS TOOLKIT

Worked study: protocol statistical considerations section (study/01_design/protocol_statistics.html) WORKED STUDY

S1–04 Analysis-set definitions (protocol level)

Specification · Biostatistics · Lead biostatistician with PK scientist, Clinical pharmacologist · Protocol development starts → Protocol final · All designs

Defines each analysis set (for example enrolled, safety, PK concentration, PK parameter, PD, and the BE or comparison set in crossover studies) and the rules that move a participant or a profile in or out. Each estimand's population attribute is linked to a concrete set; the SAP later operationalizes the derivations.

Why it matters. Unclear definitions let denominators and PK exclusions change after the data are seen, which reviewers treat as a source of bias.

NEEDS

Estimand population attributes; design structure (sentinel, cohort, crossover); PK exclusion criteria (emesis, dosing errors, missed samples, pre-dose concentrations); anticipated intercurrent events.

Depends on: S1–02

PRODUCES

Analysis-sets subsection of the protocol; basis for ADSL population flags and the SAP analysis-set rules; population footnote for every TLF.

Feeds: S1–03, S1–14

QUALITY CONTROL

Independent review of set definitions for mutual consistency and estimand linkage; clinical pharmacology review of PK exclusion rules; traceability check to the ADSL flag plan.

COMMON PITFALLS

- PK set defined as 'evaluable' without saying which profile-level exclusions remove a participant entirely.
- Participants with data from only one crossover period kept in the BE comparison without a prespecified rule.

STANDARDS AND GUIDANCE

- ICH E9: analysis sets and their roles
- Clinical Electronic Structured Harmonised Protocol (CeSHarP): protocol template analysis sets subsection
- ICH M13A: BE analysis population criteria defined in the protocol, with exclusions documented before bioanalysis
- ADaMIG v1.3: ADSL population flag variables (for example SAFFL)

IN THIS TOOLKIT

Worked study: analysis-set definitions in the protocol statistics page (study/01_design/protocol_statistics.html) WORKED STUDY

S1–05 Sample-size or precision justification (or documented rationale when there is no formal sizing, as in first-in-human cohorts)

Document · Biostatistics · Lead biostatistician with PK scientist, Clinical pharmacologist · Protocol development starts → Protocol final · All designs

For SAD and MAD cohorts, a documented rationale that cohort size is chosen for safety observation and PK characterization rather than power, with the precision expected for geometric-mean PK parameters. For comparative and special-population studies, the power or precision calculation is documented in the design module's design and sample-size basis (BE-01, BA-01, FE-01, DDI-01, QT-01, RI-01 or HI-01) and summarized here with the method and software version.

Why it matters. Without it, an underpowered comparison or an unexplained cohort size cannot be defended at review.

NEEDS

Design (crossover or parallel, cohort size, active-to-placebo ratio); within- and between-participant CV assumptions from prior or class data; assumed geometric mean ratio and acceptance limits for comparative studies; target probability of observing an adverse event of a given incidence; precision targets.

Depends on: S1–02, S1–10

PRODUCES

Sample-size subsection of the protocol; assumptions carried into the SAP; input to feasibility and enrollment planning.

Feeds: S1–03, S1–06, S1–13

QUALITY CONTROL

Independent recalculation with separate code or software; review of assumptions against their sources; method, software version and any seed recorded.

COMMON PITFALLS

- CV taken from one small study without a sensitivity range, so power collapses if variability is higher.
- Dropout allowance omitted, leaving fewer evaluable participants than the calculation required.
- Extra cohorts added after bioanalysis results are known instead of being prespecified in the protocol.

STANDARDS AND GUIDANCE

- ICH E9: items to specify for a sample-size calculation, and justification when sizing rests on another basis
- ICH M13A: sample size for prespecified power and type 1 error; at least 12 evaluable participants; 90% CI within 80.00% to 125.00%
- Clinical Electronic Structured Harmonised Protocol (CeSHarP): protocol template sample size determination subsection
- Statistical Approaches to Establishing Bioequivalence: average bioequivalence analysis on log-transformed data

IN THIS TOOLKIT

Worked study: sample size, sensitivity grids and dropout inflation (study/01_design/protocol_statistics.html) · Worked study: design script (sample size and randomization) (study/01_design/design.R) WORKED STUDY

S1–06 Randomization design and specification (including sentinel dosing and active-to-placebo allocation)

Specification · Biostatistics · Lead biostatistician with Randomization and supply (IRT) vendor, Clinical operations · Protocol development starts → Protocol final · All designs

CONDITIONAL If treatment or sequence allocation is randomized

Defines the allocation method (for example permuted blocks within cohort to keep the active-to-placebo ratio), stratification, the blinding scheme, the sentinel-dosing scheme for first-in-human cohorts, and sequence randomization for crossover studies, plus who generates and who holds the schedule. Details that make allocation predictable, such as block size, belong in the restricted randomization specification, not in the protocol.

Why it matters. A loosely specified design can break the intended ratio, unbalance sequences or make allocation predictable, undermining both the comparison and the blind.

NEEDS

Design (cohorts, sequences, sentinel scheme, ratio); blinding plan; IRT capabilities; dose-escalation structure; number of cohorts or periods; sample size.

`Depends on: S1–05`

PRODUCES

Randomization and blinding text in the protocol; design inputs for the restricted randomization specification, IRT setup and the code-break plan.

`Feeds: S1–12`

QUALITY CONTROL

Independent biostatistician review; check that block, ratio and sentinel logic enforce the intended allocation; segregation of duties (the specification author does not generate the schedule); blinding-integrity review.

COMMON PITFALLS

- Block size written into the protocol or site documents, making the last allocations in each block predictable.
- Sentinel pair not built into the cohort's allocation, so the remaining participants no longer give the intended ratio.
- Crossover sequences not balanced within each cohort or dosing group.

STANDARDS AND GUIDANCE

- ICH E9: randomization, blocking, and keeping block length out of the protocol
- EMEA/CHMP/SWP/28367/07 Rev. 1: sentinel dosing within a cohort
- ICH M13A: randomized crossover design; stratified randomization for parallel designs

IN THIS TOOLKIT

Worked study: randomization method in the protocol statistics page (study/01_design/protocol_statistics.html) · Worked study: design script (blocks, seed, balance check) (study/01_design/design.R) WORKED STUDY

S1–07 Safety review committee or DMC charter: statistical decision rules and dose-escalation criteria

Document · Safety & committees · Lead biostatistician with Medical monitor, Clinical pharmacologist, PK scientist, Safety Review Committee · Protocol development starts → First participant dosed · Single ascending dose, Multiple ascending dose

Biostatistics content of the safety review committee (or DMC) charter: the escalation algorithm (rule-based or model-based), the safety, tolerability and PK data reviewed before each escalation, stopping and de-escalation rules, exposure limits tied to nonclinical margins, and review cadence. It also defines data flow to the committee, blinded versus unblinded views, and the firewall.

Why it matters. Ambiguous or unconservative escalation rules put participants at risk and invite inconsistent decisions between cohorts.

NEEDS

Design and dose levels; nonclinical exposure margins and NOAEL; predicted human PK; stopping-rule approach; escalation method; operating-characteristics results when model-based; committee composition and governance; unblinding and firewall rules.

`Depends on: S1–08, S1–12`

PRODUCES

Statistical and escalation sections of the charter; specification of the data packages and summaries the committee receives at each review.

`Feeds: S1–11, ESC–03`

QUALITY CONTROL

Independent review of decision rules for clarity and conservatism; consistency check against protocol dose levels and stopping rules; committee and regulatory review; review of operating characteristics when the design is model-based.

COMMON PITFALLS

- Escalation rule depends on PK or safety data that cannot be available within the review timeline.
- Rules allow escalation on incomplete cohorts without stating the minimum number of evaluable participants.
- Committee receives unblinded summaries without a defined firewall, compromising the blinded study team.

STANDARDS AND GUIDANCE

- EMEA/CHMP/SWP/28367/07 Rev. 1: sentinel dosing, precautions between cohorts, data review for escalation decisions and stopping rules
- Guidance for Clinical Trial Sponsors: Establishment and…: establishment and operation of data monitoring committees
- ICH E9: formal procedures for early stopping for safety reasons

IN THIS TOOLKIT

Example safety review committee data packet (synthetic MAD cohort) (sasr_monitoring_wiki/trialmon_quarto/trialmon-src-packet.html)

S1–08 Dose-escalation operating-characteristics simulation (design stage)

Output · Biostatistics · Lead biostatistician with Clinical pharmacologist, PK scientist · Protocol development starts → Protocol final · Single ascending dose, Multiple ascending dose

CONDITIONAL If a model-based escalation design is used or the escalation rules need simulation support

Simulation of the chosen escalation design before the study starts. For rule-based designs it gives the probability of escalating or stopping under assumed dose-toxicity and dose-exposure scenarios; for model-based designs it gives the probability of selecting each dose, the risk of overdosing, expected sample size and sensitivity to the prior, with seeds, software versions and the scenario grid recorded.

Why it matters. Without operating characteristics, the team and regulators cannot judge whether the escalation rules are safe and efficient across plausible scenarios.

NEEDS

Escalation rule or model and priors; dose levels and cohort sizes; assumed dose-toxicity and dose-exposure scenarios; target toxicity interval or escalation thresholds; nonclinical predictions.

Depends on: S1–01

PRODUCES

Operating-characteristics tables and figures supporting the protocol design section and the committee charter.

Feeds: S1–07

QUALITY CONTROL

Independent re-run with the documented seed and environment; code review; review of the scenario grid for completeness; sanity check of results against design intuition.

COMMON PITFALLS

- Scenarios limited to the expected dose-toxicity curve, so overdosing risk under steeper curves is never shown.
- Simulation code not archived with seeds and versions, so results cannot be reproduced at inspection.

STANDARDS AND GUIDANCE

- Adaptive Design Clinical Trials for Drugs and Biologics:…: simulation to evaluate operating characteristics of complex designs
- EMEA/CHMP/SWP/28367/07 Rev. 1: dose-escalation and stopping rules that the simulation evaluates

S1–09 Interim and adaptive design elements specification (if any)

Specification · Biostatistics · Lead biostatistician with Clinical pharmacologist, Regulatory affairs · Protocol development starts → Protocol final · All designs

Specifies any adaptive features beyond simple sequential escalation, such as sample-size re-estimation, adaptive dose or cohort selection, optional cohorts triggered by emerging PK, or a two-stage BE design: decision points, methods, error-rate and operating-characteristic control, independence of the analysis, and information flow. When none are planned, it is a documented not-applicable statement.

Why it matters. Adaptations that are not prespecified look ad hoc, can inflate error rates, and can expose comparative interim data to the study team.

NEEDS

Design objectives; escalation and interim structure; nonclinical and emerging-PK expectations; regulatory feedback on adaptation.

Depends on: S1–01

PRODUCES

Adaptive-design subsection of the protocol and SAP; input to interim output specifications, the unblinding and firewall plan, and simulations.

Feeds: S1–03, S1–13

QUALITY CONTROL

Independent statistical review of adaptation rules and error-rate control; simulation of operating characteristics where needed; firewall and independence review; documented not-applicable decision when no adaptation is planned.

COMMON PITFALLS

- Optional cohort added mid-study without a prespecified trigger, so the decision looks data-driven.
- Staff who saw unblinded interim results later take part in the blind review or SAP changes.

STANDARDS AND GUIDANCE

- Adaptive Design Clinical Trials for Drugs and Biologics:…: prespecification, error-rate control and trial integrity for adaptive designs
- ICH E9: interim analysis, early stopping and sample-size adjustment
- Clinical Electronic Structured Harmonised Protocol (CeSHarP): protocol template interim analyses subsection

IN THIS TOOLKIT

Worked study: interim analyses statement ('None') in the protocol statistics page (study/01_design/protocol_statistics.html) WORKED STUDY

S1–10 PK precision and variability assumptions memo (CV, BLQ and LLOQ, sampling-schedule adequacy)

Document · PK & pharmacometrics · Lead biostatistician with PK scientist, Bioanalytical laboratory · Protocol development starts → Protocol final · All designs

Supporting memo for the sample-size and PK rationale: assumed within- and between-participant CV for key PK parameters with their sources, the assay LLOQ and expected BLQ pattern, and an assessment that the sampling schedule characterizes Cmax, tmax, AUC and the terminal phase. For BE studies it checks the ICH M13A expectations that AUC(0-t) covers at least 80% of AUC(0-inf) and that sampling spans at least three terminal half-lives.

Why it matters. Wrong CV or sampling assumptions produce imprecise PK estimates, unreliable terminal-phase parameters and sample sizes that do not hold.

NEEDS

Predicted human PK (nonclinical, allometry or PBPK); assay LLOQ; literature or class CV; proposed sampling times; AUC extrapolation acceptance criteria.

Depends on: S1–01

PRODUCES

Assumptions for the sample-size rationale and the SAP NCA and BLQ rules; justification of the sampling schedule in the protocol; input to ADPC and ADPP planning.

Feeds: S1–05, BA–01, BE–01, DDI–01, FE–01, HI–01, QT–01, RI–01

QUALITY CONTROL

Joint biostatistician and PK scientist review; check of sampling adequacy against the predicted half-life; sensitivity of precision to the CV assumptions; source of every assumption documented.

COMMON PITFALLS

- LLOQ too high for the lowest doses, so the terminal phase and AUC extrapolation fail in early cohorts.
- Sampling ends too early relative to half-life, so AUC(0-inf) is largely extrapolated.

STANDARDS AND GUIDANCE

- ICH M13A: sampling schedule: AUC(0-t) covering at least 80% of AUC(0-inf) and sampling for at least three terminal half-lives
- Bioavailability Studies Submitted in NDAs or INDs:…: BA study design and PK measures

IN THIS TOOLKIT

Worked study: planning assumptions (CV, ratio, washout, dropout) (study/01_design/design_parameters.json) WORKED STUDY

S1–11 Biostatistics protocol review and sign-off

Review record · Biostatistics · Lead biostatistician with Independent statistician, Medical writer · Protocol final → Database lock · All designs

Documented biostatistics review and approval of the final protocol and each amendment: internal consistency of objectives, estimands, endpoints, analysis sets and planned analyses, and correctness of the sample-size, randomization, blinding and escalation text. It is captured as tracked comments with resolutions and a signature in the protocol approval record.

Why it matters. Inconsistencies left in the protocol become amendments, SAP conflicts or analyses that cannot answer the stated objectives.

NEEDS

Final draft protocol and amendments; estimand framework; sample-size and randomization specifications; committee charter; the study's protocol-approval SOP.

`Depends on: S1–03, S1–07, S1–13, S1–14`

PRODUCES

Signed biostatistics approval record; comment-resolution log; clearance for protocol finalization and the start of SAP and specification work.

`Feeds: S2–01, S2–16, S2–19, S2–23, S2–26, S2–27, S2–29`

QUALITY CONTROL

Comment tracking and resolution log; second-biostatistician consistency check of objectives, estimands and endpoints; signed, audit-trailed approval; amendment version control.

COMMON PITFALLS

- Amendment changes dose levels or cohorts, but the sample-size and escalation sections are not re-reviewed.
- Sign-off given on a draft that differs from the version filed.

STANDARDS AND GUIDANCE

- ICH E9: principal features of the analysis described in the protocol
- Clinical Electronic Structured Harmonised Protocol (CeSHarP): harmonized protocol template structure
- 21 CFR Part 312: protocol amendments to an IND (312.30)
- 21 CFR Part 11: electronic signatures and audit trails for electronic approvals

S1–12 Emergency unblinding and code-break plan (design-stage definition)

Specification · Safety & committees · Lead biostatistician with Medical monitor, Randomization and supply (IRT) vendor, Clinical operations · Protocol development starts → Protocol final · All designs

CONDITIONAL If the study is blinded

Protocol- and charter-level definition of the emergency-unblinding pathway and the blinding firewall: who may unblind and when, the IRT mechanism, which unblinded roles (committee, PK or safety reviewers, expedited safety reporting) work behind a controlled barrier, and how unblinding events are logged and reported. It sets the governance that later code-break execution must follow.

Why it matters. Without a defined pathway, unblinding is either delayed when medically needed or spreads beyond the people who need it, and the analysis team loses its blind.

NEEDS

Blinding scheme and randomization design; committee firewall needs; IRT capabilities; expedited safety-reporting requirements; the study's unblinding SOP.

`Depends on: S1–06`

PRODUCES

Unblinding and firewall section of the protocol and committee charter; requirements for the operational code-break plan and the unblinding event log.

`Feeds: S1–07`

QUALITY CONTROL

Independent review of the firewall and segregation of duties; check that the pathway is auditable and minimally disruptive to the blind; cross-check against the charter and randomization design.

COMMON PITFALLS

- Expedited safety reporting needs case-level unblinding, but the plan routes it through the blinded study team.
- IRT unblinding rights granted by role without periodic log review, so inadvertent unblinding goes undetected.

STANDARDS AND GUIDANCE

- ICH E9: breaking the blind only when essential for care, with the procedure and timing documented
- ICH E2A: breaking the blind for an expedited case while keeping biometrics personnel blinded
- ICH E6(R3): safeguarding blinding in data governance; investigator unblinding procedures
- 21 CFR Part 312: IND safety reporting (312.32)

S1–13 Blinded sample-size re-estimation (BSSR) plan: design-stage provision

Plan · Biostatistics · Lead biostatistician with PK scientist, Clinical pharmacologist · Protocol development starts → Protocol final · All designs

CONDITIONAL If a blinded sample-size re-estimation is planned

Prespecified provision for blinded sample-size re-estimation, for example re-estimating within-participant PK variability or a PD variance from pooled blinded data. It defines the trigger, who performs it, what is recomputed, the maximum sample size, and how blinding and the type 1 error are protected; it is separate from the initial sample-size justification.

Why it matters. Without a prespecified rule, a mid-study change in cohort size or enrollment looks data-driven to reviewers and inspectors.

NEEDS

Protocol design; PK and PD variability assumptions memo; estimand definitions; initial sample-size calculation.

`Depends on: S1–05, S1–09`

PRODUCES

Re-estimation section in the protocol or SAP; reference to the procedure that keeps the data blinded.

`Feeds: S1–11`

QUALITY CONTROL

Biostatistician review and sign-off; confirmation that only blinded data are used; documented calculation.

COMMON PITFALLS

- Re-estimation done by a statistician who has seen unblinded PK or safety summaries.
- No cap on the re-estimated sample size, so supply and budget cannot be planned.

STANDARDS AND GUIDANCE

- ICH E9: sample-size re-estimation from blinded data, envisaged in the protocol
- Adaptive Design Clinical Trials for Drugs and Biologics:…: adaptations to the sample size

S1–14 Estimand-to-design traceability and assumptions log (intercurrent-event strategy register)

Log or record · Biostatistics · Lead biostatistician with Medical monitor, Clinical pharmacologist · Protocol development starts → Protocol final · All designs

A register that records, for each estimand, the five attributes, the strategy for each intercurrent event with its rationale and assumptions, and the design element and analysis set that implement it. It is kept current through amendments and carried into the SAP.

Why it matters. When the reasons behind each intercurrent-event strategy are not recorded, later SAP changes and analysis-set decisions cannot be traced to the protocol's intent.

NEEDS

Protocol objectives; estimand definitions; endpoint definitions; analysis-set definitions.

Depends on: S1–04

PRODUCES

Estimand attribute table with intercurrent-event strategies, rationale and assumptions; reference for the SAP estimand and analysis-set sections.

Feeds: S1–11

QUALITY CONTROL

Biostatistician and medical reviewer sign-off; check that every intercurrent event in the protocol has a strategy and an implementing rule.

COMMON PITFALLS

- Register kept separately from the protocol and not updated after amendments.
- Hypothetical strategy chosen without stating the scenario it assumes.

STANDARDS AND GUIDANCE

- ICH E9(R1): estimand attributes and intercurrent-event strategies

IN THIS TOOLKIT

Worked study: estimand tables with intercurrent-event strategies (study/02_sap/sap_excerpt.html) WORKED STUDY

S1–15 Statistics input to critical-to-quality factors and quality tolerance limits

Document · Quality & compliance · Lead biostatistician with Clinical operations, Data manager, Medical monitor, Quality assurance · Protocol development starts → Protocol final · All designs

Statistics contribution to the study's quality-by-design work: names the data and processes critical to the primary estimands (for example dosing and sampling clock times, pre-dose samples, emesis and meal timing, ECG timing) and proposes trial-level acceptable ranges where useful, so that monitoring, edit checks and QC focus on them.

Why it matters. If critical data are not named early, monitoring and cleaning effort is spread evenly and errors in the data that drive the primary analysis go undetected.

NEEDS

Objectives and estimands; design and analysis overview; known risks of the design (for example carryover, emesis, sampling-time deviations).

Depends on: S1–02

PRODUCES

List of critical data and processes with proposed acceptable ranges; input to the risk assessment, monitoring and data management plans, edit checks and risk-based QC tiering.

Feeds: S2–33

QUALITY CONTROL

Cross-functional review; each critical item mapped to a control (edit check, monitoring step or QC level).

COMMON PITFALLS

- Declaring every variable critical, which dilutes monitoring and QC effort.
- Setting tolerance limits without a statistical rationale or a defined action when they are exceeded.

STANDARDS AND GUIDANCE

- ICH E6(R3): quality by design: critical-to-quality factors and their risks identified before trial initiation, prespecified acceptable ranges such as quality tolerance limits, and the quality management approach described in the CSR (3.10)
- ICH E8(R1): critical-to-quality factors identified for each study, including a prespecified SAP and systems that protect critical data (3.2, 7)

BA–01 Relative bioavailability or formulation-bridging design and precision basis

Document · Biostatistics · Lead biostatistician with Clinical pharmacologist, PK scientist, Regulatory affairs · Protocol development starts → Protocol final · Relative bioavailability

Protocol statistics for a study comparing formulations, or a formulation with an intravenous or oral solution reference: whether the objective is to estimate relative bioavailability or to bridge formulations against predefined confidence-interval limits, the design (crossover, parallel for a long half-life, extra periods for several formulations), the reference, dose normalization when strengths differ, and a precision- or power-based sample size.

Why it matters. If the objective and decision criteria are not set in advance, a study sized for estimation may later be judged against limits it cannot meet.

NEEDS

Formulation development plan; prior PK and variability; intended use of the results (dose selection, bridging a clinical formulation to the to-be-marketed one); half-life.

Depends on: S1–10

PRODUCES

Protocol objectives, design and sample-size sections; inputs to the BA analysis section of the SAP and the randomization specification.

Feeds: S1–03

QUALITY CONTROL

Independent review of objective, decision criteria and sample-size calculation against the development plan.

COMMON PITFALLS

- Leaving decision criteria undefined, then judging results against limits chosen afterward.
- Comparing different strengths without prespecified dose normalization.

STANDARDS AND GUIDANCE

- Bioavailability Studies Submitted in NDAs or INDs:…: crossover usual, parallel for a long half-life, and a 90% CI with predefined boundaries when comparing the to-be-marketed and clinical trial formulations (II.A); an intravenous reference defines absolute BA (II), and single-period isotopic labeling is an acceptable alternative for absolute BA (III.A.3)
- ICH M13A: relative BA and formulation bridging without the need to show BE are outside M13A; design and decision criteria follow the study objective (1.3)

BE–01 BE design and sample-size basis (2x2, replicate or parallel)

Document · Biostatistics · Lead biostatistician with PK scientist, Clinical pharmacologist · Protocol development starts → Protocol final · Bioequivalence

Protocol statistics for a bioequivalence study: a randomized single-dose crossover, a partial or full replicate crossover for a highly variable drug, or a parallel design for a long half-life, with washout and fasting or fed conditions. The sample size is calculated for the chosen design and analysis, with the assumed GMR, within-participant CV, power, dropout allowance and sensitivity to the assumptions stated.

Why it matters. A design or sample size that does not match the planned analysis can leave the study underpowered or impossible to analyze as planned.

NEEDS

Within-participant CV and GMR from earlier studies or literature; half-life; comparator and strength; fasting or fed requirement; planned analysis (average or reference-scaled BE).

`Depends on: S1–10`

PRODUCES

Protocol design and sample-size sections; sequences for the randomization specification; inputs to the BE analysis section of the SAP.

`Feeds: S1–03`

QUALITY CONTROL

Independent re-calculation of the sample size with the stated software and assumptions; check that design, sequences and planned analysis agree.

COMMON PITFALLS

- Sizing a replicate or parallel study with the standard 2x2 crossover formula.
- Planning spare participants whose data count only if others drop out; M13A does not accept them.

STANDARDS AND GUIDANCE

- ICH M13A: randomized single-dose crossover recommended; washout of, for example, at least 5 half-lives; parallel design for long half-lives (2.1.2); at least 12 evaluable participants (12 per treatment group in a parallel design); spare participants not acceptable; added cohorts only if prespecified and before bioanalysis results are known (2.1.3)
- Statistical Approaches to Establishing Bioequivalence: power of 0.8 or higher at a target GMR (for example 0.95); sample size specific to the design, by simulation when no analytical solution exists; sensitivity to assumptions; at least 24 participants recommended for highly variable drug products (II.A.3; final May 2026)

IN THIS TOOLKIT

Worked study: protocol statistics with sample size, sensitivity grid and dropout inflation (study/01_design/protocol_statistics.html) `WORKED STUDY`

DDI–01 Drug interaction study design and sample-size basis

Document · Biostatistics · Lead biostatistician with Clinical pharmacologist, PK scientist · Protocol development starts → Protocol final · Drug-drug interaction

Protocol statistics for a clinical drug interaction study: the investigational drug's role (object or precipitant), a one-sequence or randomized crossover (preferred) or a parallel design for long half-lives, the precipitant's dose, duration and timing (for example multiple-dose pretreatment for inducers), substrate dosing, and a sample size giving a reliable estimate of the size and variability of the interaction.

Why it matters. An underdosed or too-short precipitant regimen, or an imprecise estimate, can miss or understate an interaction that labeling must address.

NEEDS

In vitro and PBPK predictions; precipitant and substrate PK (half-lives, time to steady state, enzyme turnover); within-participant variability; safety of combined dosing.

`Depends on: S1–10`

PRODUCES

Protocol design, dosing schedule and sample-size sections; inputs to the DDI analysis section.

`Feeds: S1–03`

QUALITY CONTROL

Independent review of the dosing schedule against precipitant and substrate PK; sample-size rationale re-derived.

COMMON PITFALLS

- Precipitant dosing too short to cover the substrate profile, especially when inhibition prolongs its half-life.
- Giving a mixed inhibitor and inducer at the same time, so inhibition masks maximal induction.

STANDARDS AND GUIDANCE

- ICH M12: crossover (one-sequence or randomized) preferred, parallel when a long half-life rules out crossover (3.2.1.5); maximum precipitant dose at the shortest interval (3.2.1.2); inducer pretreatment typically 7 to 14 days, and precipitant dosing long enough to cover at least 90% of the substrate's concentration-time curve (3.2.1.3); enough participants for a reliable estimate of magnitude and variability (5.3.1)

FE–01 Food-effect design and sample-size basis

Document · Biostatistics · Lead biostatistician with Clinical pharmacologist, PK scientist · Protocol development starts → Protocol final · Food effect

Protocol statistics for a food-effect study: randomized, balanced, single-dose, two-period crossover of fed versus fasted (three periods when two formulations are compared), washout, a parallel design for a long half-life, the test meal, the dose, and a sample size that characterizes the food effect given PK variability. It separates a pilot assessment (for example within first-in-human) from the definitive study with the to-be-marketed formulation.

Why it matters. A design or meal that does not follow the guidance, or a pilot result treated as definitive, can leave the food statement in labeling unsupported.

NEEDS

Formulation status (clinical or to-be-marketed); PK variability and half-life; tolerability of a high-fat meal; safety of the planned dose.

`Depends on: S1–10`

PRODUCES

Protocol design, meal specification and sample-size sections; inputs to the randomization specification and the food-effect analysis section.

`Feeds: S1–03`

QUALITY CONTROL

Independent review of design, meal composition and sample-size rationale against the guidance.

COMMON PITFALLS

- Sizing only to show no effect when a large food effect is expected and its size must be estimated.
- Treating a pilot cohort on the clinical formulation as definitive for a different to-be-marketed formulation.

STANDARDS AND GUIDANCE

- Assessing the Effects of Food on Drugs in INDs and NDAs:…: randomized, balanced, single-dose, two-treatment, two-period crossover; washout of 3 to 5 half-lives; parallel design when half-life exceeds 24 hours; sample size depends on PK variability; high-fat meal of 800 to 1000 kcal with more than 50% of calories from fat; definitive study with the to-be-marketed formulation (III, IV; Revision 1, May 2026)
- CPMP/EWP/560/95/Rev. 1 Corr. 2**: EU standard procedure for food interaction studies: 10-hour fast, meal started 30 minutes before dosing, high-fat meal of 800 to 1000 kcal with 500 to 600 kcal from fat (Appendix I, which continues to apply after ICH M12)

HI-01 Hepatic impairment design and sample-size basis

Document · Biostatistics · Lead biostatistician with Clinical pharmacologist, PK scientist, Medical monitor · Protocol development starts → Protocol final · Hepatic impairment

Protocol statistics for a hepatic impairment PK study: mild, moderate and severe groups by Child-Pugh class, or a staged design that starts with one impairment group; a within-study control group matched on factors that affect PK; stability requirements before dosing; unbound sampling for highly bound drugs; and a precision-based sample size per group.

Why it matters. Unmatched or historical controls and unstable classification can hide or exaggerate the effect of impairment on exposure.

NEEDS

Hepatic elimination and protein binding; expected population and liver disease etiology; prior variability; safety of dosing in severe impairment.

Depends on: S1-10

PRODUCES

Protocol design, classification, matching and sample-size sections; inputs to the hepatic analysis section and enrollment tracking.

Feeds: S1-03

QUALITY CONTROL

Independent review of classification, staging rule, matching and sample-size calculation.

COMMON PITFALLS

- Using historical controls instead of a matched within-study control group.
- Assuming the mild group is impaired; participants in Child-Pugh class A may have normal hepatic function.

STANDARDS AND GUIDANCE

- Pharmacokinetics in Patients with Impaired Hepatic…: mild, moderate and severe groups plus a control, or severe impairment first; Child-Pugh scores 5 to 6 (A), 7 to 9 (B) and 10 to 15 (C); controls similar on factors affecting PK; example sample size targeting a 95% CI within 60% to 140% with at least 80% power (IV; draft, September 2026)
- CPMP/EWP/2339/02: moderate impairment and controls as a screening design, expanded if an effect is found; within-study control group recommended and historical controls discouraged; clinically relevant difference prespecified (3.2)

QT-01 Cardiac safety (QT) design and sample-size basis

Document · Biostatistics · Lead biostatistician with Pharmacometrician, Clinical pharmacologist, ECG core laboratory · Protocol development starts → Protocol final · Concentration-QTc or thorough QT, Single ascending dose, Multiple ascending dose

CONDITIONAL If a thorough QT study or a concentration-QTc analysis is planned

Protocol statistics for QT assessment: a thorough QT study (randomized, blinded, placebo-controlled, with a positive control for assay sensitivity, crossover or parallel) or a concentration-QTc analysis of first-in-human or MAD data with doses giving exposures well above the high clinical exposure. Either way it fixes replicate baseline ECGs, ECG times matched to PK samples around Cmax, and a sample-size or precision rationale.

Why it matters. If exposures do not reach a sufficient multiple of the high clinical exposure, or ECGs are not matched to PK times, a dedicated thorough QT study may still be needed.

NEEDS

Predicted therapeutic and high clinical exposures (including interaction and organ-impairment scenarios); nonclinical QT findings; PK profile and half-life; dose limits for safety.

Depends on: S1-10

PRODUCES

Protocol ECG schedule, dose plan, positive-control decision and sample-size sections; inputs to the cardiodynamic analysis plan and the ECG core-laboratory specification.

Feeds: S1-03

QUALITY CONTROL

Independent review of exposure coverage and the ECG schedule; sample-size or precision calculation re-derived.

COMMON PITFALLS

- Planning concentration-QTc without exposures covering the high clinical exposure, such as interaction or organ impairment.
- ECG times that do not match PK sampling, preventing concentration-matched analysis.

STANDARDS AND GUIDANCE

- ICH E14: thorough QT study with randomization, blinding, placebo and a positive control, multiple baseline ECGs and ECGs around Cmax (2.2)
- ICH E14/S7B Q&As (R1): concentration-QTc data may come from first-in-human or MAD studies; no separate positive control when responses are characterized at a sufficient multiple, commonly 2x, of the high clinical exposure, or, when only the high clinical exposure is reached, with a justified nonclinical integrated risk assessment as supplementary evidence (Q&A 5.1)

RI-01 Renal impairment design and sample-size basis

Document · Biostatistics · Lead biostatistician with Clinical pharmacologist, PK scientist · Protocol development starts → Protocol final · Renal impairment

Protocol statistics for a renal impairment PK study: a full design across renal function groups, or a reduced worst-case design (severe impairment versus control) with a staged expansion rule; one renal function method for enrollment; a control group matched on factors that affect PK; dialysis periods if needed; and a precision-based sample size per group.

Why it matters. Mixed renal function methods, unmatched controls or too few participants at low function make the dose recommendation unreliable.

NEEDS

Fraction excreted renally and elimination pathways; predicted exposure change; expected population and dialysis use; prior PK variability.

Depends on: S1-10

PRODUCES

Protocol design, group definitions, matching criteria and sample-size sections; inputs to the renal analysis section and enrollment tracking.

Feeds: S1-03

QUALITY CONTROL

Independent review of group definitions, matching and sample-size calculation; enrollment method checked against the planned analysis.

COMMON PITFALLS

- Enrolling on eGFR normalized to body surface area but analyzing absolute mL/minute, or the reverse.
- Filling the severe group near its upper cutoff, leaving the lowest renal function underrepresented.

STANDARDS AND GUIDANCE

- Pharmacokinetics in Patients with Impaired Renal…: full design for drugs mainly excreted by the kidneys, reduced design otherwise; groups by renal function in mL/minute (control at least 90; mild 60 to under 90; moderate 30 to under 60; severe, or kidney failure not on dialysis, under 30) using one common method, with eGFR preferred over eCLcr; example sample size targeting a 95% CI within 60% to 140% of the geometric mean with at least 80% power (IV; final March 2024)
- Guideline on the evaluation of the pharmacokinetics of…: full-range, reduced or staged designs; 6 to 8 participants per group usually sufficient in a full-range study, with adequate representation at low GFR; groups comparable on factors affecting PK (5, 5.1)

STAGE 2 · CLOSES AT SAP AND SPECIFICATIONS FINAL

Planning & specifications

Everything is specified before data arrive: the SAP, table shells, dataset specifications, the randomization schedule, the data-review plan and the QC plan. 41 deliverables.

ID	DELIVERABLE	OWNER	FROM → TO	DESIGNS
S2-01	Statistical analysis plan (SAP)	Lead biostatistician	G1 → G2	All
S2-02	SAP amendment and version-control log	Lead biostatistician	G2 → G5	All
S2-03	Analysis-set definitions and derivation rules (SAP level)	Lead biostatistician	G1 → G2	All
S2-04	TLF shells (tables, listings and figures)	Lead biostatistician	G1 → G2	All
S2-05	TLF inventory (table of contents)	Lead biostatistician	G1 → G2	All
S2-06	ADaM specification: ADSL (Subject-Level Analysis Dataset)	Statistical programmer	G1 → G2	All
S2-07	ADaM specification: ADPC (PK concentrations)	Statistical programmer	G1 → G2	All
S2-08	ADaM specification: ADPP (PK parameters)	Statistical programmer	G1 → G2	All
S2-09	ADaM specification: ADAE (adverse events)	Statistical programmer	G1 → G2	All
S2-10	ADaM specifications: ADLB, ADVS and ADEG (laboratory, vital signs, ECG)	Statistical programmer	G1 → G2	All
S2-11	ADaM specification: ADPD (pharmacodynamics) CONDITIONAL	Statistical programmer	G1 → G2	All
S2-12	SDTM mapping specification	Statistical programmer	G1 → G2	All
S2-13	Annotated CRF (aCRF): preparation and review	Statistical programmer	G1 → G2	All
S2-14	Programming and derivation specifications	Statistical programmer	G1 → G2	All
S2-15	PK analysis plan (NCA detail)	PK scientist	G1 → G2	All
S2-16	Population PK and exposure-response modeling analysis plan (PMAP) CONDITIONAL	Pharmacometrician	G1 → G3	All
S2-17	PK/PD modeling dataset specification CONDITIONAL	Pharmacometrician	G1 → G3	All
S2-18	Randomization schedule generation CONDITIONAL	Independent statistician	G1 → M1	All
S2-19	Randomization specification (restricted, pre-generation) CONDITIONAL	Lead biostatistician	G1 → M1	All
S2-20	Unblinding and code-break plan CONDITIONAL	Lead biostatistician	G1 → M1	All
S2-21	Data review plan (blinded data review)	Lead biostatistician	G1 → G2	All
S2-22	Define-XML planning and metadata strategy	Statistical programmer	G1 → G2	All
S2-23	Controlled terminology plan and dictionary version pinning	Statistical programmer	G1 → G2	All
S2-24	Validation and QC plan (independent double-programming strategy)	Lead biostatistician	G1 → G2	All
S2-25	Dummy data for shell and program testing	Statistical programmer	G1 → M3	All
S2-26	Data-transfer specifications (external data interface review)	Data manager	G1 → M1	All
S2-27	Dose-escalation and operating-characteristics simulation plan and outputs CONDITIONAL	Lead biostatistician	G1 → G2	All
S2-28	Interim analysis plan: committee statistical output specifications and firewall CONDITIONAL	Independent statistician	G1 → M2	All
S2-29	Bioanalytical data-transfer and PK sample reconciliation specification	PK scientist	G1 → M1	All
S2-30	Value-level metadata (VLM) specification for Define-XML	Lead statistical programmer	G1 → G2	All
S2-31	Software and computing environment validation and qualification records	Lead statistical programmer	G1 → G2	All
S2-32	IRT user acceptance testing: randomization and cohort-assignment scripts and sign-off CONDITIONAL	Lead biostatistician	G1 → M1	All
S2-33	eCRF design review for analysis-critical data	Lead biostatistician	G1 → M1	All
BA-02	BA analysis section: comparisons, reference and data-removal rules	Lead biostatistician	G1 → G2	BA
BE-02	BE analysis section: model, acceptance limits and data-removal rules	Lead biostatistician	G1 → G2	BE
DDI-02	Drug interaction analysis section: no-effect boundaries and model	Lead biostatistician	G1 → G2	DDI
FE-02	Food-effect analysis section	Lead biostatistician	G1 → G2	FE
HI-02	Hepatic impairment analysis section	Lead biostatistician	G1 → G2	HI
QT-02	Cardiodynamic (concentration-QTc) analysis plan CONDITIONAL	Pharmacometrician	G1 → G2	QT, SAD, MAD
QT-03	ECG core-laboratory data transfer and blinded-reading specification CONDITIONAL	Data manager	G1 → M1	QT, SAD, MAD
RI-02	Renal impairment analysis section	Lead biostatistician	G1 → G2	RI

S2–01 Statistical analysis plan (SAP)

Plan · Biostatistics · Lead biostatistician with PK scientist, Clinical pharmacologist, Medical monitor, Statistical programmer · Protocol final → SAP and specifications final · All designs

The signed plan that operationalizes the protocol: estimands, analysis sets, all PK, PD and safety methods, any inferential models (BE, food-effect, DDI, concentration-QTc), missing-data and intercurrent-event handling, multiplicity, BLQ rules cross-referenced to the PK analysis plan, and the planned outputs. It is finalized before database lock and unblinding.

Why it matters. Analyses specified after the data are seen are not credible, and programmers and reviewers have no authoritative source for derivations.

NEEDS

Final protocol and amendments; estimand framework; committee charter; regulatory feedback; scope of ADaM and TLF work.

`Depends on: S1–11`

PRODUCES

Authoritative source for analysis-set rules, ADaM and programming specifications, TLF shells and Stage 4 analyses; CSR Appendix 16.1.9.

`Feeds: S2–02, S2–03, S2–15`

QUALITY CONTROL

Senior biostatistician peer review; cross-check against the protocol and estimands; documented sign-off; version control of all amendments.

COMMON PITFALLS

- SAP finalization date not recorded clearly, so finalization before unblinding cannot be shown.
- Methods copied from a template that does not match the protocol's design or analysis sets.
- BLQ, time and exclusion rules stated differently in the SAP and the PK analysis plan.

STANDARDS AND GUIDANCE

- ICH E9: prespecified analysis; SAP finalized before breaking the blind, with formal records of when
- ICH E9(R1): estimands and sensitivity analyses
- ICH E3: CSR Section 9.7.1 and Appendix 16.1.9 report the planned statistical methods
- ICH M13A: BE model prespecified; no post hoc or data-driven adjustments to the primary analysis

IN THIS TOOLKIT

Worked study: SAP excerpt (estimand, analysis sets, BE model, BLQ rules) (study/02_sap/sap_excerpt.html) · Worked study: SAP settings read by the analysis code (study/02_sap/analysis_parameters.json) `WORKED STUDY`

S2–02 SAP amendment and version-control log

Log or record · Quality & compliance · Lead biostatistician with Quality assurance · SAP and specifications final → CSR draft · All designs

Change-control record for every SAP revision: what changed (estimands, methods, analysis sets, TLF scope), why, who approved it, and when each change was made relative to the blind review and unblinding. Changes made after unblinding are flagged so the CSR can describe them.

Why it matters. If the timing of changes cannot be shown, reviewers may treat the primary analysis as not prespecified.

NEEDS

Prior SAP versions; protocol amendments; committee or regulatory input; change requests; blind review outcomes.

`Depends on: S2–01`

PRODUCES

Audit trail for CSR Section 9.8 and Appendix 16.1.9; evidence that the primary analyses were prespecified.

`Feeds: S5–01`

QUALITY CONTROL

Independent review of each change; change dates checked against lock and unblinding dates; audit trail retained.

COMMON PITFALLS

- Post-unblinding changes recorded as routine clarifications instead of being flagged as post hoc.
- Version numbers in the log do not match the signed documents.

STANDARDS AND GUIDANCE

- ICH E3: Section 9.8: changes in planned analyses and their timing relative to blind breaking
- ICH E9: formal records of when the SAP was finalized and when the blind was broken
- 21 CFR Part 11: audit trails for electronic records

S2–03 Analysis-set definitions and derivation rules (SAP level)

Specification · Biostatistics · Lead biostatistician with PK scientist, Statistical programmer · Protocol final → SAP and specifications final · All designs

Precise, programmable rules for each analysis set (for example enrolled, randomized, safety, PK concentration, PK parameter, PD, per-protocol or completers, and BE, food-effect or DDI comparison sets), including profile-level PK exclusions and who decides them and when, normally at the blind review.

Why it matters. Ambiguous rules make population flags depend on programmer interpretation and change TLF denominators late.

NEEDS

Protocol analysis sets; SAP; estimand intercurrent-event handling; PK exclusion rules.

`Depends on: S2–01`

PRODUCES

ADSL population flags and their derivation rules; denominators for every TLF; decision list for the blind review.

`Feeds: S2–05, S2–06, S2–21`

QUALITY CONTROL

Independent reviewer confirms flag logic against the SAP; flags double-programmed in ADSL; set membership reconciled at the blind review.

COMMON PITFALLS

- A participant excluded from one PK parameter is removed from the whole PK set by an over-broad flag.
- Exclusion decisions made after bioanalysis or unblinding, with no record of their timing.

STANDARDS AND GUIDANCE

- ADaMIG v1.3: ADSL population flag variables
- ICH E9: analysis sets, and exclusion decisions taken at the blind review
- ICH M13A: BE analysis includes participants with evaluable data for both products; exclusions documented before sample analysis

IN THIS TOOLKIT

Worked study: analysis sets and ADaM flags in the SAP excerpt (study/02_sap/sap_excerpt.html) `WORKED STUDY`

S2–04 TLF shells (tables, listings and figures)

Specification · Biostatistics · Lead biostatistician with Statistical programmer, Medical writer, PK scientist · Protocol final → SAP and specifications final · All designs

Mock shells for every planned table, listing and figure with final titles, column headers, row structure, population footnotes, decimal conventions, programmer notes and placeholder values. Coverage spans disposition, demographics, exposure, deviations, PK concentrations and mean profiles (linear and semi-log), NCA parameters, dose proportionality, comparative PK, concentration-QTc, safety, PD, immunogenicity and listings.

Why it matters. Without agreed shells, programmers build outputs that do not match the CSR, and late layout changes cause rework and repeated QC.

NEEDS

SAP; PK analysis plan; TLF inventory; analysis-set definitions; the study's output conventions.

`Depends on: S2–05, S2–15`

PRODUCES

Programming target for Stage 4 TLF production; basis for medical-writing review of CSR tables.

`Feeds: S2–14, S4–02, S4–03, S4–04, S4–05, S4–06, S4–08, S4–09, S4–10, S4–11, S4–13, S4–14, S4–15, S4–16, S4–17, S4–18, HI–04, RI–04`

QUALITY CONTROL

Peer, medical-writer and clinical review; titles and footnotes checked against the SAP; layout tested with dummy data.

COMMON PITFALLS

- Mean concentration figures planned on actual times or individual plots on nominal times.
- Decimal conventions not set per parameter, so summaries show more precision than the assay supports.
- Shells lack population footnotes, so denominators are guessed during programming.

STANDARDS AND GUIDANCE

- ICH E3: tables and figures in the report text and in Section 14
- ICH M13A: individual and mean concentration-time graphs (linear and log-linear) and PK summary statistics

IN THIS TOOLKIT

Worked study: mock table shells (study/02_sap/mock_shells.html) · TLF program library by design (R) (r_tlf_pseudocode/index.html) `WORKED STUDY`

S2–05 TLF inventory (table of contents)

Specification · Biostatistics · Lead biostatistician with Lead statistical programmer, Medical writer · Protocol final → SAP and specifications final · All designs

Master numbered index of all tables, listings and figures with output number, title, analysis set and SAP cross-reference, marking in-text versus appendix placement and the CSR section each output supports. It drives scope, production tracking and QC completeness checks.

Why it matters. Without one inventory, outputs are missed or duplicated and QC completeness cannot be demonstrated.

NEEDS

SAP; analysis-set definitions; ICH E3 report structure.

`Depends on: S2–03`

PRODUCES

Production tracking list; CSR section mapping; QC completeness ledger.

`Feeds: S2–04, S2–24`

QUALITY CONTROL

Cross-check against SAP analyses (no orphan or missing outputs); peer review.

COMMON PITFALLS

- Inventory not updated after SAP amendments, so completeness checks pass against an outdated list.
- Output numbers reused after deletions, breaking CSR cross-references.

STANDARDS AND GUIDANCE

- ICH E3: report structure: Section 14 tables and figures and Section 16.2 listings

IN THIS TOOLKIT

Worked study: list of table numbers and titles (study/04_analysis/tlf/tlf_titles.csv) · TLF program library by design (R), with its TLF inventory (r_tlf_pseudocode/index.html) `WORKED STUDY`

S2–06 ADaM specification: ADSL (Subject-Level Analysis Dataset)

Specification · Programming & CDISC · Statistical programmer with Lead biostatistician · Protocol final → SAP and specifications final · All designs

Specification of the one-record-per-participant dataset: population flags, planned and actual treatment variables (TRTxxP, TRTxxA), reference dates (for example TRTSDT), demographics, baseline characteristics, disposition and stratification, each with its derivation and metadata. Every other ADaM dataset merges from it.

Why it matters. Errors in ADSL propagate to every dataset and every denominator, so they are the costliest to find late.

NEEDS

SAP; analysis-set definitions; SDTM mapping for DM, EX, DS and SV; controlled terminology.

`Depends on: S2–03, S2–12`

PRODUCES

ADSL dataset (Stage 3); merge source for all ADaM datasets; population denominators.

`Feeds: S2–07, S2–09, S2–10, S3–18`

QUALITY CONTROL

Specification peer review; independent double programming of ADSL; CDISC conformance checks.

COMMON PITFALLS

- Period treatment variables (TRTxxA) not corrected when a participant is dosed out of the randomized sequence.
- Reference dates derived from planned rather than actual first dose, shifting treatment-emergent flags.

STANDARDS AND GUIDANCE

- ADaMIG v1.3: ADSL structure and variables
- Define-XML v2.1: variable metadata and derivation methods
- CDISC Controlled Terminology: codelists for flags and categories

IN THIS TOOLKIT

Worked study: ADaM specification (ADSL, ADPC, ADPP, ADAE) (study/03_adam/spec/adam_spec.xlsx) `WORKED STUDY`

S2-07 ADaM specification: ADPC (PK concentrations)

Specification · Programming & CDISC · Statistical programmer with PK scientist, Lead biostatistician · Protocol final → SAP and specifications final · All designs

BDS specification for PK concentration analysis data: nominal and actual relative times (NFRLT and AFRLT from first dose, NRRLT and ARRLT from the reference dose), analyte, concentration and units, BLQ and LLOQ flags and handling rules, profile and record exclusion flags, and pre-dose handling. It is the analysis-ready bridge from SDTM PC and EX to the NCA input.

Why it matters. Errors in times or BLQ handling flow straight into Cmax, AUC and half-life and are hard to detect once NCA has run.

NEEDS

SAP and PK analysis plan (BLQ, time and exclusion rules); SDTM PC and EX mapping; bioanalytical LLOQ; ADSL specification.

`Depends on: S2-06, S2-15`

PRODUCES

ADPC dataset (Stage 3); NCA input; concentration tables and figures.

`Feeds: S2-08, S2-11, S2-17, S3-19`

QUALITY CONTROL

Specification review against the PK analysis plan (BLQ and time rules); independent double programming of ADPC; reconciliation to SDTM PC.

COMMON PITFALLS

- Relative times computed from the first dose instead of the period's dose in crossover or multiple-dose studies.
- BLQ values overwritten in ADPC, so listings no longer show what the laboratory reported.
- Pre-dose samples drawn after dosing kept at time zero without a flag.

STANDARDS AND GUIDANCE

- ADNCA v1.0: NCA input dataset structure and relative-time variables (NFRLT, AFRLT, NRRLT, ARRLT)
- ADaMIG v1.3: BDS structure
- ICH M13A: BE studies: PK parameters calculated on actual sampling times; BLQ concentrations set to zero and omitted from kel

IN THIS TOOLKIT

Worked study: ADaM specification including ADPC (study/03_adam/spec/adam_spec.xlsx) `WORKED STUDY`

S2-08 ADaM specification: ADPP (PK parameters)

Specification · Programming & CDISC · Statistical programmer with PK scientist, Lead biostatistician · Protocol final → SAP and specifications final · All designs

BDS specification for NCA parameters carried into analysis (for example Cmax, tmax, AUC(0-t), AUC(0-inf), half-life, CL/F, Vz/F, percent AUC extrapolated, lambda-z and accumulation ratios): parameter codes and terminology, units, dose, period and treatment structure, and the rules that link validated NCA output to the dataset. Reported values come from the validated NCA software, not from re-derivation in the dataset program.

Why it matters. Parameter naming, unit or linkage errors break dose-proportionality, comparative PK and exposure-response analyses downstream.

NEEDS

SAP and PK analysis plan (parameter list and acceptance rules); ADPC specification; structure of the NCA output; ADSL specification.

`Depends on: S2-07`

PRODUCES

ADPP dataset; NCA summary tables, dose-proportionality, comparative PK and exposure-response inputs.

`Feeds: S2-14, S3-20`

QUALITY CONTROL

Specification review against the parameter list; ADPP reconciled to validated NCA output; independent double programming of the structure.

COMMON PITFALLS

- Parameters failing acceptance rules (for example adjusted R-squared) deleted instead of flagged, losing traceability.
- AUC or clearance units inconsistent with the concentration and dose units.

STANDARDS AND GUIDANCE

- ADaMIG v1.3: BDS structure for parameter-level analysis data
- CDISC Controlled Terminology: PK parameter codelists
- ICH M13A: primary and additional PK parameters to tabulate for BE

IN THIS TOOLKIT

Worked study: ADaM specification including ADPP (study/03_adam/spec/adam_spec.xlsx) `WORKED STUDY`

S2-09 ADaM specification: ADAE (adverse events)

Specification · Programming & CDISC · Statistical programmer with Lead biostatistician, Medical monitor · Protocol final → SAP and specifications final · All designs

Occurrence data structure specification for adverse events: MedDRA SOC and PT, treatment-emergent flag derivation relative to dosing (and to each period in crossover studies), severity, relationship and seriousness, and occurrence flags for first-occurrence summaries.

Why it matters. Treatment-emergence and period-attribution errors misstate safety by dose or treatment, which drives escalation decisions and the CSR.

NEEDS

SAP; SDTM AE mapping; MedDRA version; ADSL specification; exposure data.

`Depends on: S2-06`

PRODUCES

ADAE dataset; TEAE, SAE and death tables; AE listings.

`Feeds: S2-14, S3-21`

QUALITY CONTROL

Specification review of treatment-emergent logic; independent double programming; reconciliation to SDTM AE.

COMMON PITFALLS

- Events starting on a dosing day without a time are assigned inconsistently to pre- or post-dose.
- Crossover AEs assigned to a period without a rule for events that start during washout.

STANDARDS AND GUIDANCE

- ADaM OCCDS v1.1: occurrence data structure and occurrence flags
- MedDRA Version 29.1: SOC and PT hierarchy and dictionary version
- Define-XML v2.1: variable metadata and derivation methods

IN THIS TOOLKIT

Worked study: ADaM specification including ADAE (study/03_adam/spec/adam_spec.xlsx) `WORKED STUDY`

S2–10 ADaM specifications: ADLB, ADVS and ADEG (laboratory, vital signs, ECG)

Specification · Programming & CDISC · Statistical programmer with Lead biostatistician, Medical monitor · Protocol final → SAP and specifications final · All designs

BDS specifications for safety laboratory (chemistry, hematology, urinalysis), vital signs and ECG data: baseline and change from baseline, analysis visits and windows, standardized units, reference ranges, abnormality and toxicity-grade flags, and for ECG the heart-rate-corrected QT variables and categorical outlier flags.

Why it matters. Baseline, window or unit errors distort shift tables, outlier counts and the QTc data used for concentration-QTc analysis.

NEEDS

SAP; SDTM LB, VS and EG mapping; controlled terminology; reference ranges and toxicity grading scale; QT correction method from the protocol or SAP.

`Depends on: S2–06`

PRODUCES

ADLB, ADVS and ADEG datasets; safety summary and shift tables; concentration-QTc input.

`Feeds: S2–14, S3–22, S3–23, S3–24`

QUALITY CONTROL

Specification review of baseline, window and unit logic; independent double programming; reconciliation to SDTM.

COMMON PITFALLS

- Baseline taken as the last pre-dose value overall instead of per period in crossover studies.
- Replicate ECGs averaged differently at baseline and post-dose time points.
- Local and central laboratory ranges mixed without a rule for abnormality flags.

STANDARDS AND GUIDANCE

- ADaMIG v1.3: BDS baseline, change-from-baseline and analysis visit variables
- ICH E14: heart-rate corrections (QTcF, QTcB) and categorical QTc analyses
- CTCAE v6.0 (MedDRA 28.0): toxicity grading of laboratory abnormalities
- Drug-Induced Liver Injury: Premarketing Clinical Evaluation: Hy's law laboratory criteria

S2–11 ADaM specification: ADPD (pharmacodynamics)

Specification · Programming & CDISC · Statistical programmer with Clinical pharmacologist, Pharmacometrician · Protocol final → SAP and specifications final · All designs

`CONDITIONAL` If PD or biomarker endpoints are collected

BDS specification for PD and biomarker endpoints: parameter definitions, baseline, change and percent change, analysis windows, and the keys that link PD records to time-matched concentrations for PK/PD analysis.

Why it matters. Without time-matched keys and consistent baselines, PK/PD and exposure-response analyses cannot be built from the dataset.

NEEDS

SAP or PD analysis plan; SDTM PD or biomarker mapping; ADSL and ADPC specifications.

`Depends on: S2–07`

PRODUCES

ADPD dataset; PD tables and figures; exposure-response inputs.

`Feeds: S2–14, S3–25`

QUALITY CONTROL

Specification review against the PD plan; independent double programming; reconciliation to SDTM.

COMMON PITFALLS

- Percent change computed when baseline is zero or near the assay limit.
- PD and PK samples matched on nominal time although actual times differ.

STANDARDS AND GUIDANCE

- ADaMIG v1.3: BDS structure for PD parameters
- Define-XML v2.1: value-level metadata for each PD parameter

S2–12 SDTM mapping specification

Specification · Programming & CDISC · Statistical programmer with Data manager · Protocol final → SAP and specifications final · All designs

Source-to-SDTM mapping that defines how each CRF field and external transfer maps to SDTM domains and variables (for example DM, EX, PC, PP, AE, LB, VS, EG, DS, MH, CM, DV), including controlled-terminology assignments, derivations, --SEQ and RELREC linkage, and supplemental qualifiers. It drives the SDTM build and the annotated CRF.

Why it matters. Unmapped or mis-mapped fields surface as conformance findings and traceability gaps at submission, when they are expensive to fix.

NEEDS

Final CRF and EDC structure; SDTMIG; controlled terminology plan; protocol; data-transfer specifications.

`Depends on: S2–22, S2–26, S2–29, S2–33`

PRODUCES

SDTM datasets (Stage 3); annotated CRF; SDTM Define-XML metadata.

`Feeds: S2–06, S2–13, S3–04, S3–05, S3–07, S3–08, S3–09, S3–10, S3–11, S3–12, S3–13, S3–14, S3–15`

QUALITY CONTROL

Independent review against SDTMIG and controlled terminology; conformance rule check; raw-to-SDTM traceability check.

COMMON PITFALLS

- PC and EX mapped without a shared key, so time after dose cannot be derived reliably.
- Sponsor-defined test codes used where a CDISC codelist value exists.

STANDARDS AND GUIDANCE

- SDTMIG v3.4: domain models, supplemental qualifiers and RELREC
- CDISC Controlled Terminology: codelists applied in mapping
- Define-XML v2.1: origin and derivation metadata

S2–13 Annotated CRF (aCRF): preparation and review

Document · Programming & CDISC · Statistical programmer with Data manager · Protocol final → SAP and specifications final · All designs

The blank CRF annotated with SDTM domains and variables, the human-readable link between collected data and SDTM. It is prepared and reviewed against the final CRF and the mapping specification and is expected in a standardized-data submission.

Why it matters. An aCRF that disagrees with the mapping or Define-XML confuses reviewers and generates conformance and reviewer questions.

NEEDS

Final blank CRF; SDTM mapping specification; SDTMIG and controlled terminology.

`Depends on: S2–12`

PRODUCES

Submission aCRF; origin metadata for Define-XML; input to the SDTM reviewer guide.

`Feeds: S3–03`

QUALITY CONTROL

Page-by-page review against the mapping specification; consistency with Define-XML origins; peer review.

COMMON PITFALLS

- Fields collected but not submitted left unannotated instead of marked as not submitted.
- aCRF built from an older CRF version than the one used in the study.

STANDARDS AND GUIDANCE

- Study Data Technical Conformance Guide: Technical…: annotated CRF expectations for submitted study data
- SDTMIG v3.4: SDTM variable names used in annotations

S2–14 Programming and derivation specifications

Specification · Programming & CDISC · Statistical programmer with Lead statistical programmer, Lead biostatistician · Protocol final → SAP and specifications final · All designs

Programmer-facing specifications behind ADaM datasets and TLFs: algorithmic derivation logic, variable- and value-level metadata, formats and lengths, and explicit rules for complex derivations such as windowing, baseline, treatment emergence, BLQ handling and dose normalization.

Why it matters. Without explicit derivation rules, production and QC programmers interpret the SAP differently and reconciliation fails late.

NEEDS

SAP; ADaM specifications; SDTM mapping; TLF shells; controlled terminology.

`Depends on: S2–04, S2–08, S2–09, S2–10, S2–11`

PRODUCES

Targets for production and independent QC programming in Stages 3 and 4.

`Feeds: S2–25, S2–30`

QUALITY CONTROL

Independent specification review; basis for double-programming reconciliation; traceability to the SAP.

COMMON PITFALLS

- QC programmer reads the production code instead of the specification, so shared errors pass reconciliation.
- Rounding rules not stated, so production and QC differ only in the last decimal.

STANDARDS AND GUIDANCE

- ADaMIG v1.3: traceability of derivations to source data
- Define-XML v2.1: computational methods and value-level metadata

IN THIS TOOLKIT

Worked study: ADaM specification that drives and checks the dataset programs (study/03_adam/spec/adam_spec.xlsx) `WORKED STUDY`

S2–15 PK analysis plan (NCA detail)

Plan · PK & pharmacometrics · PK scientist with Lead biostatistician, Clinical pharmacologist · Protocol final → SAP and specifications final · All designs

Detailed non-compartmental analysis specification, standalone or as an SAP appendix: full parameter list, BLQ handling by position in the profile, nominal versus actual time rules, lambda-z selection and acceptance criteria, profile and parameter exclusion rules, the AUC calculation method, and the validated NCA software and settings.

Why it matters. Unwritten NCA conventions let parameter values change with the analyst, and rules changed after seeing data cannot be defended.

NEEDS

Protocol PK schedule; SAP; bioanalytical LLOQ; sampling-time conventions; PK precision memo.

`Depends on: S2–01`

PRODUCES

ADPC and ADPP specifications; NCA setup; PK concentration and parameter tables.

`Feeds: S2–04, S2–07, S2–21, S3–42, ESC–02`

QUALITY CONTROL

PK scientist and biostatistician review; rules cross-checked against the SAP; NCA software settings and templates qualified.

COMMON PITFALLS

- Lambda-z points chosen by eye without documented criteria, so half-life is not reproducible.
- BLQ rules written only for pre-dose samples, leaving embedded and trailing BLQ values to analyst choice.
- AUC calculation method (linear or linear-up log-down) not stated.

STANDARDS AND GUIDANCE

- ICH M13A: BE studies: PK parameters, BLQ set to zero and omitted from kel, actual sampling times, and reporting of NCA methods
- Bioavailability Studies Submitted in NDAs or INDs:…: BA study design and PK measures

IN THIS TOOLKIT

Worked study: PK parameter rules in the SAP excerpt (study/02_sap/sap_excerpt.html) · Worked study: NCA settings, including the BLQ rule versus the software default (study/04_analysis/nca_settings.json) `WORKED STUDY`

S2–16 Population PK and exposure-response modeling analysis plan (PMAP)

Plan · PK & pharmacometrics · Pharmacometrician with Clinical pharmacologist, PK scientist, Lead biostatistician · Protocol final → Database lock · All designs

CONDITIONAL If population PK or exposure-response modeling is planned

Separate modeling plan for population PK and exposure-response: objectives, data and BLQ handling for modeling, structural, statistical and covariate model building, estimation methods, model evaluation (goodness of fit, visual predictive checks, bootstrap), simulation objectives and reporting. It is written before modeling starts.

Why it matters. Model choices made after seeing the data look data-driven and weaken dose-selection conclusions.

NEEDS

Protocol objectives and sampling design; SAP; PK and PD scope; prior models and literature.

Depends on: S1–11

PRODUCES

Modeling run plan; modeling dataset specification; pharmacometrics report (Stage 4); dosing recommendations.

Feeds: S2–17

QUALITY CONTROL

Independent pharmacometrics review; model-evaluation criteria prespecified; run records reproducible.

COMMON PITFALLS

- BLQ handling for modeling differs from the NCA rules and the difference is not explained.
- Covariate search strategy left open, inviting data-driven selection.

STANDARDS AND GUIDANCE

- Population Pharmacokinetics: Guidance for Industry: data, model development and model evaluation in population PK analyses
- Exposure-Response Relationships: Study Design, Data…: exposure-response study design and data analysis
- CHMP/EWP/185990/06: reporting of population PK analyses
- ICH M15: model analysis plan (MAP) pre-defined before accessing the data, with objectives, data, methods and planned model evaluation (4.1)

S2–17 PK/PD modeling dataset specification

Specification · PK & pharmacometrics · Pharmacometrician with Statistical programmer · Protocol final → Database lock · All designs

CONDITIONAL If population PK or exposure-response modeling is planned

Specification for the modeling-ready dataset: dose and observation event records (for example EVID, MDV, AMT, RATE, DV and time), covariates, BLQ handling for modeling, units, and traceability to ADPC, ADSL and exposure data.

Why it matters. Dosing and time errors in the modeling dataset bias every model parameter and are hard to detect from model diagnostics.

NEEDS

Modeling analysis plan; ADPC, ADSL and exposure data; PD data when exposure-response is in scope.

Depends on: S2–07, S2–16

PRODUCES

Modeling dataset; dataset appendix to the pharmacometrics report.

Feeds: S4–12

QUALITY CONTROL

Independent check against the ADaM sources; structure and covariate completeness verified; reproducible build.

COMMON PITFALLS

- Dose records built from planned rather than actual dosing times.
- Concentrations below the LLOQ dropped silently instead of flagged for the chosen modeling method.

STANDARDS AND GUIDANCE

- ADaM popPK IG v1.0: ADPPK dataset structure and variables
- Population Pharmacokinetics: Guidance for Industry: data used in population PK analyses

S2–18 Randomization schedule generation

Log or record · Biostatistics · Independent statistician with Randomization and supply (IRT) vendor · Protocol final → First participant dosed · All designs

CONDITIONAL If treatment or sequence allocation is randomized

Generation of the validated, reproducible randomization schedule, including any sentinel, cohort or sequence allocation: block and stratification structure per the specification, documented seed and software, restricted storage, and transfer to the IRT. In blinded studies it is produced under access control by a statistician outside the blinded study team.

Why it matters. An allocation error or an exposed schedule undermines the comparison and the blind and cannot be corrected after dosing.

NEEDS

Randomization specification (ratio, blocks, strata, sentinel and cohort rules); validated generation software; documented seed.

Depends on: S2–19

PRODUCES

Randomization list for the IRT and drug supply; treatment-assignment source for ADSL; CSR Appendix 16.1.7.

Feeds: S3–36

QUALITY CONTROL

Independent QC of the allocation against the specification (ratios, blocks, strata); seed and reproducibility verified; secured storage.

COMMON PITFALLS

- Schedule generated by the blinded study statistician who also writes the SAP.
- Test schedule used for IRT acceptance testing loaded to production by mistake.

STANDARDS AND GUIDANCE

- ICH E9: reproducible randomization schedule filed securely to protect the blind
- ICH E3: Appendix 16.1.7 randomization scheme and codes
- ICH M13A: randomization scheme provided with the BE report
- 21 CFR Part 11: access controls and audit trails for electronic records

IN THIS TOOLKIT

Worked study: synthetic randomization schedule (study/01_design/randomization_schedule.csv) · Worked study: balance check of the schedule (study/01_design/balance_check.csv)

WORKED STUDY

S2–19 Randomization specification (restricted, pre-generation)

Specification · Biostatistics · Lead biostatistician with Randomization and supply (IRT) vendor, Clinical operations · Protocol final → First participant dosed · All designs

CONDITIONAL If treatment or sequence allocation is randomized

Restricted specification written before generation: allocation ratio, block sizes, stratification, sentinel and cohort handling, IRT configuration requirements, and reproducibility (software, seed management). It is the controlled basis for schedule generation and IRT user acceptance testing.

Why it matters. If the IRT configuration and the schedule are not built from one approved specification, allocation and supply errors appear at first dosing.

NEEDS

Final protocol randomization design; dose-escalation rules; drug-supply constraints.

Depends on: S1–11

PRODUCES

Instructions for schedule generation; IRT setup and user acceptance testing; ADSL treatment derivation logic.

Feeds: S2–18, S2–20, S2–32

QUALITY CONTROL

Independent review against the protocol; alignment with IRT user acceptance testing; documented approval.

COMMON PITFALLS

- Specification circulated to the whole study team, exposing block size.
- Stratification factors used in randomization but missing from the analysis model.

STANDARDS AND GUIDANCE

- ICH E9: block length withheld from investigators; stratification factors accounted for in the analysis
- 21 CFR Part 11: controls for electronic records in randomization systems

IN THIS TOOLKIT

IRT user acceptance testing guide (allocation and unblinding test examples) (uat_irt/quarto/uat-guide.html)

S2–20 Unblinding and code-break plan

Plan · Safety & committees · Lead biostatistician with Medical monitor, Randomization and supply (IRT) vendor, Clinical operations · Protocol final → First participant dosed · All designs

CONDITIONAL If the study is blinded

Operational plan for planned unblinding after database lock and for emergency code-breaks: who may unblind and when, the lock-then-unblind sequence, how the blind is kept for analysis programming, firewall roles, the emergency procedure and its log, and committee access.

Why it matters. Unplanned or undocumented unblinding compromises blind decisions and the credibility of the analysis.

NEEDS

Protocol and design-stage unblinding provisions; committee charter; randomization specification; the study's SOPs.

Depends on: S2–19

PRODUCES

Controlled unblinding at lock (Stage 4); unblinding event log; firewall definition for analysis.

Feeds: S2–28, S3–37

QUALITY CONTROL

Alignment review against SOPs and the charter; role and firewall verification; emergency-unblinding procedure and log tested.

COMMON PITFALLS

- Unblinded randomization files stored where the analysis team can read them.
- Database unblinded before blind decisions are signed, so they can no longer be shown to be blind.

STANDARDS AND GUIDANCE

- ICH E9: procedure and timing for revealing treatment assignments documented
- ICH E2A: case-level unblinding for expedited reports while keeping biometrics personnel blinded
- ICH E6(R3): safeguarding blinding in data governance

IN THIS TOOLKIT

IRT user acceptance testing guide (emergency unblinding test example) (uat_irt/quarto/uat-guide.html)

S2–21 Data review plan (blinded data review)

Plan · Data management interface · Lead biostatistician with Data manager, PK scientist, Medical monitor · Protocol final → SAP and specifications final · All designs

Plan for the blind review of cleaned data before lock and unblinding: listings and summaries to produce, data-quality and protocol-deviation checks, PK profile and sample review, criteria and decision authority for analysis-set and PK exclusions, and how meeting decisions are documented.

Why it matters. Without a plan, exclusion decisions drift past unblinding and look outcome-driven.

NEEDS

SAP; analysis-set definitions; PK exclusion rules; protocol-deviation conventions; data management plan.

Depends on: S2–03, S2–15

PRODUCES

Blind review outputs and meeting (Stage 3); documented exclusion and analysis-set decisions before unblinding.

Feeds: S3–02, S3–38

QUALITY CONTROL

Peer review; decisions recorded with timestamps before unblinding; consistency with the SAP.

COMMON PITFALLS

- Protocol-violation exclusions for BE left to the pre-lock review, although ICH M13A expects them documented before bioanalysis.
- Review listings reveal treatment, for example concentrations shown for placebo participants.

STANDARDS AND GUIDANCE

- ICH E9: blind review before unblinding, with its decisions described in the report
- ICH M13A: exclusions from the BE analysis documented before bioanalytical analysis

S2–22 Define-XML planning and metadata strategy

Specification · Programming & CDISC · Statistical programmer with Lead statistical programmer · Protocol final → SAP and specifications final · All designs

Up-front plan for SDTM and ADaM Define-XML: dataset, variable and value-level metadata conventions, how origins and derivations are documented, controlled-terminology references and computational methods, built into the mapping and ADaM specifications so define.xml is produced by design rather than retrofitted.

Why it matters. Metadata retrofitted at submission produces define files that disagree with the data and the specifications.

NEEDS

Controlled terminology plan; Define-XML version; the study's metadata conventions; submission plans.

Depends on: S2–23

PRODUCES

Metadata conventions applied in the SDTM mapping and ADaM specifications; SDTM and ADaM Define-XML (Stage 3); reviewer guide content.

Feeds: S2–12

QUALITY CONTROL

Metadata completeness against the specifications; Define-XML checks; value-level metadata review.

COMMON PITFALLS

- Origins recorded as derived without the method, so reviewers cannot trace derivations.
- Specifications kept in a layout that cannot generate define.xml, forcing manual re-entry.

STANDARDS AND GUIDANCE

- Define-XML v2.1: dataset, variable and value-level metadata, origins, methods and codelists
- Study Data Technical Conformance Guide: Technical…: define.xml expectations for submitted study data

IN THIS TOOLKIT

Worked study: define-ready ADaM specification (define.xml not authored) (study/03_adam/spec/adam_spec.xlsx) WORKED STUDY

S2–23 Controlled terminology plan and dictionary version pinning

Specification · Programming & CDISC · Statistical programmer with Data manager · Protocol final → SAP and specifications final · All designs

Records which CDISC Controlled Terminology, MedDRA and WHODrug versions the study uses, how sponsor-defined or extensible terms are handled, and when versions may change, pinned at planning for reproducibility and conformance across SDTM and ADaM.

Why it matters. Version drift between coding, datasets and define.xml causes conformance findings and irreproducible counts.

NEEDS

SDTMIG and ADaMIG versions; CDISC Controlled Terminology releases; MedDRA and WHODrug versions; mapping conventions.

Depends on: S1–11

PRODUCES

Terminology applied in SDTM and ADaM; Define-XML codelist references; reviewer guide notes.

Feeds: S2–22, S3–35

QUALITY CONTROL

Version-pin verification; terminology conformance checks; review of extensible codelist values.

COMMON PITFALLS

- MedDRA version changed mid-study without re-coding or an impact assessment.
- Define.xml cites a terminology version different from the one applied in the datasets.

STANDARDS AND GUIDANCE

- CDISC Controlled Terminology: versioned controlled terminology releases
- MedDRA Version 29.1: dictionary versions for adverse event and medical history coding
- WHODrug Global (UMCDD): dictionary versions for medication coding

S2–24 Validation and QC plan (independent double-programming strategy)

Plan · Quality & compliance · Lead biostatistician with Lead statistical programmer, PK scientist, Quality assurance · Protocol final → SAP and specifications final · All designs

Risk-based plan that assigns a QC method to each ADaM dataset and TLF (independent double programming, independent review or code review), defines reconciliation and discrepancy logging, and sets software and environment qualification expectations. It requires every reported number, including NCA results, to come from a validated program with the software and version recorded.

Why it matters. Without a plan, QC effort is spent unevenly and critical outputs can reach the CSR without independent verification.

NEEDS

SAP; TLF inventory; dataset list; risk assessment; the study's SOPs.

Depends on: S2–05

PRODUCES

Stage 4 QC execution; reconciliation and discrepancy logs; validation records for the CSR and inspection.

Feeds: S2–31

QUALITY CONTROL

QC-level assignments reviewed and approved; double programming mandatory for key outputs; plan kept audit-ready.

COMMON PITFALLS

- QC assigned by output count rather than risk, so primary PK and comparison tables get only review-level QC.
- Same programmer writes production and QC code for a key output.

STANDARDS AND GUIDANCE

- ICH E9: reliable software with documented software testing
- ICH E6(R3): risk-proportionate quality management

S2–25 Dummy data for shell and program testing

Dataset · Programming & CDISC · Statistical programmer · Protocol final → Blinded data review · All designs

Synthetic datasets that follow the planned ADaM structures, used to test TLF shells, exercise programming specifications and dry-run derivations before real data arrive, without touching participant data.

Why it matters. Programs first run on real data at lock leave no time to fix defects, and edge cases such as BLQ values or missing windows go untested.

NEEDS

ADaM and programming specifications; TLF shells; CDISC structures.

`Depends on: S2–14`

PRODUCES

Shell renders and dry-run outputs; specification defect findings; production code ready before lock.

`Feeds: S3–31`

QUALITY CONTROL

Structure conformance to the ADaM specifications; coverage of edge cases (BLQ values, missing data, window boundaries).

COMMON PITFALLS

- Dummy data too clean, so BLQ, missing and out-of-window cases never exercise the code.
- Dry-run outputs with dummy treatment assignments circulated without a clear dummy-data label.

STANDARDS AND GUIDANCE

- ADaMIG v1.3: dataset structures the dummy data must follow

IN THIS TOOLKIT

Study-lifecycle sandbox: synthetic ADaM-shaped data (sim_lifecycle/index.html)

S2–26 Data-transfer specifications (external data interface review)

Specification · Data management interface · Data manager with Statistical programmer, Central laboratory, ECG core laboratory, Bioanalytical laboratory · Protocol final → First participant dosed · All designs

Biostatistics and programming input to transfer agreements for external data (for example central laboratory, ECG core laboratory, bioanalytical concentrations, biomarkers): file structure, units, LLOQ and BLQ conventions, identifiers and keys, transfer frequency and reconciliation approach, so vendor data load cleanly into SDTM.

Why it matters. Transfers designed without analysis input arrive with missing keys, unit changes or undocumented flags that delay lock.

NEEDS

Vendor agreements; CRF and data management plan; laboratory and bioanalytical specifications; SDTM targets.

`Depends on: S1–11`

PRODUCES

Agreed transfer specifications; clean loads into SDTM PC, LB and EG; reconciliation basis for Stage 3.

`Feeds: S2–12, QT–03, S3–01`

QUALITY CONTROL

Test-transfer review; key, format and unit verification; reconciliation procedure agreed.

COMMON PITFALLS

- Vendor changes units or reference ranges mid-study without a versioned transfer specification.
- Sample identifiers in the transfer cannot be matched to visit and time point in the CRF.

STANDARDS AND GUIDANCE

- SDTMIG v3.4: target domains for transferred data (PC, LB, EG)
- ICH E6(R3): agreements with service providers; data transfer, exchange and migration

S2–27 Dose-escalation and operating-characteristics simulation plan and outputs

Plan · Biostatistics · Lead biostatistician with Clinical pharmacologist, PK scientist · Protocol final → SAP and specifications final · All designs

CONDITIONAL If simulation supports escalation rules, sample size or adaptive elements

Planning-stage simulation specification with archived, reproducible code that characterizes operating characteristics where needed (dose-escalation rules, CV-driven precision or sample size, adaptive elements), supporting the committee's decision rules and the protocol's statistical justification. It is re-run when amendments change the rules.

Why it matters. Undocumented simulations cannot be reproduced when an amendment or a regulator questions the design.

NEEDS

Final protocol design and escalation rules; assumed PK and CV; prior data; design-stage operating-characteristics results.

`Depends on: S1–11`

PRODUCES

Operating-characteristics summaries supporting the committee rules and the SAP or protocol justification; archived code and seeds.

QUALITY CONTROL

Independent review of assumptions and code; seeds and environment documented; results sanity-checked.

COMMON PITFALLS

- Simulation re-run with different seeds or package versions, and protocol figures change without explanation.

STANDARDS AND GUIDANCE

- Adaptive Design Clinical Trials for Drugs and Biologics:…: simulation to evaluate operating characteristics of complex designs
- ICH E9: reliable software with documented software testing

S2–28 Interim analysis plan: committee statistical output specifications and firewall

Specification · Safety & committees · Independent statistician with Lead biostatistician, Statistical programmer · Protocol final → Dose-escalation or interim reviews · All designs

CONDITIONAL If interim analyses or unblinded committee reviews are planned

Operational specification of interim or committee outputs: the exact tables and listings, data cutoff and snapshot procedure, any formal stopping boundaries, the split between open and closed session outputs, and the statistical firewall that keeps the independent statistician separate from the blinded study team.

Why it matters. Interim outputs produced ad hoc are not reproducible and can leak comparative results to the study team.

NEEDS

Committee charter; escalation rules; unblinding and firewall plan.

Depends on: S2–20

PRODUCES

Interim output specifications and shells; firewall procedure; committee output packages.

Feeds: ESC–01

QUALITY CONTROL

Independent statistician produces outputs from documented snapshots; firewall access reviewed; outputs reproducible from archived code.

COMMON PITFALLS

- Data snapshot not archived, so committee outputs cannot be regenerated later.
- Open-session outputs show results by treatment group.

STANDARDS AND GUIDANCE

- ICH E9: staff involved in unblinded interim analyses kept out of the blind review and SAP changes
- Guidance for Clinical Trial Sponsors: Establishment and…: establishment and operation of data monitoring committees

IN THIS TOOLKIT

Example safety review committee data packet (synthetic MAD cohort) (sasr_monitoring_wiki/trialmon_quarto/trialmon-src-packet.html)

S2–29 Bioanalytical data-transfer and PK sample reconciliation specification

Specification · Data management interface · PK scientist with Bioanalytical laboratory, Statistical programmer, Data manager · Protocol final → First participant dosed · All designs

Dedicated transfer specification for the bioanalytical laboratory feed: concentration results and units, LLOQ and BLQ conventions, analytical run metadata, links from sample identifiers to collection times and nominal time points, and handling of reanalysis and incurred-sample reanalysis results, with rules for reconciling collected, shipped and assayed samples.

Why it matters. Bioanalytical fields missing from a generic transfer specification surface as PK gaps or mismatched samples at lock.

NEEDS

Bioanalytical laboratory data dictionary; sampling schedule; LLOQ definition; sample-handling manual.

Depends on: S1–11

PRODUCES

Bioanalytical transfer specification; sample reconciliation rules; BLQ and LLOQ handling conventions.

Feeds: S2–12, S3–29, ESC–02

QUALITY CONTROL

Double check against the laboratory data dictionary; PK scientist sign-off.

COMMON PITFALLS

- Reanalysis results replace original values without a reason code, so the reported concentration cannot be traced.
- BLQ sent as text in the numeric result field, breaking loads.

STANDARDS AND GUIDANCE

- ICH M10: study sample analysis, reanalysis of study samples, incurred sample reanalysis and bioanalytical reporting
- SDTMIG v3.4: PC domain structure
- ICH M13A: drug concentrations in BE studies measured according to ICH M10

S2–30 Value-level metadata (VLM) specification for Define-XML

Specification · Programming & CDISC · Lead statistical programmer with Statistical programmer · Protocol final → SAP and specifications final · All designs

Value-level metadata for parameter-based data: where-clauses, origins, derivations, codelists and computational methods for each PARAMCD in ADPC, ADPP, ADLB, ADVS, ADEG and ADPD, and for each test code in SDTM PC and PP, as Define-XML requires.

Why it matters. Missing value-level metadata leaves parameter-specific derivations and origins undocumented for reviewers.

NEEDS

ADaM and SDTM specifications; controlled terminology; derivation specifications.

Depends on: S2–14

PRODUCES

Value-level metadata tables that feed define.xml.

Feeds: S3–16, S3–26

QUALITY CONTROL

Independent review against the datasets; Define-XML validation.

COMMON PITFALLS

- Where-clauses written for parameters that never occur in the data.
- Units differ by parameter, but value-level metadata gives one unit for the variable.

STANDARDS AND GUIDANCE

- Define-XML v2.1: value-level metadata and where-clauses
- ADaMIG v1.3: parameter-level BDS structure

IN THIS TOOLKIT

Worked study: ADaM specification with value-level and where-clause tabs (study/03_adam/spec/adam_spec.xlsx) WORKED STUDY

S2–31 Software and computing environment validation and qualification records

Log or record · Quality & compliance · Lead statistical programmer with Quality assurance, PK scientist, Pharmacometrician · Protocol final → SAP and specifications final · All designs

Evidence of the validated state of the analysis software and computing environment: installation and operational qualification records, versions of statistical, NCA, modeling and conformance software, change control, and a validated-environment attestation for the study.

Why it matters. Reported numbers are not defensible at inspection if the software that produced them cannot be shown to be validated and version-controlled.

NEEDS

Software inventory; vendor qualification documents; environment configuration; validation and QC plan.

`Depends on: S2–24`

PRODUCES

Software validation summary; version register; validated-environment attestation.

`Feeds: S4–01`

QUALITY CONTROL

QA review; reported numbers produced only in qualified environments.

COMMON PITFALLS

- Packages or software updated mid-study without re-qualification or a version record.
- Qualification covers the server but not the open-source packages actually used.

STANDARDS AND GUIDANCE

- GAMP 5 Guide, 2nd Edition: risk-based approach to computerized system validation
- 21 CFR Part 11: validation of systems that hold electronic records
- ICH E6(R3): validation of computerized systems
- EMA/INS/GCP/112288/2023: computerized systems and electronic data in clinical trials

IN THIS TOOLKIT

Worked study: run manifest recording R and package versions (study/run_manifest.json) `WORKED STUDY`

S2–32 IRT user acceptance testing: randomization and cohort-assignment scripts and sign-off

Review record · Quality & compliance · Lead biostatistician with Randomization and supply (IRT) vendor, Clinical operations, Data manager · Protocol final → First participant dosed · All designs

`CONDITIONAL` If an IRT system assigns treatment or dose

Biostatistics test scripts, run in the IRT test environment with a dummy schedule, confirming that the system allocates as the randomization specification requires (ratio, blocks, strata, sentinel and cohort sequencing, dose-level assignment), handles emergency unblinding and passes assignments to EDC correctly. Includes the defect log and the go-live sign-off.

Why it matters. An IRT configuration error found after go-live misallocates participants and can compromise the blind and the analysis.

NEEDS

Randomization specification; IRT requirements and configuration; dummy randomization schedule; sentinel and dose-escalation rules.

`Depends on: S2–19`

PRODUCES

Executed test scripts, requirement-to-test traceability, defect log with retests and signed go-live approval, filed as system validation records.

`Feeds: S3–36`

QUALITY CONTROL

Second tester or reviewer for critical scripts; sign-off only when all critical tests pass and no critical or major defect is open.

COMMON PITFALLS

- Testing with the production randomization list, which exposes treatment codes.
- Skipping sentinel, cohort and dose-level logic specific to escalation designs.

STANDARDS AND GUIDANCE

- ICH E6(R3): protocol-specific configurations validated, especially randomization, dosing and dose titration (4.3.4); trial-specific computerized system validation records are essential records (Appendix C)
- 21 CFR Part 11: validation of systems to ensure accuracy, reliability and consistent intended performance (11.10(a))

IN THIS TOOLKIT

IRT/RTSM user acceptance testing tracker (apps/uat_irt/index.html) · IRT UAT templates: test scripts, traceability matrix, defect log (uat_irt/templates/index.html)

S2–33 eCRF design review for analysis-critical data

Review record · Data management interface · Lead biostatistician with Data manager, Statistical programmer, PK scientist · Protocol final → First participant dosed · All designs

Documented biostatistics review of the eCRF before the EDC build, confirming that every item the analysis needs is collected with the right precision and structure (for example actual clock times of dosing, PK samples and ECGs, emesis and meal times, reasons for missed samples), using standard collection structures that trace to SDTM.

Why it matters. Data the analysis needs but the eCRF never collected cannot be recovered after the study, and nonstandard structures complicate SDTM mapping.

NEEDS

Draft eCRF and completion guidelines; protocol schedule of activities; statistical considerations and analysis sets; critical-data list.

`Depends on: S1–03, S1–15`

PRODUCES

Review comments and sign-off; inputs to the SDTM mapping specification and the annotated CRF.

`Feeds: S2–12`

QUALITY CONTROL

Comments tracked to resolution; final eCRF checked against the analysis data needs before go-live.

COMMON PITFALLS

- Collecting PK sample times as nominal only, so actual elapsed times cannot be derived.
- Free-text fields where coded responses are needed for analysis.

STANDARDS AND GUIDANCE

- ICH E6(R3): data to be collected prespecified, and data acquisition tools fit for purpose, capturing the information the protocol requires and validated before use (3.16.1(c), (d))
- Clinical Data Acquisition Standards Harmonization…: standard data collection structures that trace submission data to SDTM

BA–02 BA analysis section: comparisons, reference and data-removal rules

Plan · Biostatistics · Lead biostatistician with PK scientist, Statistical programmer · Protocol final → SAP and specifications final · Relative bioavailability

Design-specific SAP section listing each test-versus-reference comparison, the model and whether each comparison uses all treatments or only the two compared, log-transformed exposure measures with GMRs and 90% CIs, absolute bioavailability from dose-normalized AUC when an intravenous reference is used, and the rules for pre-dose concentrations, emesis and outliers.

Why it matters. Without explicit comparisons and removal rules, results of multi-formulation studies become selective and hard to defend.

NEEDS

BA design basis; protocol estimands; formulations and strengths; NCA conventions.

`Depends on: S1–03`

PRODUCES

SAP BA section; specifications for BA analysis flags and outputs.

`Feeds: BA–03`

QUALITY CONTROL

Independent statistician review; each comparison mapped to an objective and an output.

COMMON PITFALLS

- Copying M13A removal rules into a BA study; the FDA BA guidance words some rules differently.
- Testing several formulations without stating which comparison is primary.

STANDARDS AND GUIDANCE

- Bioavailability Studies Submitted in NDAs or INDs:…: confidence-interval approach on log-transformed exposure measures with GMRs and 90% CIs; a participant is dropped from all PK evaluations when the pre-dose concentration exceeds 5% of Cmax; immediate-release data deleted when emesis occurs at or before 2 times median Tmax; outliers removed only with coinciding documentation of a protocol violation; CI values rounded to two decimals (III.A.10; Appendix A)
- Statistical Approaches to Establishing Bioequivalence: its general approaches may also be useful for equivalence comparisons in clinical pharmacology studies (section I.A)

BE–02 BE analysis section: model, acceptance limits and data-removal rules

Plan · Biostatistics · Lead biostatistician with PK scientist, Statistical programmer · Protocol final → SAP and specifications final · Bioequivalence

Design-specific SAP section: primary parameters (Cmax and AUC(0-t), or AUC(0-72h) for long half-lives), log-transformed analysis by a prespecified GLM or mixed model (typically with sequence, participant within sequence, period and formulation), and the 90% CI of the test/reference GMR judged against 80.00% to 125.00%. It lists every prespecified reason for removing data and when each is documented.

Why it matters. Removal rules or model choices that were not prespecified are not acceptable for the primary analysis and invite challenge of the BE conclusion.

NEEDS

Protocol design and estimand; BE design and sample-size basis; protocol data-removal criteria; NCA conventions (BLQ handling, actual times).

`Depends on: S1–03`

PRODUCES

SAP BE section; specification for BE analysis-set flags and BE outputs; methods text for the CSR.

`Feeds: BE–03`

QUALITY CONTROL

Independent statistician review against M13A; check that each removal rule is prespecified and has a documentation point before bioanalysis.

COMMON PITFALLS

- Choosing GLM or mixed model after seeing the data, which changes which participants are analyzed.
- Removing statistical outliers; M13A allows removal only for prespecified protocol violations, pre-dose concentrations above 5% of Cmax or very low exposure.

STANDARDS AND GUIDANCE

- ICH M13A: 90% CI of the GMR within 80.00% to 125.00% for Cmax and AUC(0-t) (2.2.4); AUC(0-72h) when half-life is 24 hours or longer (2.1.8.2); log transformation and a prespecified model; effects of sequence, participant within sequence, period and formulation tested; no post hoc or data-driven adjustment (2.2.3)
- ICH M13A Guideline: Bioequivalence for Immediate-Release…: examples of protocol deviations that may justify removal, such as emesis within 2 times the expected median tmax; reasons prespecified and documented before bioanalysis (Q&A 2.9)
- Statistical Approaches to Establishing Bioequivalence: five estimand components in the protocol; missing-data handling and GLM versus mixed model prespecified, with prespecified sensitivity analyses (II.A.2, II.B.2)

IN THIS TOOLKIT

Worked study: SAP excerpt (estimand, analysis sets, M13A model) (study/02_sap/sap_excerpt.html) `WORKED STUDY`

DDI–02 Drug interaction analysis section: no-effect boundaries and model

Plan · Biostatistics · Lead biostatistician with Clinical pharmacologist, PK scientist, Statistical programmer · Protocol final → SAP and specifications final · Drug-drug interaction

Design-specific SAP section: GMR of object exposure (AUC, Cmax and, when applicable, Cmin) with versus without the precipitant, its 90% CI from a model suited to the design, prespecified no-effect boundaries (preferably based on exposure-response), handling of dropouts and incomplete profiles, and summaries of metabolites and precipitant concentrations where collected.

Why it matters. Boundaries or exclusions chosen after the results undermine the interaction conclusion and any dosing recommendation.

NEEDS

DDI design basis; exposure-response information for the object drug; protocol estimands.

`Depends on: S1–03`

PRODUCES

SAP DDI section; specifications for treatment-condition derivations and DDI outputs.

`Feeds: DDI–03`

QUALITY CONTROL

Independent statistician review; boundaries justified and the model matched to the design.

COMMON PITFALLS

- Reading a one-sequence result without noting that treatment is confounded with period and time trends.
- Excluding incomplete profiles without considering that the interaction itself may explain them.

STANDARDS AND GUIDANCE

- ICH M12: GMR with and without the precipitant with its 90% CI (5.2); no-effect boundaries preferably from exposure-response, with a default 90% CI of 80% to 125% acceptable but considered overly conservative for most drugs (5.3.1); results shown with and without participants proposed for exclusion (5)

IN THIS TOOLKIT

R template: one-sequence exposure ratio (fixed order) (r_tlf_pseudocode/single_sequence/t_be_anova.R)

FE-02 Food-effect analysis section

Plan · Biostatistics · Lead biostatistician with Clinical pharmacologist, PK scientist, Statistical programmer · Protocol final → SAP and specifications final · Food effect

Design-specific SAP section: log-transformed AUC(0-inf), AUC(0-t) and Cmax compared fed versus fasted with the fasted condition as reference, the 90% CI of the GMR, the no-effect criterion (80% to 125% unless exposure-response supports other limits), descriptive comparison of Tmax and Tlag, and rules for periods where meal or fasting conditions were not met.

Why it matters. Without a prespecified criterion and reference condition, a labeling statement of no food effect cannot be supported.

NEEDS

Food-effect design basis; protocol estimands; meal and fasting rules; exposure-response information.

Depends on: S1-03

PRODUCES

SAP food-effect section; specifications for condition-compliance flags and food-effect outputs.

Feeds: FE-03

QUALITY CONTROL

Independent statistician review; criterion, reference condition and compliance rules checked for ambiguity.

COMMON PITFALLS

- Using fed as the reference condition, which inverts the ratio the guidance expects.
- No rule for participants who did not finish the meal within the specified time.

STANDARDS AND GUIDANCE

- Assessing the Effects of Food on Drugs in INDs and NDAs:…: fasted treatment as reference; 90% CI of the fed/fasted GMR for AUC(0-inf), AUC(0-t) and Cmax on log-transformed data; no food effect if the CI lies within 80% to 125% unless exposure-response supports other criteria; clinical relevance of Tmax and Tlag differences stated (VI.A)

HI-02 Hepatic impairment analysis section

Plan · Biostatistics · Lead biostatistician with Pharmacometrician, PK scientist, Statistical programmer · Protocol final → SAP and specifications final · Hepatic impairment

Design-specific SAP section: GMRs of total and unbound AUC and Cmax for each impairment group versus control with 90% CIs, regression on continuous components of the classification and other markers, dose normalization if doses differ, the staged-design decision rule, and the method for dose recommendations.

Why it matters. Without prespecified comparisons and a staging rule, stopping after one group or adjusting the dose looks post hoc.

NEEDS

Hepatic design basis; protocol estimands; exposure-response information.

Depends on: S1-03

PRODUCES

SAP hepatic section; specifications for classification derivations and outputs.

Feeds: HI-03

QUALITY CONTROL

Independent statistician review.

COMMON PITFALLS

- Analyzing total concentrations only for a highly bound, highly extracted drug.

STANDARDS AND GUIDANCE

- Pharmacokinetics in Patients with Impaired Hepatic…: ratio of impaired to control means for total and unbound AUC and Cmax with 90% CI; regression on continuous components of the classification; estimates with standard errors or CIs; dosing by exposure matching or no-effect boundaries (V; draft, September 2026)
- CPMP/EWP/2339/02: descriptive statistics by group; modeling of hepatic markers against PK when a relevant marker is found; dose normalization when doses differ (4.2)

QT-02 Cardiodynamic (concentration-QTc) analysis plan

Plan · PK & pharmacometrics · Pharmacometrician with Lead biostatistician, ECG core laboratory, Clinical pharmacologist · Protocol final → SAP and specifications final · Concentration-QTc or thorough QT, Single ascending dose, Multiple ascending dose

CONDITIONAL If a thorough QT study or a concentration-QTc analysis is planned

Prespecified plan for the QT analysis: heart-rate correction, baseline and placebo adjustment, the concentration-QTc model and its components, model-selection criteria, checks for hysteresis and goodness of fit, the decision rule at the highest clinically relevant exposure, categorical analyses and the handling of pooled studies.

Why it matters. Models chosen after seeing the data can give discordant results and weaken the case that a dedicated thorough QT study is not needed.

NEEDS

QT design basis; ECG core-laboratory methods; PK sampling schedule; definitions of therapeutic and high clinical exposure.

Depends on: S1-03

PRODUCES

Approved cardiodynamic plan (SAP appendix or standalone); specifications for the concentration-QTc analysis, assumption checks and categorical tables.

Feeds: QT-04

QUALITY CONTROL

Independent pharmacometrics and statistics review; plan approved before treatment-unblinded ECG data are analyzed.

COMMON PITFALLS

- Testing by time point in a study not powered for it; the Q&A calls this inappropriate for concentration-QTc designs.
- Pooling studies with different ECG methods without a heterogeneity check.

STANDARDS AND GUIDANCE

- ICH E14/S7B Q&As (R1): model, assumptions, selection criteria and pooling specified before analysis; hysteresis plots and goodness of fit documented; upper bound of the two-sided 90% CI below 10 ms at the highest clinically relevant exposure; heterogeneity tested when pooling (Q&A 5.1)
- ICH E14: uncorrected QT, RR and heart rate data, and QTcB and QTcF data, submitted (3.1)

IN THIS TOOLKIT

C-QTc hysteresis study (synthetic operating characteristics) (cqtc_hysteresis/index.html)

QT-03 ECG core-laboratory data transfer and blinded-reading specification

Specification · Data management interface · Data manager with ECG core laboratory, Lead biostatistician, Statistical programmer · Protocol final → First participant dosed · Concentration-QTc or thorough QT, Single ascending dose, Multiple ascending dose

CONDITIONAL If a thorough QT study or a concentration-QTc analysis is planned

Study-specific agreement with the ECG core laboratory: extraction of replicate ECGs at each nominal time, blinded central reading, lead and measurement method, transferred variables (RR, QT, QTcF, QTcB, PR, QRS, heart rate, morphology), keys for matching ECGs to PK samples, and reader-variability checks.

Why it matters. Without it, unblinded or inconsistent readings and unmatched ECG and PK times can invalidate the concentration-QTc analysis.

NEEDS

Protocol ECG schedule; cardiodynamic analysis plan; generic data-transfer specification; core-laboratory procedures.

Depends on: S2–26

PRODUCES

Signed ECG transfer and reading specification; test-transfer results; reconciliation keys for ADEG and the concentration-QTc dataset.

Feeds: S3–11

QUALITY CONTROL

Test transfer before the first production transfer; variables, units and keys checked; ECG and PK times reconciled.

COMMON PITFALLS

- Readers given time or treatment labels, which breaks the blinded reading.
- Transferring only corrected QT values, so other corrections cannot be recomputed.

STANDARDS AND GUIDANCE

- ICH E14: a few skilled readers in a central laboratory, blinded to time, treatment and participant, one reader for all of a participant's ECGs, reader variability measured by blinded rereads, and a consistent lead approach (2.5.1)
- ICH E14/S7B Q&As (R1): ECG quality for concentration-QTc analysis of studies run for other purposes as rigorous as for a dedicated study (Q&A 5.1)

RI-02 Renal impairment analysis section

Plan · Biostatistics · Lead biostatistician with Pharmacometrician, PK scientist, Statistical programmer · Protocol final → SAP and specifications final · Renal impairment

Design-specific SAP section: regression of PK parameters (CL/F, AUC, Cmax, unbound values where relevant) on continuous renal function as the primary approach, GMRs versus the control group with CIs, the reduced-design decision rule, dialysis clearance, sensitivity to the renal function equation, and the method for dose recommendations.

Why it matters. An unplanned choice between regression and group comparisons, or between equations, can move the renal function threshold for dose adjustment.

NEEDS

Renal design basis; protocol estimands; renal function equations; exposure-response information.

Depends on: S1–03

PRODUCES

SAP renal section; specifications for renal derivations and outputs.

Feeds: RI–03

QUALITY CONTROL

Independent statistician review; primary approach, decision rule and equations checked for ambiguity.

COMMON PITFALLS

- Leaving the reduced-design expansion rule vague, so stopping or expanding looks post hoc.

STANDARDS AND GUIDANCE

- Pharmacokinetics in Patients with Impaired Renal…: regression with renal function and PK parameters as continuous variables usually preferred to categorical analysis; confounding by baseline covariates evaluated; estimates reported with standard errors or CIs; dosing by exposure matching to a reference group; impact of different estimating equations evaluated (V)
- Guideline on the evaluation of the pharmacokinetics of…: reduced-design criteria for not expanding the study may use CIs, possibly one-sided or at a lower confidence level, with justification; GMRs versus the control group (5, 6.2.1)

STAGE 3 · CLOSES AT DATABASE LOCK

Data & CDISC

Collected data become standardized, conformant, traceable CDISC datasets with reviewer guides and Define-XML, and the database is locked under documented controls. 52 deliverables.

ID	DELIVERABLE	OWNER	FROM → TO	DESIGNS
S3–01	EDC raw data extracts and vendor data transfers (data transfer agreement interface)	Data manager	G1 → G3	All
S3–02	Biostatistics review of edit checks, data queries and data-review listings	Lead biostatistician	G1 → G3	All
S3–03	Annotated CRF (aCRF), final submission version	Lead statistical programmer	M1 → G3	All
S3–04	SDTM dataset: DM (Demographics)	Lead statistical programmer	M1 → G3	All
S3–05	SDTM dataset: PC (Pharmacokinetic Concentrations)	Lead statistical programmer	M1 → G3	All
S3–06	SDTM dataset: PP (Pharmacokinetic Parameters)	Lead statistical programmer	G3 → G4	All
S3–07	SDTM dataset: EX (Exposure)	Lead statistical programmer	M1 → G3	All
S3–08	SDTM dataset: AE (Adverse Events)	Lead statistical programmer	M1 → G3	All
S3–09	SDTM dataset: LB (Laboratory)	Lead statistical programmer	M1 → G3	All
S3–10	SDTM dataset: VS (Vital Signs)	Lead statistical programmer	M1 → G3	All
S3–11	SDTM dataset: EG (ECG)	Lead statistical programmer	M1 → G3	All
S3–12	SDTM dataset: DS (Disposition)	Lead statistical programmer	M1 → G3	All
S3–13	SDTM dataset: MH (Medical History)	Lead statistical programmer	M1 → G3	All
S3–14	SDTM dataset: CM (Concomitant and Prior Medications)	Lead statistical programmer	M1 → G3	All
S3–15	SDTM datasets: pharmacodynamic findings, DV and other supporting domains (PE, SU, QS, CO, trial design, SUPPQUAL, RELREC)	Lead statistical programmer	M1 → G3	All
S3–16	SDTM Define-XML (define.xml and stylesheet)	Lead statistical programmer	G2 → G4	All
S3–17	SDTM reviewer guide (cSDRG)	Lead statistical programmer	G2 → G4	All
S3–18	ADaM dataset: ADSL (Subject-Level Analysis Dataset)	Lead statistical programmer	M1 → M4	All
S3–19	ADaM dataset: ADPC (PK concentrations analysis)	Lead statistical programmer	M1 → M4	All
S3–20	ADaM dataset: ADPP (PK parameters analysis)	Lead statistical programmer	G3 → G4	All
S3–21	ADaM dataset: ADAE (adverse events analysis)	Lead statistical programmer	M1 → M4	All
S3–22	ADaM dataset: ADLB (laboratory analysis)	Lead statistical programmer	M1 → M4	All
S3–23	ADaM dataset: ADVS (vital signs analysis)	Lead statistical programmer	M1 → M4	All
S3–24	ADaM dataset: ADEG (ECG analysis)	Lead statistical programmer	M1 → M4	All
S3–25	ADaM datasets: ADPD and other analysis datasets (ADCM, ADMH, deviation or exposure analysis datasets as applicable) CONDITIONAL	Lead statistical programmer	G2 → M4	All
S3–26	ADaM Define-XML (define.xml and stylesheet)	Lead statistical programmer	G2 → G4	All
S3–27	ADaM reviewer guide (ADRG)	Lead statistical programmer	G2 → G4	All
S3–28	CDISC conformance report: SDTM, ADaM and Define-XML	Lead statistical programmer	G2 → G4	All
S3–29	PK bioanalytical sample reconciliation (concentration data load and BLQ handling)	PK scientist	M1 → G3	All

ID	DELIVERABLE	OWNER	FROM → TO	DESIGNS
S3–30	Blinded data review outputs (data-review listings and review meeting package)	Lead biostatistician	G2 → M3	All
S3–31	Dry-run datasets and dry-run TLF execution	Statistical programmer	G2 → G3	All
S3–32	Data cleaning and soft lock	Data manager	M3 → G3	All
S3–33	Database hard lock	Data manager	G3 → G3	All
S3–34	Traceability matrix: raw data to SDTM to ADaM	Lead statistical programmer	G2 → G4	All
S3–35	Controlled terminology and coding-dictionary version management (CDISC CT, MedDRA, WHODrug)	Data manager	M1 → G3	All
S3–36	Randomization QC and verification report (schedule versus IRT and EDC reconciliation) CONDITIONAL	Independent statistician	G1 → G3	All
S3–37	Emergency unblinding and code-break event log with reconciliation CONDITIONAL	Clinical operations	M1 → G3	All
S3–38	Protocol-deviation classification and reconciliation log (important versus non-important)	Clinical operations	M1 → G3	All
S3–39	Analysis-population derivation and participant-disposition reconciliation (population-flag QC)	Lead biostatistician	M3 → M4	All
S3–40	Data review meeting minutes and blinded data-review sign-off (pre-lock)	Data manager	M3 → G3	All
S3–41	Database lock authorization and 21 CFR Part 11 audit-trail review records	Data manager	M3 → G3	All
S3–42	PK sampling-time deviation and actual-versus-nominal time reconciliation (PK timing QC)	PK scientist	M1 → G3	All
BE–03	BE analysis-set and period-exclusion derivations	Statistical programmer	M3 → M4	BE
DDI–03	Drug interaction treatment-condition derivations	Statistical programmer	M3 → M4	DDI
ESC–01	Per-cohort dose-escalation data package	Lead biostatistician	M1 → M2	SAD, MAD
ESC–02	Preliminary PK analysis and next-dose exposure projection	PK scientist	M1 → M2	SAD, MAD
ESC–03	Sentinel-dosing review record CONDITIONAL	Medical monitor	M1 → M2	SAD, MAD
ESC–04	Dose-escalation decision record and escalation log	Safety Review Committee	M2 → M2	SAD, MAD
ESC–05	Cumulative safety and PK summary for part transitions CONDITIONAL	Lead biostatistician	M2 → M2	SAD, MAD
FE–03	Meal-compliance and dosing-condition derivations	Statistical programmer	M1 → G3	FE
HI–03	Child-Pugh classification and unbound PK derivations	Statistical programmer	M3 → M4	HI
RI–03	Renal function classification, unbound PK and dialysis derivations	Statistical programmer	M3 → M4	RI

S3–01 EDC raw data extracts and vendor data transfers (data transfer agreement interface)

Dataset · Data management interface · Data manager with Statistical programmer, Lead biostatistician, Bioanalytical laboratory, Central laboratory, ECG core laboratory, Randomization and supply (IRT) vendor · Protocol final → Database lock · All designs

Scheduled EDC extracts plus external vendor transfers (central laboratory, bioanalytical concentrations, ECG core laboratory, IRT randomization) received under transfer agreements that fix file format, frequency, variable layout, units, code lists and merge keys. Statistical programming and biostatistics confirm that each incoming structure supports the planned SDTM mapping and analysis.

Why it matters. Without controlled, reconciled transfers, SDTM and PK datasets are built on incomplete or misaligned records that surface only at lock.

NEEDS

Protocol; eCRF design; SAP; vendor specifications (central laboratory, bioanalytical laboratory, ECG core laboratory); IRT design; data transfer specifications

Depends on: S2–26

PRODUCES

Raw extracts and vendor files feeding SDTM mapping; reconciliation keys for PK, laboratory, ECG and randomization data

Feeds: S3–02, S3–29, S3–31, S3–35, S3–36, S3–37, S3–38, S3–42

QUALITY CONTROL

Transfer-specification conformance check on every load; record counts and checksums reconciled; test transfer before the first production transfer.

COMMON PITFALLS

- A vendor changes units or column order mid-study without a new specification version, and loads still succeed with wrong values.
- Merge keys differ between EDC and vendor files (visit labels, time points), leaving unmatched records that look like missing data.
- No test transfer before the first production transfer, so format defects first appear in a safety review data cut.

STANDARDS AND GUIDANCE

- ICH E6(R3): validated or reconciled data transfers between computerized systems (Section 4.2.5)
- EMA/INS/GCP/112288/2023: ALCOA++ principles for electronic trial data
- 21 CFR Part 11: controls on electronic records, including audit trails

S3–02 Biostatistics review of edit checks, data queries and data-review listings

Review record · Data management interface · Lead biostatistician with Statistical programmer, Data manager, PK scientist · Protocol final → Database lock · All designs

Biostatistics and programming review of the edit-check specification and standing cross-form listings so that analysis-critical fields are clean before lock: dosing versus PK sampling times, nominal versus actual times, BLQ flags, randomization versus treatment received, AE coding, laboratory units and ranges, and ECG intervals. Raises queries and analysis-oriented listings beyond the standard data management checks.

Why it matters. Analysis-critical errors that standard edit checks miss surface after lock, forcing database unlocks or unplanned exclusions.

NEEDS

Raw EDC extracts; edit-check specification; SAP analysis needs; PK analysis plan time-window rules

`Depends on: S2–21, S3–01`

PRODUCES

Queries on analysis-critical fields; review listings feeding blinded data review and data cleaning

`Feeds: S3–30`

QUALITY CONTROL

Independent review of listing logic; queries tracked to closure; critical variables reconciled against the SAP list.

COMMON PITFALLS

- Edit checks reviewed once at build; later protocol amendments add time points that no check covers.
- Listings check that dosing and sampling times exist but not their order, so pre-dose samples drawn after dosing pass.
- Queries on PK sampling-time deviations raised only after bioanalytical data arrive, too late for site correction.

STANDARDS AND GUIDANCE

- ICH E6(R3): planned review of trial data, audit trails and metadata (Section 4.2.3)

S3–03 Annotated CRF (aCRF), final submission version

Submission component · Programming & CDISC · Lead statistical programmer with Statistical programmer, Data manager · First participant dosed → Database lock · All designs

The blank CRF annotated with SDTM domain and variable names and controlled-terminology mappings, showing the source of every collected SDTM variable. Submitted as acrf.pdf; it is the human-readable companion to the SDTM define.xml and the cSDRG.

Why it matters. Without an accurate aCRF, define.xml origins cannot point to CRF pages and reviewers cannot trace collected data to SDTM.

NEEDS

Final blank eCRF; SDTM mapping specification; SDTMIG; controlled terminology

`Depends on: S2–13`

PRODUCES

acrf.pdf for the submission package (eCTD Module 5); page references for define.xml origins and the cSDRG

`Feeds: S3–16`

QUALITY CONTROL

Page-by-page review against the final CRF and SDTM specifications; check that every annotated origin resolves in define.xml.

COMMON PITFALLS

- Annotations made on a draft CRF; final page numbers shift and define.xml origin links point to the wrong pages.
- Annotations not updated after a mapping change, so the aCRF and SDTM disagree on variable names.

STANDARDS AND GUIDANCE

- Study Data Technical Conformance Guide: Technical…: expectations for the annotated CRF in study data submissions
- Define-XML v2.1: origin metadata that links collected variables to aCRF pages
- SDTMIG v3.4: domain and variable names used in the annotations

S3–04 SDTM dataset: DM (Demographics)

Dataset · Programming & CDISC · Lead statistical programmer with Statistical programmer · First participant dosed → Database lock · All designs

Participant-level demographics tabulation: age, sex, race, ethnicity, country, planned and actual arm, and reference dates (RFSTDTC, RFENDTC). Anchors participant-level traceability and supplemental qualifiers.

Why it matters. Every SDTM and ADaM dataset keys on DM, so an error in arm or reference dates propagates into treatment assignment and study-day derivations everywhere.

NEEDS

Raw EDC demographics; IRT randomization data; SDTM mapping specification; aCRF

`Depends on: S2–12, S3–33`

PRODUCES

Source for ADSL and all ADaM datasets; submission tabulation dataset

`Feeds: S3–18`

QUALITY CONTROL

Independent double programming and conformance checks; reconciliation to raw data and to the randomization schedule.

COMMON PITFALLS

- ACTARM populated from the planned assignment instead of the treatment actually received, hiding dosing errors.
- RFSTDTC defined inconsistently (consent versus first dose), shifting study days across domains.
- Screen failures handled differently in DM and DS, so participant counts differ between domains.

STANDARDS AND GUIDANCE

- SDTMIG v3.4: DM domain structure and required variables
- CDISC Controlled Terminology: codelists for demographic values

IN THIS TOOLKIT

Worked study: SDTM-like domain build (DM, EX, PC, AE, VS, DS) (study/03_adam/01_sdtm.R) · Worked study: SDTM-like DM (synthetic) (study/03_adam/sdtm/dm.csv) `WORKED STUDY`

S3–05 SDTM dataset: PC (Pharmacokinetic Concentrations)

Dataset · Programming & CDISC · Lead statistical programmer with PK scientist, Statistical programmer · First participant dosed → Database lock · All designs

Drug and metabolite concentration findings: analyte, result and units, LLOQ and BLQ indicators, specimen, nominal time point and actual collection date and time. It must align with EX dosing records and reconcile to the bioanalytical transfer.

Why it matters. PC is the source of every concentration and sampling time used in NCA, so a mapping or reconciliation gap here changes reported PK parameters.

NEEDS

Bioanalytical concentration transfer; sampling-time CRF data; LLOQ and BLQ conventions from the PK analysis plan; SDTM mapping specification

`Depends on: S2–12, S3–33`

PRODUCES

Source for ADPC and NCA; submission tabulation dataset

`Feeds: S3–19`

QUALITY CONTROL

Independent double programming and conformance checks; every record reconciled to the bioanalytical transfer (record count, value, BLQ flag); nominal versus actual time check.

COMMON PITFALLS

- BLQ results stored as a numeric zero instead of following the study's BLQ convention, losing the distinction from a true zero.
- Pre-dose samples assigned to the wrong period or dose when the reference dose time is ambiguous.
- Reassayed values overwrite the originals without keeping the reason or the analytical run.

STANDARDS AND GUIDANCE

- SDTMIG v3.4: PC domain structure, including LLOQ and timing variables
- CDISC Controlled Terminology: codelists for specimen type and units

IN THIS TOOLKIT

Worked study: SDTM-like PC (synthetic) (study/03_adam/sdtm/pc.csv) · Worked study: SDTM-like domain build (includes late, missed and BLQ sample scenarios) (study/03_adam/01_sdtm.R) `WORKED STUDY`

S3–06 SDTM dataset: PP (Pharmacokinetic Parameters)

Dataset · Programming & CDISC · Lead statistical programmer with PK scientist, Statistical programmer · Database lock → Outputs QC-passed · All designs

Tabulation of NCA parameters (for example Cmax, Tmax, AUClast, AUCinf, t1/2, CL/F, Vz/F) with PPTESTCD and results, linked to the contributing PC records through RELREC. Values come from the validated NCA run of record, so PP is finalized after NCA.

Why it matters. Without PP, ADPP cannot trace to SDTM and the submission lacks the tabulated parameters that reviewers check against the NCA output.

NEEDS

NCA output from validated software; PC dataset; parameter list from the PK analysis plan

`Depends on: S4–07`

PRODUCES

Source for ADPP; PK parameter listings; submission tabulation dataset

`Feeds: S3–16, S3–20`

QUALITY CONTROL

Reconciliation to the NCA output of record; independent double programming of the SDTM structure; conformance checks.

COMMON PITFALLS

- Parameters excluded by NCA rules (for example an unreliable half-life) dropped instead of kept with a status and reason.
- RELREC built at a different level (sample versus profile) than the study's documented convention.
- Parameter units copied from the NCA output without mapping to the controlled terminology that define.xml declares.

STANDARDS AND GUIDANCE

- SDTMIG v3.4: PP domain structure and relating PP records to PC records
- CDISC Controlled Terminology: PK parameter and PK unit codelists

IN THIS TOOLKIT

Worked study: SDTM-like PP written by the NCA program (study/03_adam/sdtm/pp.csv) `WORKED STUDY`

S3–07 SDTM dataset: EX (Exposure)

Dataset · Programming & CDISC · Lead statistical programmer with Statistical programmer · First participant dosed → Database lock · All designs

Actual study drug administration: treatment, dose, units, route, frequency, and start and end date and time of each administration. Critical for multiple-dose accumulation and PK time after dose; it must reconcile to randomization and to PC sampling.

Why it matters. Errors in dosing times shift every time-after-dose value and can misclassify exposure and treatment-emergent events.

NEEDS

Dosing CRF; IRT randomization data; SDTM mapping specification; aCRF

`Depends on: S2–12, S3–33`

PRODUCES

Source for ADSL treatment variables, ADPC time after dose and exposure tables; submission tabulation dataset

`Feeds: S3–18, FE–03`

QUALITY CONTROL

Independent double programming and conformance checks; reconciliation to randomization and to PC time-after-dose derivations.

COMMON PITFALLS

- Dosing time recorded to the minute but sampling time only to the hour, distorting early time-after-dose values.
- Interrupted or partial doses entered as full doses, overstating exposure and accumulation.
- EX populated from the planned schedule instead of the administration record, hiding dosing deviations.

STANDARDS AND GUIDANCE

- SDTMIG v3.4: EX domain for protocol-specified study treatment administration
- CDISC Controlled Terminology: route, dosage form and frequency codelists

IN THIS TOOLKIT

Worked study: SDTM-like EX (synthetic) (study/03_adam/sdtm/ex.csv) · Worked study: SDTM-like domain build (DM, EX, PC, AE, VS, DS) (study/03_adam/01_sdtm.R) `WORKED STUDY`

S3–08 SDTM dataset: AE (Adverse Events)

Dataset · Programming & CDISC · Lead statistical programmer with Data manager, Statistical programmer · First participant dosed → Database lock · All designs

Adverse events with MedDRA coding (LLT, PT, SOC), seriousness, severity or grade, causality, onset and resolution, action taken and outcome, plus the dates needed to derive treatment emergence. The core safety tabulation behind dose-escalation decisions.

Why it matters. Coding or date errors here change the TEAE counts that a safety review uses to decide whether to escalate.

NEEDS

AE CRF; MedDRA-coded terms; dosing records for treatment-emergent timing; SDTM mapping specification

Depends on: S2–12, S3–33

PRODUCES

Source for ADAE, safety tables and safety review committee data; submission tabulation dataset

Feeds: S3–16, S3–21

QUALITY CONTROL

Independent double programming and conformance checks; MedDRA coding QC; reconciliation of serious adverse events with the safety database.

COMMON PITFALLS

- Partial onset dates imputed in SDTM instead of ADaM, hiding the value actually collected.
- Serious adverse event reconciliation misses events reported to the safety database after the last EDC entry.
- MedDRA version upgraded mid-study without recoding earlier terms, splitting one concept across PTs.

STANDARDS AND GUIDANCE

- SDTMIG v3.4: AE domain structure
- MedDRA Version 29.1: coding hierarchy from LLT to SOC
- CDISC Controlled Terminology: severity, outcome and action-taken codelists

IN THIS TOOLKIT

Worked study: SDTM-like AE (synthetic) (study/03_adam/sdtm/ae.csv) · Worked study: SDTM-like domain build (DM, EX, PC, AE, VS, DS) (study/03_adam/01_sdtm.R) WORKED STUDY

S3–09 SDTM dataset: LB (Laboratory)

Dataset · Programming & CDISC · Lead statistical programmer with Statistical programmer, Central laboratory · First participant dosed → Database lock · All designs

Central and local laboratory findings: test, original and standardized results and units, reference ranges, normal-range indicators, fasting status and baseline flag. Safety laboratories (liver, renal, hematology) are central to Phase 1 monitoring.

Why it matters. Unit or range errors here create false abnormal flags and shifts in safety reviews and in the CSR laboratory section.

NEEDS

Central laboratory transfer; local laboratory CRF; SDTM mapping specification; controlled terminology

Depends on: S2–12, S3–33

PRODUCES

Source for ADLB and laboratory safety tables; submission tabulation dataset

Feeds: S3–16, S3–22, HI–03, RI–03

QUALITY CONTROL

Independent double programming and conformance checks; every record reconciled to the laboratory transfer; unit-conversion QC.

COMMON PITFALLS

- Local laboratory results arrive without reference ranges, so normal-range flags are blank or use the wrong range.
- Conversion factors applied to tests already reported in standard units, converting them twice.
- Repeat or unscheduled samples mapped to a scheduled visit, overwriting the planned value.

STANDARDS AND GUIDANCE

- SDTMIG v3.4: LB domain structure, including original and standardized results
- CDISC Controlled Terminology: laboratory test codes and unit codelists

S3–10 SDTM dataset: VS (Vital Signs)

Dataset · Programming & CDISC · Lead statistical programmer with Statistical programmer · First participant dosed → Database lock · All designs

Vital signs findings (blood pressure, heart rate, respiratory rate, temperature, weight, height, BMI) with position and nominal time points relative to dosing, for early-phase hemodynamic safety.

Why it matters. Missing position or time-point detail prevents valid change-from-baseline and orthostatic assessments.

NEEDS

Vital signs CRF; SDTM mapping specification; controlled terminology

Depends on: S2–12, S3–33

PRODUCES

Source for ADVS and vital signs tables; submission tabulation dataset

Feeds: S3–16, S3–23

QUALITY CONTROL

Independent double programming and conformance checks; reconciliation to raw data.

COMMON PITFALLS

- Replicate readings replaced by their mean in SDTM rather than kept as individual records.
- Position not mapped, so supine and standing values mix in one summary.
- BMI derived from weight and height taken at different visits.

STANDARDS AND GUIDANCE

- SDTMIG v3.4: VS domain structure
- CDISC Controlled Terminology: vital signs test, position and unit codelists

IN THIS TOOLKIT

Worked study: SDTM-like VS (synthetic) (study/03_adam/sdtm/vs.csv) · Worked study: SDTM-like domain build (DM, EX, PC, AE, VS, DS) (study/03_adam/01_sdtm.R) WORKED STUDY

S3–11 SDTM dataset: EG (ECG)

Dataset · Programming & CDISC · Lead statistical programmer with ECG core laboratory, PK scientist, Statistical programmer · First participant dosed → Database lock · All designs

ECG findings, including RR, PR, QRS, QT and corrected QT (QTcF, QTcB) intervals and interpretation, with nominal time points aligned to PK sampling where a concentration-QTc analysis is planned.

Why it matters. If ECG time points cannot be matched to concentrations, a concentration-QTc analysis cannot be run from the study data.

NEEDS

ECG core laboratory transfer or ECG CRF; SDTM mapping specification; controlled terminology

`Depends on: QT–03, S2–12, S3–33`

PRODUCES

Source for ADEG and concentration-QTc analysis; submission tabulation dataset

`Feeds: S3–16, S3–24`

QUALITY CONTROL

Independent double programming and conformance checks; reconciliation to the ECG core laboratory transfer; QC of time matching to PK samples.

COMMON PITFALLS

- Machine-read and core-laboratory intervals mixed in one dataset with no method variable to separate them.
- Replicate ECGs at a time point not identified, so the planned averaging is impossible downstream.
- Machine-reported QTc used when the SAP specifies recalculation from QT and RR.

STANDARDS AND GUIDANCE

- SDTMIG v3.4: EG domain structure
- CDISC Controlled Terminology: ECG test and result codelists
- ICH E14: submitting RR, heart rate and QT data corrected by both the QTcB and QTcF methods

S3–12 SDTM dataset: DS (Disposition)

Dataset · Programming & CDISC · Lead statistical programmer with Statistical programmer · First participant dosed → Database lock · All designs

Participant disposition events and protocol milestones: informed consent, randomization, completion or discontinuation with reason, and epoch transitions. Drives analysis-set derivation and the disposition table.

Why it matters. Disposition errors change analysis-set denominators and the reported reasons for discontinuation.

NEEDS

Disposition and completion CRF; SDTM mapping specification; controlled terminology

`Depends on: S2–12, S3–33`

PRODUCES

Source for ADSL disposition and population flags and for the disposition table; submission tabulation dataset

`Feeds: S3–18`

QUALITY CONTROL

Independent double programming and conformance checks; reconciliation to enrollment and randomization counts.

COMMON PITFALLS

- Discontinuation reasons captured only as free text, so standard categories cannot be summarized.
- Crossover studies record one disposition per study when the SAP needs completion by period.
- Randomization milestone missing for participants randomized but never dosed.

STANDARDS AND GUIDANCE

- SDTMIG v3.4: DS domain for disposition events and protocol milestones
- CDISC Controlled Terminology: disposition event codelists

IN THIS TOOLKIT

Worked study: SDTM-like DS (synthetic) (study/03_adam/sdtm/ds.csv) · Worked study: SDTM-like domain build (DM, EX, PC, AE, VS, DS) (study/03_adam/01_sdtm.R) `WORKED STUDY`

S3–13 SDTM dataset: MH (Medical History)

Dataset · Programming & CDISC · Lead statistical programmer with Data manager, Statistical programmer · First participant dosed → Database lock · All designs

Medical and surgical history with MedDRA coding and timing relative to the study; supports eligibility review, baseline characterization and special-population studies.

Why it matters. Uncoded or mistimed history weakens eligibility checks and the baseline description in the CSR.

NEEDS

Medical history CRF; MedDRA coding; SDTM mapping specification; controlled terminology

`Depends on: S2–12, S3–33`

PRODUCES

Baseline characterization in ADSL and listings; submission tabulation dataset

`Feeds: S3–16, S3–25`

QUALITY CONTROL

Independent double programming and conformance checks; coding QC.

COMMON PITFALLS

- Ongoing conditions lack an end-relative-to-reference value, so baseline status cannot be derived.
- History coded with a different MedDRA version than adverse events.
- Pre-dose adverse events entered as medical history, or the reverse.

STANDARDS AND GUIDANCE

- SDTMIG v3.4: MH domain structure
- MedDRA Version 29.1: coding of medical history terms

S3–14 SDTM dataset: CM (Concomitant and Prior Medications)

Dataset · Programming & CDISC · Lead statistical programmer with Data manager, Statistical programmer · First participant dosed → Database lock · All designs

Prior and concomitant medications with WHODrug coding (including ATC classes), indication, dose and timing; supports DDI interpretation and checks of protocol-permitted medications.

Why it matters. Uncoded or mistimed medications hide prohibited-medication deviations and confound DDI interpretation.

NEEDS

Concomitant medication CRF; WHODrug coding; SDTM mapping specification; controlled terminology

Depends on: S2–12, S3–33

PRODUCES

Concomitant medication listings; DDI context; submission tabulation dataset

Feeds: S3–16, S3–25

QUALITY CONTROL

Independent double programming and conformance checks; coding QC.

COMMON PITFALLS

- Drugs with several ATC classes resolved arbitrarily, so class-level counts change between runs.
- Prior versus concomitant status based on consent date instead of first dose date.
- Partial start dates imputed in SDTM instead of in the analysis dataset.

STANDARDS AND GUIDANCE

- SDTMIG v3.4: CM domain structure
- WHODrug Global (UMCDD): medication coding with ATC classification

S3–15 SDTM datasets: pharmacodynamic findings, DV and other supporting domains (PE, SU, QS, CO, trial design, SUPPQUAL, RELREC)

Dataset · Programming & CDISC · Lead statistical programmer with PK scientist, Clinical pharmacologist, Data manager, Statistical programmer · First participant dosed → Database lock · All designs

Pharmacodynamic and biomarker findings in the SDTMIG domain that fits each measurement (for example LB), protocol deviations (DV), physical examination (PE), substance use (SU), questionnaires (QS), comments (CO), special-purpose and trial design domains (SE, SV, TA, TE, TV, TS, TI), and SUPPQUAL and RELREC records.

Why it matters. Missing trial design or deviation domains leave reviewers unable to see the planned design or which participants deviated, and conformance checks fail.

NEEDS

CRF and vendor data for each domain; classified deviation log; trial design metadata from the protocol; SDTM mapping specification; controlled terminology

Depends on: S2–12, S3–33

PRODUCES

Source for ADPD and supporting listings; deviation data for analysis-set decisions; trial design metadata for define.xml; submission tabulation datasets

Feeds: S3–16, S3–25, S4–05, S4–16

QUALITY CONTROL

Independent double programming and conformance checks; RELREC integrity checks; DV reconciled to the classified deviation log.

COMMON PITFALLS

- Biomarker results placed in a custom domain when an SDTMIG domain such as LB fits.
- Trial summary (TS) parameters left incomplete, so required study information is missing at submission.
- SUPPQUAL used for variables that have a standard place in the parent domain.

STANDARDS AND GUIDANCE

- SDTMIG v3.4: structures for these domains and for SUPPQUAL and RELREC
- CDISC Controlled Terminology: codelists for these domains, including trial summary parameters

S3–16 SDTM Define-XML (define.xml and stylesheet)

Submission component · Programming & CDISC · Lead statistical programmer with Statistical programmer · SAP and specifications final → Outputs QC-passed · All designs

Machine-readable metadata for every SDTM dataset: dataset, variable and value-level metadata, codelists, computational methods and origins, with links to aCRF pages and the cSDRG. A required component of the tabulation package.

Why it matters. Without a correct define.xml, reviewers cannot interpret the datasets and conformance checks fail at submission.

NEEDS

Final SDTM datasets; SDTM mapping specification; aCRF; controlled terminology; value-level metadata specification

Depends on: S2–30, S3–03, S3–06, S3–08, S3–09, S3–10, S3–11, S3–13, S3–14, S3–15

PRODUCES

SDTM define.xml for the submission package (eCTD Module 5); input to conformance checks

Feeds: S3–28, S3–34

QUALITY CONTROL

Define-XML conformance check and dataset-to-define consistency check; peer review; check that origins resolve to aCRF pages.

COMMON PITFALLS

- define.xml generated from the specification rather than the final datasets, so lengths and codelists no longer match.
- Value-level metadata missing for result variables whose meaning changes by test code.
- Controlled terminology version stated in define.xml differs from the version applied to the data.

STANDARDS AND GUIDANCE

- Define-XML v2.1: dataset, variable and value-level metadata, codelists, methods and origins
- Study Data Technical Conformance Guide: Technical…: define.xml expectations for submitted study data
- CDISC Controlled Terminology: codelists referenced from define.xml

S3–17 SDTM reviewer guide (cSDRG)

Submission component · Programming & CDISC · Lead statistical programmer with Lead biostatistician · SAP and specifications final → Outputs QC-passed · All designs

Clinical Study Data Reviewer's Guide: orients a regulatory reviewer to the tabulation package (study design, dataset inventory, mapping conventions, sponsor decisions, known data issues) and explains each conformance finding that was not fixed.

Why it matters. Unexplained conformance findings and undocumented mapping decisions generate reviewer questions and delay review.

NEEDS

SDTM datasets; SDTM define.xml; conformance report; SDTM mapping specification; protocol

`Depends on: S3–28`

PRODUCES

cSDRG for the submission package (eCTD Module 5)

`Feeds: S5–09, S5–15`

QUALITY CONTROL

Peer review; consistency with define.xml and the conformance explanations; sign-off.

COMMON PITFALLS

- Conformance explanations copied from a previous study and no longer true for this data.
- Explanations written against an earlier conformance run, so finding counts do not match the final report.
- Sponsor-defined domains and SUPPQUAL choices left unexplained.

STANDARDS AND GUIDANCE

- PHUSE cSDRG Package: cSDRG template and completion guidance
- Study Data Technical Conformance Guide: Technical…: reviewer guide recommendation and explanation of unresolved validation issues

S3–18 ADaM dataset: ADSL (Subject-Level Analysis Dataset)

Dataset · Programming & CDISC · Lead statistical programmer with Lead biostatistician, Statistical programmer · First participant dosed → Unblinding · All designs

One record per participant with analysis-set flags (for example SAFFL and PK population flags), planned and actual treatment variables (TRTxxP, TRTxxA), demographics, baseline covariates and key dates. The spine merged into every other ADaM dataset.

Why it matters. Population flags and treatment variables in ADSL set the denominator and treatment column of every table, so one error repeats across the whole package.

NEEDS

SDTM DM, EX and DS; randomization data; SAP analysis-set definitions; ADSL specification

`Depends on: S2–06, S3–04, S3–07, S3–12`

PRODUCES

Merged into all ADaM datasets; population counts for every table; submission analysis dataset

`Feeds: S3–19, S3–21, S3–22, S3–23, S4–03, S4–05`

QUALITY CONTROL

Independent double programming and conformance checks; population flags reconciled to disposition, randomization and deviation decisions.

COMMON PITFALLS

- Actual treatment taken from randomization rather than the drug received, hiding dosing errors.
- Population flags hard-coded from a list instead of derived from documented rules.
- Crossover studies lack period-level treatment variables, forcing ad hoc derivations in every BDS dataset.

STANDARDS AND GUIDANCE

- ADaMIG v1.3: ADSL structure, required variables and population flags

IN THIS TOOLKIT

Worked study: ADSL program (study/03_adam/02_adsl.R) · Worked study: ADSL dataset (CSV) (study/03_adam/data/adsl.csv) · Worked study: ADaM datasets and specification (study/03_adam/index.html) `WORKED STUDY`

S3–19 ADaM dataset: ADPC (PK concentrations analysis)

Dataset · Programming & CDISC · Lead statistical programmer with PK scientist, Statistical programmer · First participant dosed → Unblinding · All designs

Analysis-ready concentration dataset with nominal and actual times relative to first dose and to the reference dose, BLQ handling per the analysis rules, and analysis flags such as inclusion in NCA. Direct input to NCA and to concentration tables and figures.

Why it matters. Time or BLQ errors in ADPC flow straight into NCA parameters, so every PK conclusion inherits them.

NEEDS

SDTM PC and EX; ADSL; PK analysis plan (BLQ, time-window and exclusion rules); ADPC specification

`Depends on: S2–07, S3–05, S3–18`

PRODUCES

NCA input; concentration tables and figures; submission analysis dataset

`Feeds: S3–24, S3–25, S4–06, S4–07, S4–12`

QUALITY CONTROL

Independent double programming and conformance checks; QC of time after dose and BLQ rules; reconciliation to PC.

COMMON PITFALLS

- Actual relative times computed from the first dose in multiple-dose profiles where the reference dose was intended.
- Pre-dose samples of later periods assigned to the previous period's profile.
- Excluded samples deleted instead of flagged, so listings cannot show what NCA ignored.

STANDARDS AND GUIDANCE

- ADNCA v1.0: ADaM BDS structure for NCA input data, including nominal and actual relative-time variables
- ADaMIG v1.3: BDS structure and analysis flags

IN THIS TOOLKIT

Worked study: ADPC program (nominal and actual relative times) (study/03_adam/03_adpc.R) · Worked study: ADPC dataset (CSV) (study/03_adam/data/adpc.csv) `WORKED STUDY`

S3–20 ADaM dataset: ADPP (PK parameters analysis)

Dataset · Programming & CDISC · Lead statistical programmer with PK scientist, Statistical programmer · Database lock → Outputs QC-passed · All designs

Analysis-ready PK parameters from the validated NCA run, with analysis flags, dose-normalized values and the grouping needed for dose-proportionality, accumulation, food-effect, DDI and BA or BE analyses.

Why it matters. Statistical comparisons and parameter tables read ADPP, so a parameter with a missing flag or wrong grouping silently changes the analysis.

NEEDS

SDTM PP; ADSL; ADPC; PK analysis plan; ADPP specification

`Depends on: S2–08, S3–06`

PRODUCES

PK parameter tables; inputs to ANOVA and geometric mean ratio analyses; submission analysis dataset

`Feeds: S3–26, S3–39, BE–03, DDI–03, HI–03, RI–03, S4–08, S4–09, S4–10`

QUALITY CONTROL

Reconciliation to the NCA output of record; independent double programming; conformance checks.

COMMON PITFALLS

- Parameters failing NCA acceptance rules removed rather than kept with an exclusion flag and reason.
- Dose normalization uses the nominal dose when the SAP specifies the actual administered dose.
- Period and sequence variables missing, so the comparison program re-derives them differently from ADSL.

STANDARDS AND GUIDANCE

- ADaMIG v1.3: BDS structure and analysis flags
- CDISC Controlled Terminology: PK parameter codelists

IN THIS TOOLKIT

Worked study: ADPP program (parameters kept with exclusion reasons) (study/03_adam/05_adpp.R) · Worked study: ADPP dataset (CSV) (study/03_adam/data/adpp.csv) `WORKED STUDY`

S3–21 ADaM dataset: ADAE (adverse events analysis)

Dataset · Programming & CDISC · Lead statistical programmer with Lead biostatistician, Statistical programmer · First participant dosed → Unblinding · All designs

Analysis-ready AE dataset in the occurrence data structure with the treatment-emergent flag (TRTEMFL), severity and causality analysis variables, treatment period and first-occurrence flags. Drives all TEAE tables and dose-escalation safety review.

Why it matters. TEAE counts by dose depend on the treatment-emergent and period assignments made here.

NEEDS

SDTM AE and EX; ADSL; SAP TEAE definition; ADAE specification

`Depends on: S2–09, S3–08, S3–18`

PRODUCES

TEAE, SAE and severity tables; safety review committee data; submission analysis dataset

`Feeds: S3–26, S4–13, S4–14`

QUALITY CONTROL

Independent double programming and conformance checks; QC of TRTEMFL logic; reconciliation to SDTM AE and the safety database.

COMMON PITFALLS

- Crossover AEs attributed to a period by onset date without the washout rule the SAP defines.
- Events with partial dates excluded from treatment emergence instead of imputed conservatively per the SAP.
- Worsening of a pre-existing condition not captured, so it is missed as treatment-emergent.

STANDARDS AND GUIDANCE

- ADaM OCCDS v1.1: occurrence data structure, TRTEMFL and occurrence flags
- MedDRA Version 29.1: coding hierarchy used for summaries

IN THIS TOOLKIT

Worked study: ADAE program (study/03_adam/04_adae.R) · Worked study: ADAE dataset (CSV) (study/03_adam/data/adae.csv) `WORKED STUDY`

S3–22 ADaM dataset: ADLB (laboratory analysis)

Dataset · Programming & CDISC · Lead statistical programmer with Statistical programmer · First participant dosed → Unblinding · All designs

Analysis-ready laboratory data (BDS): baseline and change from baseline, standardized results, shift and toxicity-grade categories, and analysis flags. Central to Phase 1 safety laboratory monitoring, including the components of a Hy's law evaluation.

Why it matters. Shift tables, outlier listings and liver-safety evaluations all depend on the baseline and grade derivations made here.

NEEDS

SDTM LB; ADSL; SAP baseline, shift and grading rules; ADLB specification

`Depends on: S2–10, S3–09, S3–18`

PRODUCES

Laboratory tables (summary, shift, change, marked abnormality); submission analysis dataset

`Feeds: S3–26, S4–15`

QUALITY CONTROL

Independent double programming and conformance checks; QC of baseline, change and category derivations; reconciliation to LB.

COMMON PITFALLS

- Baseline taken from the screening visit when the SAP defines it as the last value before first dose.
- Grades derived without the baseline-relative criteria that some grading terms require.
- Multiples of the upper limit computed with a reference range from a different laboratory than the result.

STANDARDS AND GUIDANCE

- ADaMIG v1.3: BDS baseline, change and toxicity-grade variables
- CTCAE v6.0 (MedDRA 28.0): grading of laboratory-based adverse event terms when the protocol uses CTCAE

IN THIS TOOLKIT

Sandbox: ADaM-shaped synthetic ADLB (structural checks only) (sim_lifecycle/out/adlb.csv)

S3–23 ADaM dataset: ADVS (vital signs analysis)

Dataset · Programming & CDISC · Lead statistical programmer with Statistical programmer · First participant dosed → Unblinding · All designs

Analysis-ready vital signs (BDS) with baseline, change from baseline, position handling and analysis flags for hemodynamic safety summaries.

Why it matters. Without consistent baseline and position rules, vital sign changes and outlier counts are not comparable across cohorts.

NEEDS

SDTM VS; ADSL; SAP rules; ADVS specification

`Depends on: S2–10, S3–10, S3–18`

PRODUCES

Vital signs tables; submission analysis dataset

`Feeds: S3–26, S4–16`

QUALITY CONTROL

Independent double programming and conformance checks; baseline and change QC; reconciliation to VS.

COMMON PITFALLS

- Replicate readings averaged at some visits and not others.
- Orthostatic changes derived from readings not paired by time point.
- Baseline defined differently from ADLB and ADEG without an SAP reason.

STANDARDS AND GUIDANCE

- ADaMIG v1.3: BDS baseline and change-from-baseline variables

IN THIS TOOLKIT

Sandbox: ADaM-shaped synthetic ADVS (structural checks only) (sim_lifecycle/out/advs.csv)

S3–24 ADaM dataset: ADEG (ECG analysis)

Dataset · Programming & CDISC · Lead statistical programmer with PK scientist, Pharmacometrician, Statistical programmer · First participant dosed → Unblinding · All designs

Analysis-ready ECG data (BDS) with QTc baseline and change from baseline, categorical outlier flags, and concentrations time-matched from ADPC for concentration-QTc analysis.

Why it matters. Categorical QTc tables and the concentration-QTc analysis are only as good as the baseline and time matching built here.

NEEDS

SDTM EG; ADPC (time-matched concentrations); ADSL; SAP QTc rules; ADEG specification

`Depends on: S2–10, S3–11, S3–19`

PRODUCES

QTc tables and outlier summaries; concentration-QTc analysis input; submission analysis dataset

`Feeds: S3–26, S4–11, S4–16, QT–04`

QUALITY CONTROL

Independent double programming and conformance checks; QC of QTc categories and time matching; reconciliation to EG.

COMMON PITFALLS

- Concentrations matched to ECGs by nominal time when actual times differ beyond the SAP window.
- Baseline averaged over replicates at some visits but taken from single readings at others.
- Categories computed from QTcB when the SAP specifies QTcF.

STANDARDS AND GUIDANCE

- ADaMIG v1.3: BDS baseline and change-from-baseline variables
- ICH E14: categorical QTc analyses: QTc above 450, 480 and 500 ms; increases from baseline above 30 and 60 ms
- ICH E14/S7B Q&As (R1): choice of time-matched or pre-dose baseline for QTc, prospectively defined (Q&A 4.2)

IN THIS TOOLKIT

Sandbox: ADaM-shaped synthetic ADEG (structural checks only) (sim_lifecycle/out/adeg.csv)

S3–25 ADaM datasets: ADPD and other analysis datasets (ADCM, ADMH, deviation or exposure analysis datasets as applicable)

Dataset · Programming & CDISC · Lead statistical programmer with PK scientist, Clinical pharmacologist, Statistical programmer · SAP and specifications final → Unblinding · All designs

`CONDITIONAL` If PD endpoints or other SAP-specified analysis datasets apply

Pharmacodynamic and biomarker analysis dataset (BDS) with baseline, change and PK-PD linkage, plus other SAP-specified analysis datasets, for example ADCM, ADMH, exposure, deviation or immunogenicity datasets.

Why it matters. PD summaries, exposure-response inputs and several supporting tables have no traceable source without these datasets.

NEEDS

SDTM pharmacodynamic, biomarker, CM, MH and other source domains; ADSL; ADPC or ADPP for PK-PD linkage; SAP; ADaM specifications

`Depends on: S2–11, S3–13, S3–14, S3–15, S3–19`

PRODUCES

PD tables and figures; exposure-response and modeling inputs; supporting tables; submission analysis datasets

`Feeds: S3–26, S4–04, S4–17, S4–18`

QUALITY CONTROL

Independent double programming and conformance checks; QC of PK-PD linkage and of baseline and change derivations.

COMMON PITFALLS

- PK-PD pairs matched by nominal time when the PK and PD sampling schedules differ.
- PD parameters with different units or assays pooled under one PARAMCD.
- Supporting datasets built ad hoc inside table programs instead of specified, traceable ADaM.

STANDARDS AND GUIDANCE

- ADaMIG v1.3: BDS structure for PD and exposure datasets
- ADaM OCCDS v1.1: occurrence data structure for medications and medical history

IN THIS TOOLKIT

Sandbox: ADaM-shaped synthetic exposure dataset (ADEX) (sim_lifecycle/out/adex.csv)

S3–26 ADaM Define-XML (define.xml and stylesheet)

Submission component · Programming & CDISC · Lead statistical programmer with Statistical programmer · SAP and specifications final → Outputs QC-passed · All designs

Machine-readable metadata for the ADaM package: dataset, variable and value-level metadata, codelists, derivation methods and traceability to SDTM sources. A required component of the analysis package.

Why it matters. Without ADaM define.xml, reviewers cannot reproduce derivations or trace analysis values to SDTM.

NEEDS

Final ADaM datasets; ADaM specifications; derivation methods; controlled terminology; SDTM source references

Depends on: S2–30, S3–20, S3–21, S3–22, S3–23, S3–24, S3–25

PRODUCES

ADaM define.xml for the submission package (eCTD Module 5); input to conformance checks

Feeds: S3–28, S3–34

QUALITY CONTROL

Define-XML conformance and dataset-to-define consistency checks; review of derivation methods; traceability check.

COMMON PITFALLS

- Derivation text copied from the specification while the program implements a later revision.
- Parameter-level metadata missing for BDS datasets with many PARAMCD values.
- define.xml written before the final datasets, leaving stale lengths and codelists.

STANDARDS AND GUIDANCE

- Define-XML v2.1: dataset, variable and value-level metadata and derivation methods
- Study Data Technical Conformance Guide: Technical…: define.xml expectations for analysis datasets

IN THIS TOOLKIT

Worked study: ADaM specification in define-ready layout (define.xml not authored) (study/03_adam/spec/adam_spec.xlsx) WORKED STUDY

S3–27 ADaM reviewer guide (ADRG)

Submission component · Programming & CDISC · Lead statistical programmer with Lead biostatistician · SAP and specifications final → Outputs QC-passed · All designs

Analysis Data Reviewer's Guide: orients reviewers to the analysis package, including analysis-set derivations, key derivations and conventions, conformance findings and their explanations, SDTM-to-ADaM traceability, and links to the SAP and outputs.

Why it matters. Without an ADRG, reviewers must reverse-engineer derivations, and unexplained conformance findings become review questions.

NEEDS

ADaM datasets; ADaM define.xml; conformance report; SAP; analysis-set definitions; traceability matrix

Depends on: S3–28, S3–34

PRODUCES

ADRG for the submission package (eCTD Module 5)

Feeds: S5–09, S5–15

QUALITY CONTROL

Peer review; consistency with define.xml, the SAP and the conformance explanations; sign-off.

COMMON PITFALLS

- Analysis-set derivations described from the SAP text rather than from the implemented flags.
- No program-to-output inventory, so reviewers cannot find the program behind a table.
- Explanations cite rule identifiers from an older validator version than the final run.

STANDARDS AND GUIDANCE

- PHUSE ADRG Package: ADRG template and completion guidance
- Study Data Technical Conformance Guide: Technical…: reviewer guide recommendation and explanation of unresolved validation issues

S3–28 CDISC conformance report: SDTM, ADaM and Define-XML

Review record · Programming & CDISC · Lead statistical programmer with Statistical programmer, Lead biostatistician · SAP and specifications final → Outputs QC-passed · All designs

Conformance validation of the SDTM and ADaM datasets and both define.xml files against the applicable CDISC and FDA rules, with every finding triaged: fixed in the data or metadata, or explained in the reviewer guides. The gate for package readiness.

Why it matters. Unresolved or unexplained findings can trigger technical rejection or information requests after submission.

NEEDS

SDTM and ADaM datasets; define.xml files; controlled terminology version; validation rule set and version

Depends on: S3–16, S3–26

PRODUCES

Issue list with dispositions; explanations for the cSDRG and ADRG; package go or no-go

Feeds: S3–17, S3–27

QUALITY CONTROL

Validator and rule-set versions recorded; every finding dispositioned; explanations reconciled with the reviewer guides.

COMMON PITFALLS

- Validator rule version differs from the one the agency applies, so findings change at submission.
- Errors explained away in the reviewer guide when they could have been fixed in the data.
- Final run done on datasets that were regenerated afterward.

STANDARDS AND GUIDANCE

- FDA Validator Rules v1.6: FDA validator rules for SDTM and SEND data, applied after eCTD technical validation; no ADaM rules
- SDTM and SDTMIG Conformance Rules v2.0: CDISC conformance rules for SDTM v2.0 and SDTMIG v3.4
- ADaM Conformance Rules v5.0: CDISC conformance rules for ADaMIG v1.0 to v1.3, OCCDS, ADNCA and the ADaM popPK IG; study-specific review still needed
- Study Data Technical Conformance Guide: Technical…: expectations for conformance and for explaining unresolved issues

IN THIS TOOLKIT

Worked study: dataset-versus-specification checks and XPT export (not a CDISC conformance run) (study/03_adam/06_checks_xpt.R) · Worked study: metadata check results (study/03_adam/data/metadata_checks.csv) WORKED STUDY

S3–29 PK bioanalytical sample reconciliation (concentration data load and BLQ handling)

Review record · PK & pharmacometrics · PK scientist with Statistical programmer, Bioanalytical laboratory, Data manager · First participant dosed → Database lock · All designs

Reconciles the bioanalytical concentration transfer against the clinical sampling record: every collected sample accounted for (analyzed, missing, insufficient, reassayed), LLOQ and BLQ flags consistent with the analysis plan, analyte and units mapped correctly, and collection times verified. The gate for loading concentrations into PC.

Why it matters. Unreconciled samples become silent gaps or duplicates in concentration-time profiles and bias NCA parameters.

NEEDS

Bioanalytical transfer from the validated assay; PK sampling CRF; LLOQ and BLQ rules; sample accountability log

Depends on: S2–29, S3–01

PRODUCES

Reconciled concentration load; documented BLQ handling; resolved discrepancies

Feeds: S3–32

QUALITY CONTROL

Every sample accounted for; independent check of BLQ and LLOQ flags; record counts and values matched to the transfer.

COMMON PITFALLS

- Reassay results loaded without the reason, so the reported value cannot be justified under the assay's reanalysis rules.
- Samples collected but never shipped treated as missing at random instead of investigated.
- An LLOQ change between assay validations not carried into the BLQ rules.

STANDARDS AND GUIDANCE

- ICH M10: study sample analysis, reanalysis of study samples, incurred sample reanalysis and bioanalytical reports
- SDTMIG v3.4: PC conventions for LLOQ and for samples not analyzed
- 21 CFR Part 11: controls on electronic records

S3–30 Blinded data review outputs (data-review listings and review meeting package)

Meeting package · Biostatistics · Lead biostatistician with Data manager, Medical monitor, PK scientist, Statistical programmer · SAP and specifications final → Blinded data review · All designs

Pre-lock listings and summaries (data quality, candidate protocol deviations, outliers, missing data, PK time-window issues) prepared for the data review meeting where analysis-set assignments and deviation classifications are agreed before unblinding. In blinded studies the package carries no treatment information.

Why it matters. Without a structured package, analysis-set and exclusion decisions are made late, inconsistently or after results are visible.

NEEDS

Cleaned, blinded SDTM or raw data; data review plan; SAP analysis-set and deviation rules

Depends on: S3–02

PRODUCES

Meeting package; proposed analysis-set and deviation classifications; inputs to soft and hard lock

Feeds: S3–40

QUALITY CONTROL

Independent review of listing logic; decisions minuted before unblinding; reconciliation to SAP rules.

COMMON PITFALLS

- Listings sorted or grouped in a way that reveals treatment, for example by kit number in blinded cohorts.
- PK exclusion candidates reviewed after concentrations are already visible by treatment.
- Package built from an older data cut than the one the meeting discusses.

STANDARDS AND GUIDANCE

- ICH E9: blind review: checking data between trial completion and breaking the blind to finalize the planned analysis

S3–31 Dry-run datasets and dry-run TLF execution

Program · Programming & CDISC · Statistical programmer with Lead biostatistician, Lead statistical programmer · SAP and specifications final → Database lock · All designs

End-to-end run of the SDTM, ADaM and TLF programs on a pre-lock or dummy data cut to test the pipeline, the shells and program robustness before lock; surfaces specification and program defects early.

Why it matters. Defects found only at the final run compress QC into the days after lock and delay topline results.

NEEDS

Interim or dummy data cut; ADaM specifications; TLF shells; programs under development

Depends on: S2–25, S3–01

PRODUCES

Defect list for specifications and programs; confirmation that production can run at lock

Feeds: S4–20

QUALITY CONTROL

Independent programming on the dry run where planned; output-versus-shell reconciliation; defects tracked to closure.

COMMON PITFALLS

- Dry-run data lack edge cases (BLQ values, missed doses, discontinuations), so those program branches stay untested.
- Dry-run outputs circulated and read as results in a blinded study.
- Programs changed after the dry run without rerunning the affected outputs.

STANDARDS AND GUIDANCE

- ICH E9: reliable software with documented testing (Section 5.8)

S3–32 Data cleaning and soft lock

Review record · Data management interface · Data manager with Lead biostatistician · Blinded data review → Database lock · All designs

Interim freeze once queries are resolved or dispositioned, coding is complete, reconciliations (serious adverse events, PK samples, laboratory, randomization) are closed and data review decisions are captured, allowing final checks before hard lock. Signed off.

Why it matters. Without a controlled freeze, data keep changing during final checks and lock readiness cannot be shown.

NEEDS

Query status; coding status; reconciliation status; data review decisions

Depends on: S3–29, S3–35, S3–36, S3–37, S3–40, S3–42

PRODUCES

Soft-locked snapshot for final checks; readiness for hard lock; sign-off record

Feeds: S3–41

QUALITY CONTROL

Lock checklist completed; cross-functional reconciliation sign-off; audit-trail review.

COMMON PITFALLS

- Sites can still edit data during soft lock because EDC permissions were not changed.
- Serious adverse event reconciliation closed against an outdated safety database extract.
- Vendor transfers received after soft lock not reconciled before hard lock.

STANDARDS AND GUIDANCE

- ICH E6(R3): finalization of data sets before analysis (Section 4.2.6)
- 21 CFR Part 11: audit trails for changes to electronic records

S3–33 Database hard lock

Log or record · Data management interface · Data manager with Lead biostatistician, Medical monitor, Clinical operations, Sponsor · Database lock → Database lock · All designs

Formal lock: data frozen, write access removed and a final snapshot taken under change control. The stage 3 exit gate and the data freeze on which final datasets and reported outputs are based.

Why it matters. Any analysis run before a documented lock cannot be shown to use final, unchanged data.

NEEDS

Soft-lock status; closed reconciliations; final coding; data review decisions; lock authorization

Depends on: S3–41

PRODUCES

Locked database snapshot; final SDTM and ADaM regeneration; unblinding can proceed

Feeds: S3–04, S3–05, S3–07, S3–08, S3–09, S3–10, S3–11, S3–12, S3–13, S3–14, S3–15

QUALITY CONTROL

Hard-lock checklist with multi-party sign-off; verification that write access is removed; snapshot integrity check.

COMMON PITFALLS

- PK or ECG vendor data still arriving after lock and loaded without an unlock procedure.
- Snapshot not checksummed, so later extracts cannot be shown to be identical.
- Unlock and relock performed without a documented reason and re-approval.

STANDARDS AND GUIDANCE

- ICH E6(R3): restricting edit access to data acquisition tools before final analysis and unblinding (Section 3.16.1)
- 21 CFR Part 11: audit trails and controls on electronic records

S3–34 Traceability matrix: raw data to SDTM to ADaM

Document · Programming & CDISC · Lead statistical programmer with Lead biostatistician · SAP and specifications final → Outputs QC-passed · All designs

End-to-end matrix linking CRF and raw sources to SDTM variables (through the aCRF and define.xml), SDTM to ADaM derivations (through define.xml and the ADRG), and ADaM to analysis outputs, showing an unbroken chain for inspection and submission.

Why it matters. Without demonstrated traceability, reviewers and inspectors cannot confirm that reported values come from collected data.

NEEDS

aCRF; SDTM and ADaM define.xml; mapping and derivation specifications; reviewer guides

Depends on: S3–16, S3–26

PRODUCES

Traceability evidence for submission and inspection; input to the ADRG and CSR appendices

Feeds: S3–27, S5–10

QUALITY CONTROL

Cross-link verification from raw to SDTM to ADaM; peer review; consistency with define.xml and the reviewer guides.

COMMON PITFALLS

- Matrix built from specifications and never checked against the final programs.
- Derived SDTM variables listed without their derivation source.
- ADaM variables traced back to SDTM but not forward to the outputs that use them.

STANDARDS AND GUIDANCE

- ADaMIG v1.3: traceability between ADaM and SDTM
- Define-XML v2.1: origins and derivation methods

S3–35 Controlled terminology and coding-dictionary version management (CDISC CT, MedDRA, WHODrug)

Log or record · Data management interface · Data manager with Lead statistical programmer · First participant dosed → Database lock · All designs

Documented record of the CDISC CT, MedDRA and WHODrug versions used, with an impact assessment if recoding occurs, keeping SDTM, ADaM and define.xml consistent and coding reproducible.

Why it matters. Mixed dictionary versions split terms across categories and make counts irreproducible between runs.

NEEDS

Dictionary versions; coding decisions; CT subsets used in mapping

`Depends on: S2–23, S3–01`

PRODUCES

Version records for define.xml and the reviewer guides; reproducible coding

`Feeds: S3–32`

QUALITY CONTROL

Version-lock verification; CT conformance checks; coding QC against the locked dictionaries.

COMMON PITFALLS

- Dictionary upgraded mid-study for new terms only, leaving earlier terms on the old version.
- define.xml states a dictionary version different from the one used for coding.
- Sponsor-extended codelist values not documented, so conformance checks fail.

STANDARDS AND GUIDANCE

- CDISC Controlled Terminology: versioned codelist releases
- MedDRA Version 29.1: versioned dictionary releases
- WHODrug Global (UMCDD): versioned dictionary releases
- Define-XML v2.1: references to controlled terminology and external dictionary versions

S3–36 Randomization QC and verification report (schedule versus IRT and EDC reconciliation)

Review record · Biostatistics · Independent statistician with Randomization and supply (IRT) vendor, Data manager · Protocol final → Database lock · All designs

CONDITIONAL If participants are randomized

Independent verification that the randomization schedule was loaded into the IRT correctly and that as-randomized assignments in the EDC reconcile to the master schedule (allocation, kit or treatment assignment, sentinel sequence). Performed by a statistician who did not generate the schedule.

Why it matters. An undetected load or allocation error can unbalance cohorts or dose participants with the wrong treatment.

NEEDS

Master randomization schedule; IRT load file; IRT randomization records; EDC randomization data

`Depends on: S2–18, S2–32, S3–01`

PRODUCES

Randomization QC report; discrepancy log; sign-off

`Feeds: S3–32`

QUALITY CONTROL

Independent double check; segregation of duties from the schedule generator.

COMMON PITFALLS

- Verification done on the vendor's test-environment load rather than the production load.
- Sentinel participants randomized outside the IRT and never reconciled.
- Replacement participants given new numbers without checking the schedule's replacement rules.

STANDARDS AND GUIDANCE

- ICH E9: randomization schedule documentation, reproducibility and secure filing (Section 2.3.2)
- ICH E3: randomization scheme and codes in the CSR (Appendix 16.1.7)

IN THIS TOOLKIT

IRT UAT tracker: allocation and IRT-EDC reconciliation test scripts (apps/uat_irt/index.html) · IRT UAT test-script template (uat_irt/templates/test_scripts.csv)

S3–37 Emergency unblinding and code-break event log with reconciliation

Log or record · Safety & committees · Clinical operations with Randomization and supply (IRT) vendor, Medical monitor, Lead biostatistician · First participant dosed → Database lock · All designs

CONDITIONAL If the study is blinded

Conduct-phase log of each emergency unblinding (who, when, participant, reason, authorization), reconciled against IRT code-break records and the safety database, with confirmation that the blinded team was not inadvertently unblinded.

Why it matters. Undocumented unblinding undermines the blind and the credibility of safety and analysis-set decisions.

NEEDS

IRT code-break records; safety database records; unblinding authorizations

`Depends on: S2–20, S3–01`

PRODUCES

Code-break event log; reconciliation memo; blinding-integrity statement

`Feeds: S3–32`

QUALITY CONTROL

Independent verification; segregation from the blinded team.

COMMON PITFALLS

- Unblinding for expedited safety reporting not logged alongside investigator unblinding events.
- Code-break notifications sent to distribution lists that include blinded staff.
- Backup code envelopes not reconciled with IRT records at closeout.

STANDARDS AND GUIDANCE

- ICH E6(R3): documenting and explaining premature, including emergency, unblinding (Sections 2.11 and 4.1.4)
- 21 CFR Part 11: audit trails for electronic records

IN THIS TOOLKIT

IRT UAT tracker: emergency unblinding test scripts (apps/uat_irt/index.html)

S3–38 Protocol-deviation classification and reconciliation log (important versus non-important)

Log or record · Quality & compliance · Clinical operations with Lead biostatistician, Medical monitor, Data manager · First participant dosed → Database lock · All designs

Conduct-phase deviation register with documented importance classification, reconciled between the clinical operations tracker, EDC deviation data and the analysis-population impact assessment, and reviewed while blinded.

Why it matters. Unclassified or unreconciled deviations make analysis-set exclusions look arbitrary and leave the CSR deviation section incomplete.

NEEDS

Deviation tracker; EDC deviation data; protocol-defined deviation criteria

Depends on: S2–21, S3–01

PRODUCES

Classified deviation log; blinded review notes; map of deviations to analysis-set impact (source for SDTM DV)

Feeds: S3–40

QUALITY CONTROL

Blinded review with documented medical and biostatistics adjudication.

COMMON PITFALLS

- Importance assigned after unblinding, inviting bias in exclusions.
- The same deviation recorded in the tracker and in EDC with different categories.
- Dosing or sampling deviations not linked to the PK profiles they affect.

STANDARDS AND GUIDANCE

- ICH E3: describing important protocol deviations by category (Section 10.2)
- ICH E9: handling of protocol violations identified during the blind review (Section 5.2)
- ICH E3 Q&As (R1): definitions of protocol deviation and important protocol deviation, the latter set in part by study design, critical procedures and planned analyses (Question 7)

S3–39 Analysis-population derivation and participant-disposition reconciliation (population-flag QC)

Review record · Biostatistics · Lead biostatistician with Statistical programmer, PK scientist · Blinded data review → Unblinding · All designs

Documents how each analysis set (for example randomized, safety, PK concentration, PK parameter, PD, per-protocol) is derived into ADSL flags, with the blinded allocation decisions, exclusion reasons, and QC that the flags reconcile to disposition, randomization, deviations and PK data.

Why it matters. Flags that cannot be traced to documented rules and blinded decisions make every denominator open to challenge.

NEEDS

ADSL; ADPP (PK evaluability); disposition; randomization; deviation classifications; data review decisions

Depends on: S3–20

PRODUCES

Population derivation memo; ADSL flag QC; blinded allocation decision log

Feeds: S4–02, S4–23

QUALITY CONTROL

Independent double programming of the flags; blinded allocation sign-off.

COMMON PITFALLS

- PK evaluability decided after looking at treatment-level results.
- In BE studies, protocol-violation exclusions decided after sample analysis; only the pre-dose and low-exposure rules may use concentrations.
- Exclusion reasons stored as free text, so the CSR listing of excluded participants is incomplete.

STANDARDS AND GUIDANCE

- ICH E9: analysis sets (Section 5.2)
- ADaMIG v1.3: ADSL population flags
- ICH M13A: BE studies: exclusion criteria defined in the protocol and exclusions documented before bioanalysis (Section 2.2.1)

IN THIS TOOLKIT

Worked study: PK exclusion rules evaluated per participant and period (study/03_adam/data/exclusion_rules_evaluated.csv) · Worked study: analysis-set counts (study/04_analysis/analysis_sets.csv) · Worked study: QC ledger (population flags re-derived, check Q5) (study/05_qc/qc_ledger.html) WORKED STUDY

S3–40 Data review meeting minutes and blinded data-review sign-off (pre-lock)

Review record · Data management interface · Data manager with Lead biostatistician, Medical monitor, PK scientist, Clinical operations · Blinded data review → Database lock · All designs

Minutes and decisions from the pre-lock data review meeting: disposition of outstanding queries, cleaning decisions, analysis-set and deviation determinations made while blinded, attendees and roles, and the formal go or no-go to lock.

Why it matters. Undocumented decisions cannot be shown to predate unblinding, which inspectors and reviewers check.

NEEDS

Data review package; query status; deviation and analysis-set determinations

Depends on: S3–30, S3–38

PRODUCES

Signed minutes; decision log; blinded sign-off to proceed to lock

Feeds: S3–32

QUALITY CONTROL

Attendees confirm the minutes; signatures retained with dates before unblinding.

COMMON PITFALLS

- Minutes signed after unblinding, so their dates no longer show that decisions were blind.
- Decisions recorded without the rule or SAP section that justifies them.
- Open action items carried into lock without closure.

STANDARDS AND GUIDANCE

- ICH E9: blind review before breaking the blind to finalize the planned analysis
- ICH E6(R3): maintaining the blind during database review before planned unblinding (Section 4.1.1)

S3–41 Database lock authorization and 21 CFR Part 11 audit-trail review records

Log or record · Quality & compliance · Data manager with Quality assurance, Lead biostatistician, Sponsor · Blinded data review → Database lock · All designs

Formal lock authorization memo plus evidence of the EDC audit-trail review (audit-trail completeness, electronic-signature manifest, lock and unlock records, access logs) supporting Part 11 compliance.

Why it matters. Without authorization and audit-trail evidence, the integrity of the locked data cannot be demonstrated at inspection.

NEEDS

EDC audit trail; query closure status; electronic-signature records; lock checklist

Depends on: S3–32

PRODUCES

Lock authorization memo; audit-trail review report; electronic-signature manifest

Feeds: S3–33

QUALITY CONTROL

QA review of the audit trail; independent lock approval.

COMMON PITFALLS

- Audit-trail review limited to counts of changes, not who changed critical data and why.
- Investigator signatures missing for eCRF pages entered late.
- Accounts of departed staff still active at lock.

STANDARDS AND GUIDANCE

- 21 CFR Part 11: audit trails, access controls and electronic signatures
- Electronic Systems, Electronic Records, and Electronic…: electronic systems, records and signatures used in clinical investigations
- EMA/INS/GCP/112288/2023: audit trail and audit trail review
- ICH E6(R3): audit trails and planned review of data and metadata (Sections 4.2.2 and 4.2.3)

S3–42 PK sampling-time deviation and actual-versus-nominal time reconciliation (PK timing QC)

Review record · PK & pharmacometrics · PK scientist with Statistical programmer · First participant dosed → Database lock · All designs

Reconciliation and QC of actual versus nominal sampling and dosing times feeding ADPC and NCA, including out-of-window collections and missing or repeat samples.

Why it matters. AUC, Cmax and half-life depend on actual times, so unreconciled times bias parameters without any visible error.

NEEDS

Actual collection and dosing date-times; nominal schedule; sampling windows from the PK analysis plan

Depends on: S2–15, S3–01

PRODUCES

Actual-versus-nominal time QC; out-of-window listing; confirmed timing inputs for ADPC

Feeds: S3–32

QUALITY CONTROL

Independent check against source; PK scientist sign-off.

COMMON PITFALLS

- Daylight-saving clock changes not handled, shifting actual times for overnight samples.
- Reference dose time missing in multiple-dose profiles, so actual times silently fall back to nominal.
- Out-of-window samples excluded instead of analyzed at their actual time as the plan specifies.

STANDARDS AND GUIDANCE

- ADNCA v1.0: nominal and actual relative-time variables for NCA input data
- SDTMIG v3.4: PC and EX timing variables
- ICH M13A: BE studies: PK parameters calculated with the actual sampling times (Section 2.2.2.2)

IN THIS TOOLKIT

Worked study: ADPC relative times re-derived in QC (ledger check Q4) (study/05_qc/qc_ledger.html) WORKED STUDY

BE–03 BE analysis-set and period-exclusion derivations

Dataset · Programming & CDISC · Statistical programmer with PK scientist, Lead biostatistician · Blinded data review → Unblinding · Bioequivalence

ADaM derivations that flag each period and participant for the BE analysis: exclusions documented before bioanalysis (protocol violations, emesis, incomplete sampling) and exclusions that need concentrations (pre-dose concentration above 5% of that period's Cmax, very low exposure). They produce the BE analysis-set flag and the flags used by supplementary analyses.

Why it matters. Exclusions applied without a traceable flag and reason cannot be reproduced and may contradict what was documented before bioanalysis.

NEEDS

BE analysis section of the SAP; deviation classification log with dates; ADPC and NCA results; emesis and dosing times.

Depends on: BE–02, S3–20

PRODUCES

Period-level exclusion flags with reasons in ADPP, BE analysis-set flag in ADSL, and a listing of excluded data for the CSR.

Feeds: S4–10, BE–04, BE–05

QUALITY CONTROL

Independent double programming of the flags; each exclusion traced to a dated source record; check that protocol-violation exclusions predate bioanalysis.

COMMON PITFALLS

- Deciding exclusions after concentrations are known, beyond the pre-dose and very-low-exposure rules M13A allows.
- Dropping excluded data from listings; all data should still be submitted and flagged.

STANDARDS AND GUIDANCE

- ICH M13A: exclusions documented before bioanalysis (2.2.1); very low exposure means AUC below 5% of the product's geometric mean AUC computed without that participant, in general no more than 1 participant per study (2.2.1.1); a period is excluded from the primary analysis when its pre-dose concentration exceeds 5% of Cmax (2.2.3.3)
- ICH M13A Guideline: Bioequivalence for Immediate-Release…: in a 2-way crossover a participant with one excluded period is not included in the statistical analysis (Q&A 2.9)
- ADaMIG v1.3: population flags in ADSL and record-level analysis flags (ANLzzFL) in BDS datasets

IN THIS TOOLKIT

Worked study: exclusion rules evaluated per period (study/03_adam/data/exclusion_rules_evaluated.csv) · Carryover and the M13A 5% pre-dose rule (synthetic operating characteristics) (sim_lifecycle/carryover_oc/index.html) WORKED STUDY

DDI-03 Drug interaction treatment-condition derivations

Dataset · Programming & CDISC · Statistical programmer with PK scientist, Lead biostatistician · Blinded data review → Unblinding · Drug-drug interaction

ADaM derivations that assign each object PK profile to its treatment condition (object alone or with the precipitant), check precipitant dosing before and during the profile against the protocol, pair profiles within participant, and flag profiles for the interaction analysis.

Why it matters. Misassigned conditions or unflagged missed precipitant doses bias the interaction ratio.

NEEDS

EX records for both drugs; ADPC and ADPP; DDI analysis section; protocol dosing schedule.

`Depends on: DDI-02, S3-20`

PRODUCES

Treatment-condition and analysis flags with reasons in ADPP; listing of profiles with precipitant dosing deviations.

`Feeds: S4-10`

QUALITY CONTROL

Independent double programming of condition and flag logic; reconciliation to EX records.

COMMON PITFALLS

- Deriving the condition from the planned schedule instead of actual precipitant dosing.

STANDARDS AND GUIDANCE

- ICH M12: analysis includes all enrolled participants with evaluable PK, and each dropout is described (5)
- ADaMIG v1.3: period and treatment variables (APERIOD, TRTP, TRTA) and record-level analysis flags in BDS datasets

ESC-01 Per-cohort dose-escalation data package

Output · Safety & committees · Lead biostatistician with Statistical programmer, PK scientist, Medical monitor, Data manager · First participant dosed → Dose-escalation or interim reviews · Single ascending dose, Multiple ascending dose

The versioned set of safety listings and summaries, dosing data and preliminary PK that the Safety Review Committee reviews before each dose-escalation decision, built from a dated data snapshot under the charter's blinding rules. It covers every evaluable participant in the completed cohort, participants who stopped after at least one dose, and cumulative data from earlier cohorts.

Why it matters. Without a fixed, versioned package the committee may decide on incomplete or unreproducible data, and the decision cannot be reconstructed for the CSR or an inspection.

NEEDS

Interim output specifications and data-cut procedure; committee charter decision rules; dated EDC snapshot (adverse events, vital signs, ECG, laboratory, dosing); preliminary PK summary; protocol definition of evaluable participants.

`Depends on: ESC-02, S2-28`

PRODUCES

Package stamped with snapshot date and data status: cohort and cumulative safety summaries, listings, preliminary PK and exposure against protocol limits; input to the escalation decision record.

`Feeds: ESC-04`

QUALITY CONTROL

Risk-proportionate program QC (independent review or double programming of key safety summaries); snapshot date and data status on every output; participant counts reconciled to dosing records before release.

COMMON PITFALLS

- Leaving out participants who stopped after one dose; they still belong in the data review.
- Rebuilding a package from a later snapshot, so the archived version no longer matches what the committee saw.
- Unblinded summaries reaching blinded team members outside the charter's firewall.

STANDARDS AND GUIDANCE

- EMEA/CHMP/SWP/28367/07 Rev. 1: next cohort dosed only after PK, PD and clinical safety data from the preceding cohort are reviewed, with cumulative review of earlier cohorts (8.2.7); evaluable participants and the minimum number for review defined in the protocol (8.2.8)
- ICH E3: all interim analyses, formal or informal, described in the CSR; data reports reviewed at monitoring meetings may be provided in appendix 16.1.9 (11.4.2.3)
- ICH E6(R3): documented quality control of statistical programming, including analysis results for IDMC review (3.16.2(b))

IN THIS TOOLKIT

TRIALMON safety review console (synthetic dose-escalation study) (apps/trialmon/index.html)

ESC-02 Preliminary PK analysis and next-dose exposure projection

Output · PK & pharmacometrics · PK scientist with Pharmacometrician, Lead biostatistician, Bioanalytical laboratory · First participant dosed → Dose-escalation or interim reviews · Single ascending dose, Multiple ascending dose

Preliminary noncompartmental PK on the concentrations available at the data cut, summarized by dose, with a projection of exposure at the next planned dose compared with the protocol's maximum exposure. It is labeled preliminary because it may rely on nominal times and unaudited bioanalytical results, and the final NCA supersedes it.

Why it matters. Without an exposure check the next cohort could exceed the protocol's maximum exposure, and later differences from the final PK go unexplained.

NEEDS

Preliminary bioanalytical results; dosing and sampling times; PK analysis plan rules for preliminary runs; observed dose-exposure data from completed cohorts; protocol maximum exposure and dose-increment limits.

`Depends on: S2-15, S2-29`

PRODUCES

Preliminary concentration listings and PK parameter summaries by dose; projected exposure at the next dose against the exposure limit; data-status label for the escalation package.

`Feeds: ESC-01`

QUALITY CONTROL

Second-person check of preliminary NCA settings and of the projection; preliminary label on every output; preliminary-to-final comparison kept for the CSR.

COMMON PITFALLS

- Quoting preliminary PK parameters later as final results.
- Projecting exposure with a dose-proportional model when emerging data show nonlinear or plateauing exposure.
- Comparing mean projected exposure with the maximum exposure limit; EMA expects the highest individual exposure in the cohort to be considered.

STANDARDS AND GUIDANCE

- EMEA/CHMP/SWP/28367/07 Rev. 1: dose levels include an estimate of the exposure to be achieved, adapted after review of emerging data, with smaller increments for nonlinear PK and attention to plateauing exposure (7.4); maximum exposure predefined per study part (7.5); a reduced PK data set may be acceptable in some cases (8.2.7); individual rather than mean exposure considered against the exposure stopping criterion (8.2.9)

IN THIS TOOLKIT

R template: dose proportionality power model (SAD) (r_tlf_pseudocode/sad/t_dose_proportionality.R)

ESC-03 Sentinel-dosing review record

Review record · Safety & committees · Medical monitor with Lead biostatistician, Clinical pharmacologist, Clinical operations · First participant dosed → Dose-escalation or interim reviews · Single ascending dose, Multiple ascending dose

CONDITIONAL If the protocol uses sentinel dosing

Dated record of the review of all available data for the sentinel participants at the end of the protocol's observation period, and the decision to dose the rest of the cohort, hold or stop. It names the reviewers, the data and data cut reviewed and the stopping rules checked.

Why it matters. Without it there is no evidence that the rest of the cohort was dosed only after the sentinel observation period and data review the protocol requires.

NEEDS

Sentinel participants' adverse events, vital signs, ECG and laboratory data through the observation period; protocol stopping rules; dosing times.

`Depends on: S1-07`

PRODUCES

Signed, dated sentinel review record filed in the TMF; source for the CSR description of dose escalation.

`Feeds: ESC-04`

QUALITY CONTROL

Record timestamp checked against the first non-sentinel dose; reviewer roles match the protocol; any deviation documented.

COMMON PITFALLS

- Record written after the rest of the cohort was dosed, so its timestamp contradicts the protocol sequence.
- Unblinding reviewers unnecessarily to interpret a single sentinel event.

STANDARDS AND GUIDANCE

- EMEA/CHMP/SWP/28367/07 Rev. 1: first dose in a cohort given to a sentinel participant, with one active and one placebo when placebo is used; clearly defined review of all available sentinel data before the rest of the cohort is dosed (8.2.6)
- ICH E6(R3): documentation of important trial-related decisions is an essential record (Appendix C.3.1)

ESC-04 Dose-escalation decision record and escalation log

Log or record · Safety & committees · Safety Review Committee with Lead biostatistician, Medical monitor, Clinical pharmacologist, PK scientist · Dose-escalation or interim reviews → Dose-escalation or interim reviews · Single ascending dose, Multiple ascending dose

Minutes and decision for each escalation review (escalate, repeat, intermediate dose, de-escalate, stop or move to the next part), citing the package version reviewed and the charter rule applied, plus a running log of cohorts, doses, dates and decisions.

Why it matters. Without it the CSR cannot describe how doses were selected and every interim review, and inspectors cannot confirm that decisions followed the predefined rules.

NEEDS

Per-cohort escalation packages; sentinel review records; charter decision and stopping rules; attendance and quorum.

`Depends on: ESC-01, ESC-03`

PRODUCES

Signed decision minutes per review; cumulative escalation log; source for the CSR description of dose selection and interim reviews, and for appendix 16.1.9 where provided.

`Feeds: ESC-05, ESC-06`

QUALITY CONTROL

Minutes approved by the committee; each decision traced to a package version and a charter rule; log reconciled to dosing records before database lock.

COMMON PITFALLS

- Minutes that omit the data cut and package version, so the basis cannot be reconstructed.
- Escalating outside the charter's predefined rules without the required protocol amendment.

STANDARDS AND GUIDANCE

- ICH E3: basis for the doses chosen described (9.4.4); every formal or informal interim analysis described, and monitoring-group minutes may go in appendix 16.1.9 (11.4.2.3)
- EMEA/CHMP/SWP/28367/07 Rev. 1: dose changes outside predefined decision criteria may need a substantial amendment (7.4); unambiguous stopping rules, including for progression to the next part (8.2.9)
- ICH E6(R3): composition, decisions and correspondence of committees involved in trial conduct are essential records (Appendix C.3.1(g))

ESC-05 Cumulative safety and PK summary for part transitions

Output · Safety & committees · Lead biostatistician with PK scientist, Statistical programmer, Medical monitor, Clinical pharmacologist · Dose-escalation or interim reviews → Dose-escalation or interim reviews · Single ascending dose, Multiple ascending dose

CONDITIONAL If the protocol has more than one part (for example SAD followed by MAD)

Cumulative summary across all completed cohorts, prepared before a new study part starts (for example SAD to MAD, or adding a food-effect or patient cohort): adverse events by dose, laboratory, vital sign and ECG findings, and dose-exposure across cohorts against the protocol's maximum exposure.

Why it matters. Without it the next part's starting dose and regimen rest on cohort-by-cohort snapshots instead of the totality of data reviewed cumulatively.

NEEDS

All escalation packages and decision records to date; preliminary PK by dose; protocol part-transition criteria.

`Depends on: ESC-04`

PRODUCES

Cumulative tables and a dose-exposure figure with limits; input to the part-transition decision and, where needed, to a protocol amendment.

`Feeds: S5-03`

QUALITY CONTROL

Independent check of cumulative counts against the individual packages; dose-exposure figure checked against preliminary PK listings.

COMMON PITFALLS

- Mixing preliminary and final PK values in one dose-exposure plot without labeling data status.
- Missing late adverse events reported after earlier escalation decisions.

STANDARDS AND GUIDANCE

- EMEA/CHMP/SWP/28367/07 Rev. 1: before a further part starts, sufficient information from completed preceding parts or cohorts; cumulative review of earlier cohorts, including late-emerging events (8.2.7)

FE–03 Meal-compliance and dosing-condition derivations

Dataset · Programming & CDISC · Statistical programmer with Data manager, PK scientist, Lead biostatistician · First participant dosed → Database lock · Food effect

Derivations from meal and dosing records: fasting duration, meal start and end relative to dosing, meal completion and post-dose food restriction, giving flags that show whether each fed or fasted period met the protocol conditions and enters the food-effect analysis.

Why it matters. Without derived compliance flags, periods with incomplete meals or wrong timing enter the comparison and bias the food effect.

NEEDS

Meal and dosing CRF data; protocol fasting and meal-timing rules; food-effect analysis section of the SAP.

`Depends on: FE–02, S3–07`

PRODUCES

Period-level condition-compliance flags with reasons in ADaM; listing of noncompliant periods for the blinded data review and the CSR.

`Feeds: S4–10`

QUALITY CONTROL

Independent double programming of the flags; spot-check against source meal records; listing reviewed at the blinded data review.

COMMON PITFALLS

- Recording only planned meal times, so compliance cannot be derived.
- Excluding periods for meal noncompliance without a prespecified rule.

STANDARDS AND GUIDANCE

- Assessing the Effects of Food on Drugs in INDs and NDAs:…: overnight fast of at least 10 hours; meal started 30 minutes before dosing and eaten in 30 minutes or less; no food for at least 4 hours after the dose; meal composition described in the study report (IV.C, IV.F)
- ICH E6(R3): criteria for excluding participants or data from an analysis set predefined, with the rationale documented (3.16.2(d))

HI–03 Child-Pugh classification and unbound PK derivations

Dataset · Programming & CDISC · Statistical programmer with Medical monitor, Central laboratory, PK scientist · Blinded data review → Unblinding · Hepatic impairment

Derivation of the Child-Pugh score from its five components (encephalopathy, ascites, bilirubin, albumin, prothrombin time or INR), the impairment group, the assessment conditions, any reclassification closest to PK sampling, and unbound fraction and unbound PK parameters.

Why it matters. A misderived score or undocumented reclassification puts participants in the wrong group and distorts every comparison.

NEEDS

Screening and pre-dose laboratory values; clinical assessments of ascites and encephalopathy; protein-binding results; hepatic analysis section.

`Depends on: HI–02, S3–09, S3–20`

PRODUCES

Score, component points and group in ADSL; unbound PK parameters in ADPP; listing of classification changes.

`Feeds: HI–04`

QUALITY CONTROL

Independent double programming of score and group; component values checked against source.

COMMON PITFALLS

- Scoring an INR raised by anticoagulants or vitamin K deficiency as hepatic impairment.
- Using one Child-Pugh table in the protocol and another in programming; the FDA and EMA tables word components differently.

STANDARDS AND GUIDANCE

- Pharmacokinetics in Patients with Impaired Hepatic…: total score and each of the five components reported, with how ascites and encephalopathy were assessed; assessment repeated when values change substantially before dosing, and the group closest to PK sampling used (IV.A, IV.B; draft, September 2026)
- CPMP/EWP/2339/02: Child-Pugh classification table in the appendix, with total scores 5 to 6 mild (A), 7 to 9 moderate (B) and 10 to 15 severe (C)

RI–03 Renal function classification, unbound PK and dialysis derivations

Dataset · Programming & CDISC · Statistical programmer with PK scientist, Central laboratory, Bioanalytical laboratory · Blinded data review → Unblinding · Renal impairment

Derivations of baseline renal function by the protocol method (eGFR converted to mL/minute when required), renal function group, unbound fraction and unbound PK parameters, renal clearance from urine, and dialysate recovery and dialysis clearance for dialysis periods.

Why it matters. Errors in renal function or unbound derivations pass directly into the regression and the dose recommendation.

NEEDS

Serum creatinine and other markers required by the protocol equation; body size; protein-binding results; urine and dialysate concentrations and volumes; renal analysis section.

`Depends on: RI–02, S3–09, S3–20`

PRODUCES

Renal function variables and group in ADSL; unbound and renal clearance parameters in ADPP; dialysis clearance.

`Feeds: RI–04`

QUALITY CONTROL

Independent double programming of renal function and clearance derivations; spot-check against laboratory source.

COMMON PITFALLS

- Switching renal function equations between screening and analysis without documenting it.

STANDARDS AND GUIDANCE

- Pharmacokinetics in Patients with Impaired Renal…: eGFR in mL/minute obtained by multiplying the value per 1.73 m2 by body surface area and dividing by 1.73; unbound fraction measured when protein binding is altered; dialysis clearance as the amount recovered in dialysate divided by the AUC over the dialysis session (IV, V)
- Guideline on the evaluation of the pharmacokinetics of…: renal function expressed as absolute GFR in mL/min (5.1, 6.2.1)

STAGE 4 · CLOSES AT OUTPUTS QC-PASSED

Analysis & QC

Tables, listings, figures and PK analyses are produced from the locked data, independently checked, and every reported number traces to a qualified program. 42 deliverables.

ID	DELIVERABLE	OWNER	FROM → TO	DESIGNS
S4-01	Analysis environment qualification and program setup (directory structure, validated software versions, run and seed control)	Statistical programmer	G2 → G3	All
S4-02	Disposition tables (screened, randomized, dosed, completed or discontinued; analysis-set attribution)	Statistical programmer	G2 → G4	All
S4-03	Demographics and baseline characteristics tables	Statistical programmer	G2 → G4	All
S4-04	Exposure, dosing and treatment-compliance tables	Statistical programmer	G2 → G4	All
S4-05	Protocol-deviation summary tables (with importance classification)	Statistical programmer	G2 → G4	All
S4-06	PK concentration summary tables and mean concentration-time figures (linear and semi-log)	Statistical programmer	G2 → G4	All
S4-07	Noncompartmental analysis (NCA) execution in validated software (PK parameter derivation)	PK scientist	G3 → G4	All
S4-08	PK parameter (NCA) summary tables (Cmax, Tmax, AUC, t1/2, CL/F, Vz/F)	Statistical programmer	G2 → G4	All
S4-09	Dose-proportionality and accumulation analyses	Statistical programmer	G2 → G4	SAD, MAD
S4-10	Food-effect, DDI and relative BA or BE statistical analysis (ANOVA, 90% CI, geometric mean ratio)	Statistical programmer	G2 → G4	BE, BA, FE, DDI
S4-11	Concentration-QTc (C-QTc) and QT/QTc analysis CONDITIONAL	Pharmacometrician	G2 → G4	SAD, MAD, QT
S4-12	Population PK and exposure-response modeling with run records CONDITIONAL	Pharmacometrician	G3 → G4	All
S4-13	Safety TLFs: TEAEs (overview, by SOC and PT, severity, relationship, by dose)	Statistical programmer	G2 → G4	All
S4-14	Safety TLFs: deaths, SAEs and AEs leading to discontinuation	Statistical programmer	G2 → G4	All
S4-15	Safety TLFs: clinical laboratory analyses (summary, change from baseline, shift tables, outliers, Hy's law)	Statistical programmer	G2 → G4	All
S4-16	Safety TLFs: vital signs, 12-lead ECG parameters and physical examination	Statistical programmer	G2 → G4	All
S4-17	PD endpoint analyses (biomarker and target-engagement summaries and PK/PD) CONDITIONAL	Pharmacometrician	G2 → G4	All
S4-18	Immunogenicity and anti-drug antibody (ADA) analyses CONDITIONAL	Statistical programmer	G2 → G4	All
S4-19	Exploratory and post hoc analyses CONDITIONAL	Lead biostatistician	G2 → G5	All
S4-20	Statistical analysis execution (run-all production of SAP-specified analyses)	Statistical programmer	M4 → G4	All
S4-21	Independent double-programming QC and reconciliation (discrepancy resolution)	Statistical programmer	G2 → G4	All
S4-22	TLF validation and QC documentation with discrepancy log	Statistical programmer	G2 → G4	All
S4-23	Blind data review meeting and documented decisions (before unblinding)	Lead biostatistician	G3 → M4	All
S4-24	Unblinding and code-break execution CONDITIONAL	Lead biostatistician	M4 → M4	All
S4-25	Pharmacometrics (population PK) analysis report CONDITIONAL	Pharmacometrician	G3 → M5	All
S4-26	QC documentation and validation records package (analysis stage)	Lead biostatistician	G4 → G5	All
S4-27	Independent NCA verification record (parameter-derivation QC)	PK scientist	G3 → G4	All
S4-28	Pharmacometrics model qualification and validation records (covariate selection, goodness of fit, VPC, bootstrap, run log) CONDITIONAL	Pharmacometrician	G3 → M5	All
S4-29	Risk-based QC plan execution and QC-level assignment matrix (per-output QC tiering)	Lead statistical programmer	G2 → G4	All
S4-30	Concentration-QTc (C-QTc) model assumption checks and QC record CONDITIONAL	Pharmacometrician	G3 → G4	SAD, MAD, QT
S4-31	Post-unblinding data change and analysis deviation log	Lead biostatistician	M4 → G5	All
BA-03	Relative or absolute bioavailability results	Statistical programmer	M4 → G4	BA
BA-04	Clinical relevance assessment when bioavailability similarity is not shown CONDITIONAL	Pharmacometrician	G4 → G5	BA
BE-04	Reference-scaled BE analysis for a highly variable drug (replicate design) CONDITIONAL	Statistical programmer	M4 → G4	BE
BE-05	BE supplementary and sensitivity analyses	Statistical programmer	M4 → G4	BE
BE-06	BE presentation set: ANOVA tables, individual ratios and profiles	Statistical programmer	M4 → G4	BE
DDI-04	Drug interaction results: GMR, forest plot and inhibitor or inducer classification	Statistical programmer	M4 → G4	DDI
FE-04	Fed versus fasted results and labeling support	Statistical programmer	M4 → G4	FE
HI-04	Hepatic impairment results: PK by Child-Pugh group and dosing support	Statistical programmer	M4 → G4	HI
QT-04	Thorough QT by-time-point analysis and assay sensitivity CONDITIONAL	Statistical programmer	M4 → G4	QT
QT-05	Cardiodynamic analysis report CONDITIONAL	Pharmacometrician	M4 → G5	QT, SAD, MAD
RI-04	Renal impairment results: PK versus renal function and dosing support	Statistical programmer	M4 → G4	RI

S4–01 Analysis environment qualification and program setup (directory structure, validated software versions, run and seed control)

Program · Quality & compliance · Statistical programmer with Lead statistical programmer, Quality assurance · SAP and specifications final → Database lock · All designs

A qualified, version-locked execution environment for analysis programs, NCA software and modeling software, with documented versions, fixed seeds and a controlled directory structure that separates development from production. Establishes reproducibility before any production run.

Why it matters. Outputs produced in an undocumented environment cannot be reproduced or defended once software versions change.

NEEDS

Validation and QC plan; software validation records; SAP; programming specifications

`Depends on: S2–31`

PRODUCES

Reproducible environment for TLF, NCA and modeling runs; version log

`Feeds: S4–07, S4–11, S4–12`

QUALITY CONTROL

Installation and operational qualification records; environment checklist; version log; reviewer sign-off.

COMMON PITFALLS

- Package or library versions float between the dry run and production because nothing pins them.
- Development and production share folders, so unvalidated edits reach production outputs.
- Random seeds set inside programs but not recorded with each run.

STANDARDS AND GUIDANCE

- ICH E9: reliable software with documented testing for data management and analysis (Section 5.8)
- ICH E6(R3): risk-based validation of computerized systems (Section 4.3.4)
- GAMP 5 Guide, 2nd Edition: risk-based approach to computerized system validation

IN THIS TOOLKIT

Worked study: run manifest (software versions and seed) (study/run_manifest.json) · Worked study: package-version lockfile script (study/R/snapshot_lock.R) `WORKED STUDY`

S4–02 Disposition tables (screened, randomized, dosed, completed or discontinued; analysis-set attribution)

Output · Biostatistics · Statistical programmer with Lead biostatistician · SAP and specifications final → Outputs QC-passed · All designs

Participant disposition by cohort or treatment: screened, enrolled, randomized, dosed, completed and discontinued with reasons, plus the number of participants in each analysis set.

Why it matters. The disposition table sets the denominators readers check first, so an error here undermines every other table.

NEEDS

ADSL; disposition data; SAP; TLF shells

`Depends on: S2–04, S3–39`

PRODUCES

CSR disposition displays (Section 10.1 and Section 14); analysis-set denominators for later tables

`Feeds: S4–20`

QUALITY CONTROL

Independent double programming and reconciliation against the shells; analysis-set counts cross-checked to ADSL.

COMMON PITFALLS

- Screen failures counted as enrolled when the SAP defines enrollment at randomization.
- Crossover discontinuations counted per period in one table and per participant in another.
- Analysis-set counts that do not match the population flags used in later tables.

STANDARDS AND GUIDANCE

- ICH E3: disposition of participants (Section 10.1) and data sets analyzed (Section 11.1)

IN THIS TOOLKIT

Worked study: analysis-set counts (no full disposition table) (study/04_analysis/analysis_sets.csv) · R TLF library: disposition table (crossover) (r_tlf_pseudocode/crossover/t_disposition.R) `WORKED STUDY`

S4–03 Demographics and baseline characteristics tables

Output · Biostatistics · Statistical programmer with Lead biostatistician · SAP and specifications final → Outputs QC-passed · All designs

Age, sex, race, ethnicity, weight, height, BMI and other baseline characteristics by cohort or treatment and overall, in the safety set.

Why it matters. Readers judge the study population, and how far PK and safety results generalize, from this table.

NEEDS

ADSL; SDTM DM, VS and MH where needed; SAP; TLF shells

`Depends on: S2–04, S3–18`

PRODUCES

CSR demographics displays (Section 11.2 and Section 14); covariate context for population PK

`Feeds: S4–20`

QUALITY CONTROL

Independent double programming and reconciliation; statistics verified against ADSL.

COMMON PITFALLS

- Race and ethnicity categories collapsed differently from the CRF without a footnote.
- BMI recomputed in the table from rounded height and weight.
- Crossover tables summarized by treatment, counting each participant twice.

STANDARDS AND GUIDANCE

- ICH E3: demographic and other baseline characteristics (Section 11.2)

IN THIS TOOLKIT

Worked study: Table 14.1.2 demographic and baseline characteristics (study/04_analysis/tlf/t_14_1_2.html) · Worked study: TLF program (study/04_analysis/03_tlf.R) `WORKED STUDY`

S4–04 Exposure, dosing and treatment-compliance tables

Output · Biostatistics · Statistical programmer with Lead biostatistician · SAP and specifications final → Outputs QC-passed · All designs

Administered dose, dose level or cohort, duration of exposure, number of doses, dose modifications or interruptions, and compliance for multiple dosing, by treatment.

Why it matters. Safety and PK results cannot be interpreted without knowing how much drug participants actually received.

NEEDS

Exposure analysis dataset or SDTM EX; ADSL; SAP; TLF shells

Depends on: S2–04, S3–25

PRODUCES

CSR exposure and compliance sections; context for safety and PK interpretation

Feeds: S4–20

QUALITY CONTROL

Independent double programming and reconciliation; dosing records cross-checked to EX.

COMMON PITFALLS

- Duration computed from first to last dose date, ignoring interruptions.
- Placebo participants given the active dose level in cumulative dose summaries.
- Compliance derived from scheduled rather than administered doses.

STANDARDS AND GUIDANCE

- ICH E3: extent of exposure (Section 12.1) and measurements of treatment compliance (Section 11.3)

IN THIS TOOLKIT

R TLF library: exposure table (MAD) (r_tlf_pseudocode/mad/t_exposure.R)

S4–05 Protocol-deviation summary tables (with importance classification)

Output · Biostatistics · Statistical programmer with Lead biostatistician · SAP and specifications final → Outputs QC-passed · All designs

Protocol deviations by category and importance, by cohort or treatment, supporting per-protocol and analysis-set decisions.

Why it matters. Readers and inspectors use this table to judge whether deviations could have affected the results.

NEEDS

SDTM DV; classified deviation log; ADSL; SAP; TLF shells

Depends on: S2–04, S3–15, S3–18

PRODUCES

CSR protocol deviations section (Section 10.2) and Appendix 16.2.2 listing

Feeds: S4–20

QUALITY CONTROL

Independent double programming and reconciliation; classifications checked against the blinded review decisions.

COMMON PITFALLS

- Table built from EDC deviation data that were never reconciled with the classified log.
- Deviations counted by record rather than by participant without saying so.
- Deviations that led to exclusions not cross-referenced to the excluded-participant listing.

STANDARDS AND GUIDANCE

- ICH E3: summarizing important protocol deviations (Section 10.2) and listing them (Appendix 16.2.2)

IN THIS TOOLKIT

R TLF library: protocol deviations table (crossover) (r_tlf_pseudocode/crossover/t_protocol_deviations.R)

S4–06 PK concentration summary tables and mean concentration-time figures (linear and semi-log)

Output · PK & pharmacometrics · Statistical programmer with PK scientist · SAP and specifications final → Outputs QC-passed · All designs

Descriptive statistics of concentrations by nominal time and dose or treatment (n, mean, SD, CV%, median, minimum, maximum, geometric mean), with mean and individual concentration-time profiles on linear and semi-log scales. BLQ values handled per the SAP.

Why it matters. These are the first PK outputs reviewers compare with the NCA parameters, so inconsistent BLQ or time handling is visible immediately.

NEEDS

ADPC; SDTM PC; BLQ and nominal versus actual time rules from the PK analysis plan; SAP; TLF shells

Depends on: S2–04, S3–19

PRODUCES

CSR PK concentration tables and figures; individual concentration listings (Appendix 16.2.5)

Feeds: S4–20

QUALITY CONTROL

Independent double programming and reconciliation; BLQ handling verified; figures checked against the tables.

COMMON PITFALLS

- BLQ set to zero in the tables but treated as missing in the figures, so they disagree.
- Semi-log figures drop zero values without a footnote.
- Summaries by nominal time include samples drawn far outside the window.

STANDARDS AND GUIDANCE

- ICH E3: individual drug concentration data (Appendix 16.2.5)

IN THIS TOOLKIT

Worked study: Table 14.4.1.1 plasma concentrations by nominal time (study/04_analysis/tlf/t_14_4_1_1.html) · R TLF library: mean concentration-time figure, linear and semi-log (crossover) (r_tlf_pseudocode/crossover/f_pk_conc_mean.R) WORKED STUDY

S4–07 Noncompartmental analysis (NCA) execution in validated software (PK parameter derivation)

Output · PK & pharmacometrics · PK scientist with Statistical programmer · Database lock → Outputs QC-passed · All designs

NCA run in validated software to derive PK parameters (for example Cmax, Tmax, AUC0-t, AUC0-inf, percent AUC extrapolated, t1/2, lambda-z, CL/F, Vz/F) per the PK analysis plan, with documented lambda-z selection, BLQ handling, actual sampling times and exclusion rules. This run is the source of record for reported parameters.

Why it matters. Every PK parameter table, statistical comparison and exposure-response input depends on this run being correct and reproducible.

NEEDS

ADPC (concentrations and actual times); dosing; PK analysis plan (parameter list, lambda-z, BLQ and exclusion rules); qualified environment

`Depends on: S3–19, S4–01`

PRODUCES

NCA parameters for SDTM PP and ADPP; parameter tables; modeling inputs

`Feeds: S3–06, S4–27, ESC–06`

QUALITY CONTROL

Validated software is the source of record; independent recomputation of key parameters (second analyst or second tool) and reconciliation; lambda-z selection reviewed; run records archived.

COMMON PITFALLS

- Lambda-z points chosen automatically without review, including points still in the absorption phase.
- Software default BLQ handling differs from the PK analysis plan and nobody checks the setting.
- Nominal instead of actual times used for AUC because the time variable was mapped wrongly.

STANDARDS AND GUIDANCE

- ICH M13A: BE studies: parameters to tabulate, actual sampling times, and BLQ values set to zero but omitted from kel and t1/2 (Section 2.2.2.2)
- ICH E9: reliable software with documented testing (Section 5.8)

IN THIS TOOLKIT

Worked study: NCA program (study/04_analysis/01_nca.R) · Worked study: NCA settings and parameter rules (study/04_analysis/nca_settings.json) · Worked study: effect of the BLQ rule versus the software default (study/04_analysis/blq_rule_effect.csv) `WORKED STUDY`

S4–08 PK parameter (NCA) summary tables (Cmax, Tmax, AUC, t1/2, CL/F, Vz/F)

Output · PK & pharmacometrics · Statistical programmer with PK scientist · SAP and specifications final → Outputs QC-passed · All designs

Descriptive statistics (n, arithmetic and geometric mean, SD, CV% and geometric CV%, median, minimum, maximum) of NCA parameters by dose or treatment, read from ADPP.

Why it matters. These tables carry the study's headline PK results into the CSR and into the design of later studies.

NEEDS

ADPP; ADSL; SAP and PK analysis plan; TLF shells

`Depends on: S2–04, S3–20`

PRODUCES

CSR PK parameter tables; inputs to dose-proportionality, accumulation and exposure-response work

`Feeds: S4–20`

QUALITY CONTROL

Independent double programming and reconciliation; values traceable to the NCA output of record.

COMMON PITFALLS

- Geometric CV computed from the arithmetic SD instead of the SD of log values.
- Tmax summarized with mean and SD instead of median and range.
- Parameters flagged as unreliable by NCA rules included in the summaries.

STANDARDS AND GUIDANCE

- ICH E3: CSR tables and figures referred to but not included in the text (Section 14)
- ICH M13A: BE studies: summary statistics to report for PK parameters (Section 2.2.2.2)

IN THIS TOOLKIT

Worked study: Table 14.4.4.1 plasma PK parameters (study/04_analysis/tlf/t_14_4_4_1.html) `WORKED STUDY`

S4–09 Dose-proportionality and accumulation analyses

Output · PK & pharmacometrics · Statistical programmer with PK scientist, Lead biostatistician · SAP and specifications final → Outputs QC-passed · Single ascending dose, Multiple ascending dose

Dose proportionality across dose levels (for example a power model with a confidence interval for the slope, or analysis of dose-normalized exposure) and accumulation ratios for Cmax and AUC between the first dose and steady state with multiple dosing, with supporting figures.

Why it matters. Dose selection for later cohorts and studies relies on whether exposure rises proportionally and how much it accumulates.

NEEDS

ADPP; SAP and PK analysis plan; TLF shells

`Depends on: S2–04, S3–20`

PRODUCES

CSR sections on linearity, accumulation and time dependence

`Feeds: S4–20`

QUALITY CONTROL

Independent double programming and reconciliation; model specification reviewed; outputs traced to ADPP.

COMMON PITFALLS

- Proportionality concluded from a slope interval without prespecified bounds.
- Accumulation computed from AUCs over different intervals on the first dose and at steady state.
- Nominal or placebo doses entered in the model instead of actual administered doses.

STANDARDS AND GUIDANCE

- ICH E3: drug dose, drug concentration and relationships to response (Section 11.4.4)

IN THIS TOOLKIT

R TLF library: dose-proportionality power model (SAD) (r_tlf_pseudocode/sad/t_dose_proportionality.R) · R TLF library: accumulation ratios (MAD) (r_tlf_pseudocode/mad/t_accumulation.R)

S4–10 Food-effect, DDI and relative BA or BE statistical analysis (ANOVA, 90% CI, geometric mean ratio)

Output · PK & pharmacometrics · Statistical programmer with Lead biostatistician, PK scientist · SAP and specifications final → Outputs QC-passed · Bioequivalence, Relative bioavailability, Food effect, Drug-drug interaction

Crossover or parallel analysis of log-transformed Cmax and AUC with the prespecified linear or mixed model: geometric mean ratio (test or treatment over reference) with 90% confidence interval, compared with acceptance limits where the design requires them, for food-effect, drug-drug interaction, or relative bioavailability and bioequivalence studies.

Why it matters. The study's primary conclusion (equivalence, absence of a food effect, or the size of an interaction) rests on this analysis.

NEEDS

ADPP; ADSL (period and sequence); SAP; TLF shells

`Depends on: BE–03, DDI–03, FE–03, S2–04, S3–20`

PRODUCES

CSR comparison section; exposure ratios for dosing and labeling decisions

`Feeds: BA–03, BE–06, DDI–04, FE–04`

QUALITY CONTROL

Independent double programming of the model, ratio and 90% CI with reconciliation; model statement reviewed; values traced to ADPP.

COMMON PITFALLS

- Model terms chosen or changed after seeing the data, although the primary model must be prespecified.
- A carry-over test used to justify a first-period-only analysis, which ICH M13A rejects.
- Participants with evaluable data for only one product kept or dropped inconsistently with the SAP.

STANDARDS AND GUIDANCE

- ICH M13A: 90% CI of the geometric mean ratio within 80.00% to 125.00% (Section 2.2.4); prespecified model with all effects tested (Sections 2.2.3.1 and 2.2.3.2)
- Statistical Approaches to Establishing Bioequivalence: average bioequivalence analysis of log-transformed AUC and Cmax
- Assessing the Effects of Food on Drugs in INDs and NDAs:…: design and analysis of fed versus fasted comparisons
- ICH M12: design, analysis and reporting of clinical drug interaction studies
- Bioavailability Studies Submitted in NDAs or INDs:…: relative bioavailability study considerations

IN THIS TOOLKIT

Worked study: BE analysis program (study/04_analysis/02_be.R) · Worked study: Table 14.4.3.1 geometric mean ratio and 90% CI (study/04_analysis/tlf/t_14_4_3_1.html) · R TLF library: exposure ratio for a fixed-sequence DDI study (r_tlf_pseudocode/single_sequence/t_be_anova.R) `WORKED STUDY`

S4–11 Concentration-QTc (C-QTc) and QT/QTc analysis

Output · PK & pharmacometrics · Pharmacometrician with Lead biostatistician, Statistical programmer · SAP and specifications final → Outputs QC-passed · Single ascending dose, Multiple ascending dose, Concentration-QTc or thorough QT

`CONDITIONAL` If a QTc or concentration-QTc assessment is planned

Cardiodynamic analysis: categorical QTc outlier and change-from-baseline summaries plus a prespecified concentration-QTc mixed-effects model (slope and predicted placebo-corrected change in QTc at relevant exposures, with two-sided 90% confidence intervals), per the cardiodynamic analysis plan.

Why it matters. Whether a dedicated QT study or expanded ECG monitoring is needed later depends on this analysis being prespecified and adequate.

NEEDS

ADEG (time-matched QTc); ADPC; SAP or cardiodynamic analysis plan

`Depends on: S2–04, S3–24, S4–01`

PRODUCES

CSR QT and cardiodynamic section; concentration-QTc evidence for regulatory decisions

`Feeds: S4–20, S4–30`

QUALITY CONTROL

Independent re-modeling or double programming with reconciliation; model assumptions reviewed; run records archived.

COMMON PITFALLS

- Model structure chosen after seeing the data instead of prespecified.
- Observed exposures do not reach well above the expected clinical exposure, limiting the conclusion.
- Placebo participants left out of the model, removing the placebo correction.

STANDARDS AND GUIDANCE

- ICH E14: QT/QTc evaluation, including central tendency and categorical analyses
- ICH E14/S7B Q&As (R1): concentration-QTc analysis as the primary basis for decisions; upper bound of the two-sided 90% CI below 10 ms at the highest clinically relevant exposure (Q&A 5.1)

S4–12 Population PK and exposure-response modeling with run records

Output · PK & pharmacometrics · Pharmacometrician with PK scientist, Lead biostatistician · Database lock → Outputs QC-passed · All designs

`CONDITIONAL` If population PK or exposure-response modeling is planned

Population PK model building (structural, statistical and covariate models) and exposure-response or exposure-safety analyses in nonlinear mixed-effects software, with parameter estimates and precision, diagnostic plots, visual predictive checks, covariate effects and simulations, plus complete archived run records (control files, run log, seeds, datasets).

Why it matters. Dose selection and covariate claims depend on models that can be reproduced from their run records.

NEEDS

Modeling dataset built from ADPC, ADSL and dosing data per its specification; modeling analysis plan; NCA results where used

`Depends on: S2–17, S3–19, S4–01`

PRODUCES

Pharmacometrics analysis report; CSR modeling section; dose-selection inputs

`Feeds: S4–28`

QUALITY CONTROL

Run records (control files, seeds, logs) archived; independent review of model code and assumptions; key results reproduced; diagnostics reviewed.

COMMON PITFALLS

- BLQ observations dropped without assessing the bias the analysis plan said would be addressed.
- Covariate selection rules changed during modeling without documentation.
- Final model run on a dataset version different from the archived one.

STANDARDS AND GUIDANCE

- Population Pharmacokinetics: Guidance for Industry: population PK data, model development and model evaluation
- Exposure-Response Relationships: Study Design, Data…: exposure-response study design and data analysis
- CHMP/EWP/185990/06: reporting the results of population PK analyses

S4–13 Safety TLFs: TEAEs (overview, by SOC and PT, severity, relationship, by dose)

Output · Safety & committees · Statistical programmer with Lead biostatistician, Medical monitor · SAP and specifications final → Outputs QC-passed · All designs

Treatment-emergent adverse event overview and summaries by MedDRA SOC and PT, maximum severity, relationship to study drug, and dose or cohort, with adverse event listings.

Why it matters. Dose-escalation and tolerability conclusions are drawn directly from these counts.

NEEDS

ADAE; SDTM AE; ADSL; SAP; TLF shells

`Depends on: S2–04, S3–21`

PRODUCES

CSR adverse event section and Appendix 16.2.7 listings; safety review committee data

`Feeds: S4–20, ESC–06`

QUALITY CONTROL

Independent double programming and reconciliation; TEAE flag verified; counts cross-checked to ADAE.

COMMON PITFALLS

- A participant with several PTs in one SOC counted more than once in the SOC row.
- Maximum severity taken across periods in crossover tables that are meant to be by treatment.
- Placebo pooled across cohorts in one table and not in another.

STANDARDS AND GUIDANCE

- ICH E3: adverse events (Section 12.2), displays of adverse events (Section 14.3.1) and listings (Appendix 16.2.7)
- MedDRA Version 29.1: SOC and PT hierarchy used for summaries

IN THIS TOOLKIT

Worked study: Table 14.3.1.1 overview of TEAEs (study/04_analysis/tlf/t_14_3_1_1.html) · R TLF library: TEAEs by SOC and PT (SAD) (r_tlf_pseudocode/sad/t_ae_by_soc_pt.R)

`WORKED STUDY`

S4–14 Safety TLFs: deaths, SAEs and AEs leading to discontinuation

Output · Safety & committees · Statistical programmer with Lead biostatistician, Medical monitor · SAP and specifications final → Outputs QC-passed · All designs

Summaries and listings of deaths, serious adverse events and adverse events leading to dose modification or withdrawal, by dose or treatment; the source for narratives.

Why it matters. These events drive narratives, regulatory reporting checks and stopping-rule decisions.

NEEDS

ADAE (seriousness, death and action variables); ADSL; SAP; TLF shells

`Depends on: S2–04, S3–21`

PRODUCES

CSR deaths, serious and other significant adverse events section; narrative source; safety review committee data

`Feeds: S4–20`

QUALITY CONTROL

Independent double programming and reconciliation; serious adverse event and death flags reconciled with the safety database.

COMMON PITFALLS

- Serious adverse event counts differ from the safety database because reconciliation closed before late reports.
- Withdrawal due to an AE taken from disposition in one output and from the AE action field in another.
- Deaths after the reporting window omitted without a footnote.

STANDARDS AND GUIDANCE

- ICH E3: deaths, other serious and other significant adverse events (Section 12.3 and Section 14.3.2)
- 21 CFR Part 312: definition of a serious adverse event for IND safety reporting (21 CFR 312.32)

IN THIS TOOLKIT

R TLF library: deaths, SAEs and AEs leading to withdrawal (SAD) (r_tlf_pseudocode/sad/t_ae_sae_death_withdrawal.R) · R TLF library: SAE and death listing (SAD) (r_tlf_pseudocode/sad/l_sae_death.R)

S4–15 Safety TLFs: clinical laboratory analyses (summary, change from baseline, shift tables, outliers, Hy's law)

Output · Safety & committees · Statistical programmer with Lead biostatistician, Medical monitor · SAP and specifications final → Outputs QC-passed · All designs

Laboratory summaries and change from baseline by visit, shift tables (normal range or toxicity grade), markedly abnormal value listings, and liver test evaluation for potential Hy's law cases (for example a plot of peak ALT against peak total bilirubin), by dose or treatment.

Why it matters. Liver, renal and hematology signals that decide escalation are read from these outputs.

NEEDS

ADLB; SDTM LB; reference ranges; ADSL; SAP; TLF shells

`Depends on: S2–04, S3–22`

PRODUCES

CSR clinical laboratory section (Section 12.4); safety review committee data

`Feeds: S4–20`

QUALITY CONTROL

Independent double programming and reconciliation; reference-range and grade derivations verified.

COMMON PITFALLS

- Peak ALT and peak bilirubin taken from different time windows without saying so.
- Shift tables use a repeat screening value as baseline when the SAP excludes it.
- Marked-abnormality criteria differ between the table and the listing.

STANDARDS AND GUIDANCE

- ICH E3: clinical laboratory evaluation (Section 12.4)
- Drug-Induced Liver Injury: Premarketing Clinical Evaluation: evaluation of liver test abnormalities and Hy's law cases
- CTCAE v6.0 (MedDRA 28.0): grades for shift tables when the protocol uses CTCAE

IN THIS TOOLKIT

R TLF library: laboratory shift table (MAD) (r_tlf_pseudocode/mad/t_lab_shift.R) · R TLF library: peak ALT versus peak bilirubin plot (MAD) (r_tlf_pseudocode/mad/f_lab_lft_scatter.R)

S4–16 Safety TLFs: vital signs, 12-lead ECG parameters and physical examination

Output · Safety & committees · Statistical programmer with Lead biostatistician, Medical monitor · SAP and specifications final → Outputs QC-passed · All designs

Vital signs summaries, change from baseline and outliers; ECG interval summaries (heart rate, PR, QRS, QT, QTcF, QTcB) with change from baseline and categorical outlier tables; and physical examination findings; by dose or treatment.

Why it matters. Hemodynamic and ECG signals are core early-phase safety endpoints and feed escalation decisions.

NEEDS

ADVS; ADEG; SDTM VS, EG and PE; ADSL; SAP; TLF shells

`Depends on: S2–04, S3–15, S3–23, S3–24`

PRODUCES

CSR vital signs, physical findings and other safety observations section (Section 12.5)

`Feeds: S4–20`

QUALITY CONTROL

Independent double programming and reconciliation; baseline, change and outlier criteria verified.

COMMON PITFALLS

- Outlier categories applied to single replicates rather than the time-point mean the SAP defines.
- Categorical QTc counts reported without the number of participants assessed.
- Physical examination changes listed but never compared with baseline findings.

STANDARDS AND GUIDANCE

- ICH E3: vital signs, physical findings and other observations related to safety (Section 12.5)
- ICH E14: categorical QTc analyses: QTc above 450, 480 and 500 ms; increases from baseline above 30 and 60 ms

IN THIS TOOLKIT

R TLF library: vital signs summary (SAD) (r_tlf_pseudocode/sad/t_vitals_summary.R) · R TLF library: categorical QTc table (SAD) (r_tlf_pseudocode/sad/t_ecg_qtc_categorical.R)

S4–17 PD endpoint analyses (biomarker and target-engagement summaries and PK/PD)

Output · PK & pharmacometrics · Pharmacometrician with Statistical programmer, Clinical pharmacologist · SAP and specifications final → Outputs QC-passed · All designs

CONDITIONAL If PD or biomarker endpoints are collected

Descriptive or model-based summaries of pharmacodynamic endpoints and biomarkers (absolute values, change and percent change from baseline) by dose or treatment and time, plus prespecified exploratory PK/PD relationships.

Why it matters. Target engagement and early dose-response evidence come from these analyses.

NEEDS

ADPD; ADPC; SAP or PK/PD analysis plan; TLF shells

`Depends on: S2–04, S3–25`

PRODUCES

CSR pharmacodynamics section; PK/PD and dose-selection inputs

`Feeds: S4–20`

QUALITY CONTROL

Independent double programming and reconciliation; derivations and units verified; PK/PD models reviewed.

COMMON PITFALLS

- PK/PD relationships fitted post hoc but presented as prespecified.
- Assay or laboratory changes during the study not accounted for in the summaries.

STANDARDS AND GUIDANCE

- ICH E3: drug dose, drug concentration and relationships to response (Section 11.4.4)

IN THIS TOOLKIT

R TLF library: PD summary (SAD) (r_tlf_pseudocode/sad/t_pd_summary.R) · R TLF library: PK/PD overlay figure (SAD) (r_tlf_pseudocode/sad/f_pkpd_overlay.R)

S4–18 Immunogenicity and anti-drug antibody (ADA) analyses

Output · PK & pharmacometrics · Statistical programmer with PK scientist, Bioanalytical laboratory · SAP and specifications final → Outputs QC-passed · All designs

CONDITIONAL If immunogenicity is assessed, for example for therapeutic proteins

ADA incidence and prevalence (treatment-emergent ADA, titers, neutralizing antibodies) by treatment and time, and the impact of ADA status on PK, PD and safety, per the immunogenicity analysis plan.

Why it matters. Without this analysis, loss of exposure or hypersensitivity linked to ADA cannot be identified.

NEEDS

Immunogenicity analysis dataset; SDTM IS; bioanalytical ADA results; SAP

`Depends on: S2–04, S3–25`

PRODUCES

CSR immunogenicity section; assessment of ADA impact on PK

`Feeds: S4–20`

QUALITY CONTROL

Independent double programming and reconciliation; treatment-emergent ADA derivation verified.

COMMON PITFALLS

- Treatment-emergent ADA defined without a titer-boost rule for participants positive at baseline.
- Samples affected by drug interference reported as negative without a caveat.
- ADA impact on PK assessed only with group summaries, missing individual loss of exposure.

STANDARDS AND GUIDANCE

- SDTMIG v3.4: IS domain for immunogenicity specimen assessments

IN THIS TOOLKIT

R TLF library: ADA summary (SAD) (r_tlf_pseudocode/sad/t_ada_summary.R) · R TLF library: ADA impact on PK (SAD) (r_tlf_pseudocode/sad/t_ada_impact_pk.R)

S4–19 Exploratory and post hoc analyses

Output · Biostatistics · Lead biostatistician with Statistical programmer · SAP and specifications final → CSR draft · All designs

CONDITIONAL If exploratory analyses are specified or post hoc analyses are requested

Prespecified exploratory analyses and post hoc analyses (subgroups, special populations, additional PK, PD or safety explorations), each clearly labeled as exploratory or post hoc in the outputs and the CSR.

Why it matters. Unlabeled post hoc results can be mistaken for planned findings and erode trust in the whole report.

NEEDS

ADaM datasets; SAP or SAP addendum; post hoc requests documented after unblinding

Depends on: S4–20

PRODUCES

CSR exploratory section; hypotheses for later studies

Feeds: S5–03

QUALITY CONTROL

Risk-based QC, with key exploratory outputs double programmed; labeling reviewed.

COMMON PITFALLS

- Post hoc requests not documented with date, requester and rationale.
- Exploratory subgroup results shown with p-values and no multiplicity context.
- Post hoc outputs produced outside the controlled environment and not reproducible.

STANDARDS AND GUIDANCE

- ICH E9: work not planned in the protocol strictly distinguished in the report (Section 7.1)
- ICH E3: changes in the conduct of the study or planned analyses (Section 9.8)

S4–20 Statistical analysis execution (run-all production of SAP-specified analyses)

Program · Programming & CDISC · Statistical programmer with Lead statistical programmer, Lead biostatistician · Unblinding → Outputs QC-passed · All designs

Controlled production run of all SAP-specified programs against the locked, unblinded analysis datasets in the qualified environment, generating the complete TLF package with logged, reproducible runs.

Why it matters. Outputs not produced in one controlled run cannot be shown to come from the same data and program versions.

NEEDS

Locked ADaM datasets; final SAP; programming specifications; TLF shells; qualified environment

Depends on: BA–03, BE–04, BE–05, BE–06, DDI–04, FE–04, HI–04, QT–04, RI–04, S3–31, S4–02, S4–03, S4–04, S4–05, S4–06, S4–08, S4–09, S4–11, S4–13, S4–14, S4–15, S4–16, S4–17, S4–18, S4–24

PRODUCES

Full TLF package; outputs for the CSR and for QC reconciliation

Feeds: S4–19, S4–21

QUALITY CONTROL

Log review (no errors or unexplained warnings); output-versus-shell check; double-programming reconciliation before release.

COMMON PITFALLS

- Outputs regenerated one at a time after fixes, so the package mixes run dates and dataset versions.
- Log warnings accepted without review, hiding merge or conversion problems.
- Production run started before the unblinding authorization was documented.

STANDARDS AND GUIDANCE

- ICH E9: credibility of results depends on validated methods and software (Section 5.8)
- ICH E6(R3): risk-based validation of computerized systems (Section 4.3.4)

IN THIS TOOLKIT

Worked study: run-all script (study/run_all.R) · Worked study: run manifest (study/run_manifest.json) · Methods workspace: rerun the worked study in the browser (apps/methods/index.html) WORKED STUDY

S4–21 Independent double-programming QC and reconciliation (discrepancy resolution)

Review record · Quality & compliance · Statistical programmer with Lead statistical programmer, Lead biostatistician, PK scientist · SAP and specifications final → Outputs QC-passed · All designs

Independent re-creation of key outputs by a second programmer, and independent recomputation of PK parameters, with formal comparison of production and QC results until reportable values agree; discrepancies investigated and resolved.

Why it matters. Without independent reproduction, programming errors reach the CSR unchecked.

NEEDS

Production outputs; independent QC programs; validation and QC plan; SAP

Depends on: S4–20

PRODUCES

Reconciled, release-ready outputs; discrepancy log; QC sign-off

Feeds: S4–22, S5–01, S5–02, S5–05, S5–09

QUALITY CONTROL

This is the QC step: independent double programming with documented reconciliation; lead biostatistician sign-off.

COMMON PITFALLS

- The QC programmer reuses production code or shared macros, so errors replicate instead of being caught.
- Comparison at displayed precision hides differences in unrounded values.
- QC run against a different dataset version than production.

STANDARDS AND GUIDANCE

- ICH E6(R3): risk-based quality control at each stage of data handling (Section 3.11.3)
- ICH E9: reliable software with documented testing (Section 5.8)

IN THIS TOOLKIT

Worked study: QC re-derivation ledger (same author, not independent double programming) (study/05_qc/qc_ledger.html) · Worked study: QC program (study/05_qc/qc.R) WORKED STUDY

S4–22 TLF validation and QC documentation with discrepancy log

Log or record · Quality & compliance · Statistical programmer with Lead biostatistician · SAP and specifications final → Outputs QC-passed · All designs

Per-output QC records: validation method (independent double programming, peer review, log review), QC status, discrepancies found and resolved, reviewer and date; the audit-ready evidence that each output was checked.

Why it matters. Without per-output records, QC cannot be demonstrated at inspection even if it was done.

NEEDS

Double-programming results; reconciliation outcomes; validation and QC plan

Depends on: S4–21

PRODUCES

QC documentation package; inspection-ready evidence of output QC

Feeds: S4–29

QUALITY CONTROL

Completeness and consistency review of the QC records; lead biostatistician sign-off.

COMMON PITFALLS

- QC marked complete before discrepancies were closed.
- Records state the method planned, not the method actually used.
- Outputs regenerated after QC without a new QC record.

STANDARDS AND GUIDANCE

- EMA/INS/GCP/112288/2023: ALCOA++ attributes of electronic records
- 21 CFR Part 11: electronic records and electronic signatures

IN THIS TOOLKIT

Worked study: QC ledger (method, tolerance and result per check) (study/05_qc/qc_ledger.html) · Worked study: QC ledger data (CSV) (study/05_qc/qc_ledger.csv) WORKED STUDY

S4–23 Blind data review meeting and documented decisions (before unblinding)

Review record · Biostatistics · Lead biostatistician with PK scientist, Medical monitor, Data manager · Database lock → Unblinding · All designs

Final review of the locked analysis datasets before unblinding: confirms analysis-set assignments, deviation classifications, PK exclusions (for example aberrant concentrations or unreliable terminal phases) and any SAP clarifications, all decided while blinded and minuted.

Why it matters. Decisions taken after unblinding look results-driven and weaken the credibility of the analysis.

NEEDS

Blinded review listings; deviation log; PK profile review; draft analysis sets; SAP

Depends on: S3–39

PRODUCES

Signed decisions and minutes; final analysis sets and exclusions; authorization to unblind

Feeds: S4–24

QUALITY CONTROL

Decisions documented before unblinding; cross-functional sign-off; decisions locked.

COMMON PITFALLS

- Meeting held after topline outputs were already run on unblinded data.
- Terminal-phase or profile exclusions decided without the PK scientist present.
- SAP clarifications agreed verbally but never added to the SAP or its amendment log.

STANDARDS AND GUIDANCE

- ICH E9: blind review between trial completion and breaking the blind to finalize the planned analysis
- ICH E6(R3): maintaining the blind during database review before planned unblinding (Section 4.1.1)

S4–24 Unblinding and code-break execution

Log or record · Biostatistics · Lead biostatistician with Randomization and supply (IRT) vendor, Statistical programmer, Sponsor · Unblinding → Unblinding · All designs

CONDITIONAL If the study is blinded

Controlled, authorized release of the randomization codes after lock and the blinded decisions: treatment codes applied to the analysis datasets, with documented authorization, timing and access control.

Why it matters. Unblinding before decisions are locked, or without records, compromises the integrity of every analysis.

NEEDS

Locked database; signed blind review decisions; randomization schedule or code list; unblinding plan

Depends on: S4–23

PRODUCES

Unblinded analysis datasets; treatment-resolved outputs; audit-trail entry; codes for CSR Appendix 16.1.7

Feeds: S4–20, S4–31, S5–16

QUALITY CONTROL

Authorization and timing documented; access-controlled execution; audit trail; segregation of duties verified.

COMMON PITFALLS

- Codes released to the full study team when only the analysis team was authorized.
- Codes merged by randomization number without handling replacement participants.
- Unblinding time not recorded, so pre-unblinding decisions cannot be proven.

STANDARDS AND GUIDANCE

- ICH E6(R3): documenting planned unblinding and who may access unblinded information (Section 4.1)
- ICH E3: randomization scheme and codes (Appendix 16.1.7)
- 21 CFR Part 11: audit trails and access controls

S4–25 Pharmacometrics (population PK) analysis report

Document · PK & pharmacometrics · Pharmacometrician with PK scientist, Clinical pharmacologist · Database lock → CSR final · All designs

CONDITIONAL If population PK or exposure-response modeling is planned

Standalone modeling report covering objectives, data, methods, model development, results, diagnostics, simulations and conclusions, with run records appended; supports the CSR and dose selection.

Why it matters. Without a report traceable to its run records, modeling conclusions cannot be used in regulatory interactions.

NEEDS

Modeling run records and outputs; modeling analysis plan; NCA results

Depends on: S4–28

PRODUCES

CSR modeling section or appendix; dose-selection rationale; submission modeling package

Feeds: S5–01

QUALITY CONTROL

Independent review of model and report; key results reproduced; run-record completeness check.

COMMON PITFALLS

- Report figures generated from a different run than the final model.
- Data exclusions described in aggregate without a listing of excluded records.
- Simulations reported without the parameter uncertainty they assumed.

STANDARDS AND GUIDANCE

- Population Pharmacokinetics: Guidance for Industry: content of population PK study reports
- CHMP/EWP/185990/06: reporting the results of population PK analyses
- ICH M15: model analysis report (MAR) structure, with the MAP as an appendix and justified changes to planned analyses (4.2, Appendix 2)

S4–26 QC documentation and validation records package (analysis stage)

Log or record · Quality & compliance · Lead biostatistician with Lead statistical programmer, Quality assurance · Outputs QC-passed → CSR draft · All designs

Consolidated validation evidence for the analysis: environment qualification, per-output QC records, reconciliations, NCA and modeling run records, seed and version logs, and reviewer sign-offs; the inspection-ready validation file.

Why it matters. A scattered QC trail cannot be presented quickly or completely when an inspector or sponsor audit asks for it.

NEEDS

Environment qualification; per-output QC records; reconciliation logs; NCA and modeling run records

Depends on: S4–27, S4–28, S4–29, S4–30

PRODUCES

Validation file for the trial master file and inspection readiness

Feeds: S5–13

QUALITY CONTROL

Completeness and traceability review; lead biostatistician sign-off; archived per the retention policy.

COMMON PITFALLS

- Package compiled from file names without checking each record's version and date.
- NCA and modeling records held by other teams and never added.
- Superseded QC records mixed with final ones.

STANDARDS AND GUIDANCE

- EMA/INS/GCP/112288/2023: ALCOA++ attributes of electronic records
- 21 CFR Part 11: electronic records and electronic signatures

IN THIS TOOLKIT

Worked study: QC ledger (study/05_qc/qc_ledger.html) · Worked study: run manifest (versions, seed, QC counts) (study/run_manifest.json) WORKED STUDY

S4–27 Independent NCA verification record (parameter-derivation QC)

Review record · PK & pharmacometrics · PK scientist with Statistical programmer · Database lock → Outputs QC-passed · All designs

Independent QC of the NCA run itself: lambda-z point selection, percent AUC extrapolated, partial-area definitions, BLQ handling and parameter reproducibility, checked by a second PK scientist or a second tool and documented as a discrete record.

Why it matters. Reported PK parameters are only as trustworthy as the documented check of the NCA settings behind them.

NEEDS

ADPC; NCA settings and profiles; lambda-z selection rationale

Depends on: S4–07

PRODUCES

NCA QC checklist; lambda-z review; parameter reproducibility sign-off

Feeds: S4–26

QUALITY CONTROL

Independent PK reviewer; documented parameter verification.

COMMON PITFALLS

- The second tool runs with its own default BLQ and interpolation settings, so agreement proves little.
- Only some profiles checked, with no documented selection rule.
- Lambda-z review limited to fit statistics, without visual inspection of the profiles.

STANDARDS AND GUIDANCE

- ICH M13A: BE studies: report the NCA methods, such as the AUC method and the terminal-phase points used for kel (Section 2.2.2.2)
- ICH E9: reliable software with documented testing (Section 5.8)

IN THIS TOOLKIT

Worked study: NCA recomputed with a second package, side by side (same author) (study/05_qc/qc_nca_side_by_side.csv) · Worked study: PK parameter-rule and exclusion checks (study/04_analysis/pk_checks.csv) WORKED STUDY

S4–28 Pharmacometrics model qualification and validation records (covariate selection, goodness of fit, VPC, bootstrap, run log)

Review record · PK & pharmacometrics · Pharmacometrician · Database lock → CSR final · All designs

CONDITIONAL If population PK or exposure-response modeling is planned

The model-development audit trail behind the modeling report: run log, version-controlled control files, goodness-of-fit and diagnostic plots, visual predictive checks, bootstrap or other uncertainty assessments, covariate-selection justification and final-model qualification, reviewed by an independent pharmacometrician.

Why it matters. Without these records, the report's conclusions cannot be verified or reproduced.

NEEDS

Modeling dataset; control files; run records; diagnostics

Depends on: S4–12

PRODUCES

Model run log; diagnostic package; qualification memo

Feeds: S4–25, S4–26

QUALITY CONTROL

Independent pharmacometrics review; reproducible run records.

COMMON PITFALLS

- Run log reconstructed after the fact instead of kept during development.
- Bootstrap run on a different model or dataset version than the final model.
- Failed or unstable runs left out of the log.

STANDARDS AND GUIDANCE

- Population Pharmacokinetics: Guidance for Industry: population PK model evaluation and reporting

S4–29 Risk-based QC plan execution and QC-level assignment matrix (per-output QC tiering)

Log or record · Quality & compliance · Lead statistical programmer with Lead biostatistician · SAP and specifications final → Outputs QC-passed · All designs

Documented assignment of a QC level to each dataset and output (full independent double programming, review-level QC or visual check) with its rationale, plus evidence that the assigned level was carried out for each output.

Why it matters. Without a recorded rationale, lighter QC on some outputs looks like an omission rather than a risk-based choice.

NEEDS

TLF inventory; criticality and risk assessment; validation and QC plan; QC records

Depends on: S4–22

PRODUCES

QC-level assignment matrix; evidence of the QC performed per output

Feeds: S4–26

QUALITY CONTROL

QC lead sign-off; independent double programming required for outputs assigned the highest QC level.

COMMON PITFALLS

- Levels assigned at planning and not revisited when an output gains importance, for example moving into the synopsis.
- Matrix lists the planned QC but holds no evidence that it was done.
- Lowest level given to outputs that feed safety review decisions.

STANDARDS AND GUIDANCE

- ICH E6(R3): quality management and risk management (Section 3.10)
- ICH E8(R1): critical-to-quality factors

S4–30 Concentration-QTc (C-QTc) model assumption checks and QC record

Review record · PK & pharmacometrics · Pharmacometrician with Lead biostatistician · Database lock → Outputs QC-passed · Single ascending dose, Multiple ascending dose, Concentration-QTc or thorough QT

CONDITIONAL If a concentration-QTc analysis is performed

Supporting diagnostics and QC for the concentration-QTc model: tests of model assumptions, hysteresis assessment (by-time-point plots and hysteresis loop plots), goodness of fit, coverage of the observed concentration range, and QC of the mixed-effects fit.

Why it matters. A concentration-QTc conclusion is not credible if assumptions such as a direct effect without delay are untested.

NEEDS

Time-matched concentration and QTc data; prespecified model and selection criteria

Depends on: S4–11

PRODUCES

Assumption-check report; model QC record; outputs for regulatory review

Feeds: S4–26, QT–05

QUALITY CONTROL

Independent review by a second statistician or pharmacometrician.

COMMON PITFALLS

- Hysteresis checked only on pooled means, missing a delay visible in individual profiles.
- Linear model kept despite curvature at high concentrations.
- Predictions extrapolated above the observed concentration range without a caveat.

STANDARDS AND GUIDANCE

- ICH E14/S7B Q&As (R1): prespecify modeling methods; document tests of model assumptions, hysteresis and goodness of fit (Q&A 5.1)

IN THIS TOOLKIT

C-QTc hysteresis check (synthetic method study) (cqtc_hysteresis/index.html) · R TLF library: exploratory C-QTc delay diagnostic (SAD) (r_tlf_pseudocode/sad/t_cqtc_hysteresis.R)

S4–31 Post-unblinding data change and analysis deviation log

Log or record · Biostatistics · Lead biostatistician with Statistical programmer, Medical writer · Unblinding → CSR draft · All designs

Running record, from unblinding to CSR finalization, of every data change made after unblinding and every departure from the final SAP implemented without an SAP amendment: what changed, why, when, who authorized it and what data were available at the time. SAP revisions themselves stay in the SAP amendment and version-control log.

Why it matters. Undocumented changes cannot be described in the CSR as the guidelines require and suggest that results drove the analysis.

NEEDS

Final SAP and its amendment log; programming change requests; data change requests after unblinding; review comments on outputs.

Depends on: S4–24

PRODUCES

Signed change log; source for the CSR section on changes in planned analyses and for appendix 16.1.9.

Feeds: S5–01

QUALITY CONTROL

Periodic review by a second statistician; each entry traced to a program or output change; reconciled with the CSR methods before sign-off.

COMMON PITFALLS

- Recording changes only in program comments, never in a reviewable log.
- Entries without the date relative to unblinding, which reviewers check first.

STANDARDS AND GUIDANCE

- ICH E3: changes in planned analyses described with their timing relative to breaking the blind and the data available (9.8); documentation of statistical methods in appendix 16.1.9
- ICH E6(R3): deviations from planned analyses and post-unblinding data changes documented, justified and reported in the clinical trial report (3.16.2(e))
- ICH E9: careful explanation of deviations from the planned analysis (7.1)

BA–03 Relative or absolute bioavailability results

Output · Biostatistics · Statistical programmer with Lead biostatistician, PK scientist · Unblinding → Outputs QC-passed · Relative bioavailability

Presentation set built on the primary GMRs and 90% CIs: geometric and arithmetic means for AUC(0-t), AUC(0-inf) and Cmax for each prespecified comparison, within- and between-participant variability, Tmax summaries and, with an intravenous reference, absolute bioavailability.

Why it matters. Without these, formulation decisions and bridging claims rest on unreported or inconsistent comparisons.

NEEDS

ADPP; primary comparative analysis; BA analysis section of the SAP; TLF shells.

Depends on: BA–02, S4–10

PRODUCES

BA result tables and figures for the CSR; inputs to formulation decisions and to later biopharmaceutic summaries.

Feeds: S4–20, BA–04

QUALITY CONTROL

Independent double programming of the summary and variability tables; values traced to ADPP.

COMMON PITFALLS

- Reporting only the ratio and CI without geometric means and variability for each formulation.
- Using AUC(0-inf) when extrapolation is large, without the prespecified acceptance rule.

STANDARDS AND GUIDANCE

- Bioavailability Studies Submitted in NDAs or INDs:…: geometric means, arithmetic means and GMRs with 90% CIs for AUC(0-t), AUC(0-inf) and Cmax, with within-participant, between-participant and total variability where available (Appendix A)
- ICH M4E(R2): BA study reports in Module 5.3.1.1 and comparative BA and BE study reports in Module 5.3.1.2

IN THIS TOOLKIT

R template: crossover exposure ratio (test versus reference) (r_tlf_pseudocode/crossover/t_be_anova.R)

BA–04 Clinical relevance assessment when bioavailability similarity is not shown

Document · PK & pharmacometrics · Pharmacometrician with Clinical pharmacologist, Lead biostatistician · Outputs QC-passed → CSR draft · Relative bioavailability

CONDITIONAL If the predefined similarity criteria are not met

When the confidence interval for a formulation comparison falls outside the predefined limits, an assessment using available dose-response or concentration-response data of whether the difference in rate or extent of absorption matters for safety or efficacy, with the resulting recommendation.

Why it matters. Without it, a failed comparison leaves the program choosing between reformulation and more data with no documented rationale.

NEEDS

BA results; exposure-response or dose-response models and data; clinical safety data.

Depends on: BA–03

PRODUCES

Assessment memo or report section with a recommendation (no meaningful impact, dose adjustment, reformulation or further study).

Feeds: S5–03

QUALITY CONTROL

Independent pharmacometrics review; model assumptions and simulations documented and reproducible.

COMMON PITFALLS

- Arguing irrelevance from point estimates alone, ignoring the confidence interval and variability.
- Using an exposure-response model outside the exposure range it was fitted on.

STANDARDS AND GUIDANCE

- Bioavailability Studies Submitted in NDAs or INDs:…: when similarity in BA is not shown, the sponsor shows that the differences do not meaningfully affect safety and efficacy using dose-response or concentration-response data, or considers reformulation or more data (II.A)

BE–04 Reference-scaled BE analysis for a highly variable drug (replicate design)

Output · Biostatistics · Statistical programmer with Lead biostatistician, PK scientist · Unblinding → Outputs QC-passed · Bioequivalence

CONDITIONAL If a replicate design with reference scaling is prespecified for a highly variable drug

Analysis of a partial or full replicate crossover that estimates the within-participant variability of the reference product and applies the prespecified reference-scaled criterion of the target region, with its point-estimate constraint, alongside the average BE result.

Why it matters. Applying the wrong regional criterion, or scaling a parameter the method does not allow, can reverse the BE conclusion.

NEEDS

Replicate-design ADPP; SAP reference-scaling rules and regulatory constants; outlier handling plan.

Depends on: BE–03

PRODUCES

Tables of the reference within-participant SD (sWR), the scaled criterion or widened limits, GMR with 90% CI and point estimate; analyses with and without statistical outliers where requested.

Feeds: S4–20

QUALITY CONTROL

Independent double programming of sWR, the scaled criterion and the point estimate; regulatory constants checked against the cited guidance.

COMMON PITFALLS

- Treating FDA reference scaling and EMA widened limits as interchangeable; criteria and scaled parameters differ.
- Outliers inflating the reference CV; FDA encourages analyses with and without them, and EMA requires a reliable estimate.
- Deciding to scale after the results are known instead of prespecifying it in the protocol.

STANDARDS AND GUIDANCE

- Statistical Approaches to Establishing Bioequivalence: for AUC and Cmax: average BE if sWR is below 0.294; otherwise the 95% upper confidence bound of (muT - muR)^2 - theta x sWR^2 must be at or below 0, with sigma_w0 = 0.25, and the GMR point estimate within 0.8000 to 1.2500; partial or full replicate design (III.C, Appendix G); GMRs reported with 90% CIs or 95% upper confidence bounds (II.C.2)
- Guideline on the investigation of bioequivalence: Cmax limits only may be widened, by exp(+/-0.760 x sWR) up to 69.84% to 143.19%, when the reference within-participant CV exceeds 30% in a replicate design; GMR within 80.00% to 125.00%; prospectively specified with clinical justification; AUC limits not widened (4.1.10)
- Considerations regarding the implementation of ICH M13A…: the EMA guideline continues to apply, read with M13A, to replicate designs for highly variable drugs

BE–05 BE supplementary and sensitivity analyses

Output · Biostatistics · Statistical programmer with Lead biostatistician, PK scientist · Unblinding → Outputs QC-passed · Bioequivalence

Prespecified analyses that test the robustness of the BE conclusion: excluded periods restored, participants with Cmax at the first post-dose sample removed, alternative missing-data handling, group-by-formulation exploration in multi-group studies, and potency-corrected results when correction is prespecified.

Why it matters. Without them a reviewer cannot judge whether the conclusion depends on exclusions, a missed Cmax or the missing-data method.

NEEDS

BE analysis section of the SAP; BE exclusion flags; ADPP; group or site assignments; assayed potency of test and comparator batches if correction is prespecified.

Depends on: BE–03

PRODUCES

Supplementary GMR and 90% CI tables naming the participants removed or restored; text for the CSR discussion of robustness.

Feeds: S4–20

QUALITY CONTROL

Same validation level as the primary analysis for each supplementary table; each analysis marked as prespecified or post hoc.

COMMON PITFALLS

- Labeling post hoc analyses as prespecified.
- Adding the group-by-treatment term to the primary model; M13A puts it in a supportive analysis only.

STANDARDS AND GUIDANCE

- ICH M13A: BE assessed by the 90% CI of the GMR (2.2.3.1); robustness discussed when Cmax is at the first post-dose sample, possibly by removing affected participants (2.1.8.1); group-by-treatment examined in a supportive analysis, not the primary model (2.2.3.5); uncorrected and potency-corrected analyses when correction is prespecified (2.2.2.3)
- Statistical Approaches to Establishing Bioequivalence: sensitivity analyses prespecified to assess assumptions about missing data (II.B.2)

IN THIS TOOLKIT

Worked study: supplementary BE analyses (study/04_analysis/be_supplementary.csv) · Worked study: participants with Cmax at the first post-dose sample (study/04_analysis/first_point_cmax.csv) WORKED STUDY

BE–06 BE presentation set: ANOVA tables, individual ratios and profiles

Output · Programming & CDISC · Statistical programmer with PK scientist, Lead biostatistician, Medical writer · Unblinding → Outputs QC-passed · Bioequivalence

The BE-specific tables, listings and figures that accompany the primary GMR table: individual concentrations and parameters by formulation with summary statistics, individual linear and log-linear profiles on actual times, mean profiles on nominal times, ANOVA tables testing all model effects, and individual test/reference ratios by sequence.

Why it matters. Missing presentations stop a reviewer from repeating the PK and statistical analyses, which the guidelines expect the report to allow.

NEEDS

ADPC, ADPP and the primary BE analysis; BE exclusion flags; TLF shells; randomization scheme.

Depends on: S4–10

PRODUCES

BE TLFs for the CSR body and appendices; listings that let a reviewer repeat the analysis.

Feeds: S4–20

QUALITY CONTROL

Independent double programming of the ANOVA tables and individual ratios; figures checked against listings; completeness check against the M13A presentation items.

COMMON PITFALLS

- Plotting individual profiles on nominal instead of actual sampling times.
- Leaving excluded or withdrawn participants out of listings; their data should still be listed and flagged.

STANDARDS AND GUIDANCE

- ICH M13A: individual and mean linear and log-linear graphs, actual times for individual and nominal times for mean profiles; listed summary statistics (2.2.2); tests of sequence, participant within sequence, period and formulation (2.2.3.2); data sufficient to repeat the analyses and all BE studies listed in Module 2.7.1 regardless of outcome (4)
- Statistical Approaches to Establishing Bioequivalence: individual test/reference ratios tabulated side by side with the sequence shown; geometric and arithmetic means and GMRs with 90% CIs (II.C.2)

IN THIS TOOLKIT

Worked study: ANOVA effects table for ln(Cmax) and ln(AUC(0-t)) (study/04_analysis/tlf/t_14_4_3_2.html) WORKED STUDY

DDI-04 Drug interaction results: GMR, forest plot and inhibitor or inducer classification

Output · Biostatistics · Statistical programmer with Lead biostatistician, Clinical pharmacologist, PK scientist · Unblinding → Outputs QC-passed · Drug-drug interaction

DDI presentation set built on the primary GMRs and 90% CIs: the range of individual ratios; forest, spaghetti or individual-ratio plots; the proportion of participants beyond the no-effect boundary; and, with a sensitive index CYP substrate, classification of the investigational drug as a strong, moderate or weak CYP inhibitor or inducer. ICH M12 has no such classification for transporters or non-CYP enzymes.

Why it matters. Without these, labeling and concomitant-medication rules lack the magnitude, variability and classification evidence they depend on.

NEEDS

Primary comparative analysis; ADPP with treatment-condition flags; DDI analysis section; TLF shells.

Depends on: S4–10

PRODUCES

DDI TLFs and interpretation text for the CSR; inputs to labeling and to concomitant-medication rules in later protocols.

Feeds: S4–20

QUALITY CONTROL

Independent double programming of individual-ratio tables and plots; classification checked against the M12 CYP thresholds.

COMMON PITFALLS

- Classifying inhibitor strength with a substrate that is not a sensitive index substrate, without justification.
- Reporting only mean ratios and hiding participants with large individual effects.

STANDARDS AND GUIDANCE

- ICH M12: GMR with 90% CI and variability such as the range of individual ratios; forest and individual plots; proportion beyond the no-effect boundary (5.2); strong, moderate and weak CYP inhibitors raise a sensitive index CYP substrate's AUC at least 5-fold, 2- to under 5-fold and 1.25- to under 2-fold; CYP inducers lower it by at least 80%, 50% to under 80% and 20% to under 50%; no classification for transporters or non-CYP enzymes (5.3.2)

FE-04 Fed versus fasted results and labeling support

Output · Biostatistics · Statistical programmer with Lead biostatistician, Clinical pharmacologist, PK scientist · Unblinding → Outputs QC-passed · Food effect

Food-effect presentation set built on the primary fed/fasted GMRs and 90% CIs: individual values and summary statistics for all derived parameters, Tmax and Tlag summaries and overlaid mean profiles, with the interpretation needed for the effect-of-food statement in labeling.

Why it matters. Without a complete results set aligned with the guidance, the labeling statement and dosing instructions on food lack support.

NEEDS

Primary comparative analysis; ADPP with condition-compliance flags; food-effect analysis section; TLF shells.

Depends on: S4–10

PRODUCES

Food-effect TLFs and interpretation text for the CSR; input to labeling and dosing instructions.

Feeds: S4–20

QUALITY CONTROL

Independent double programming of the summary tables; interpretation checked against the prespecified criterion.

COMMON PITFALLS

- Concluding no food effect when only one of AUC and Cmax has its CI within the limits.
- Ignoring clinically relevant Tmax or Tlag changes when AUC and Cmax are similar.

STANDARDS AND GUIDANCE

- Assessing the Effects of Food on Drugs in INDs and NDAs:…: 90% CI of the fed/fasted GMR for AUC and Cmax; AUC, Cmax, Tmax, Tlag, t1/2, CL/F and Vd/F reported with individual values and summary statistics; effect of food described under the Absorption heading of labeling (VI)
- ICH M4E(R2): food-effect study reports placed with BA study reports in Module 5.3.1.1

IN THIS TOOLKIT

R template: crossover exposure ratio (test versus reference) (r_tlf_pseudocode/crossover/t_be_anova.R)

HI-04 Hepatic impairment results: PK by Child-Pugh group and dosing support

Output · Biostatistics · Statistical programmer with Lead biostatistician, Pharmacometrician, Clinical pharmacologist · Unblinding → Outputs QC-passed · Hepatic impairment

Geometric and arithmetic means and variability by group, GMRs with 90% CIs versus control for total and unbound AUC and Cmax, plots of PK against classification components, and simulations or no-effect boundary comparisons that support dosing recommendations.

Why it matters. Without these outputs the hepatic dosing statement in labeling lacks a quantitative basis.

NEEDS

ADPP and ADSL with hepatic variables; hepatic analysis section.

Depends on: HI-03, S2-04

PRODUCES

Hepatic TLFs and dosing-support analyses for the CSR and labeling.

Feeds: S4–20

QUALITY CONTROL

Independent double programming of GMR tables and plots.

COMMON PITFALLS

- Pooling the mild and moderate groups after the results to gain precision.

STANDARDS AND GUIDANCE

- Pharmacokinetics in Patients with Impaired Hepatic…: geometric and arithmetic means with variability, and impaired-to-control ratios with 90% CI for total and unbound AUC and Cmax; dosing by simulations within the 5th to 95th percentiles of reference exposure or by no-effect boundaries (V; draft, September 2026)
- CPMP/EWP/2339/02: graphical relationship between measures of hepatic function and PK parameters (4.2)

QT–04 Thorough QT by-time-point analysis and assay sensitivity

Output · Biostatistics · Statistical programmer with Lead biostatistician, Pharmacometrician · Unblinding → Outputs QC-passed · Concentration-QTc or thorough QT

CONDITIONAL If a thorough QT study with a positive control is conducted

For a thorough QT study: baseline- and placebo-adjusted QTc by time point with the largest time-matched mean effect and its one-sided 95% upper bound, the positive-control analysis that shows assay sensitivity, and categorical QTc tables.

Why it matters. Without demonstrated assay sensitivity, a negative thorough QT result cannot be accepted.

NEEDS

ADEG with replicate-averaged QTc; cardiodynamic analysis plan; TLF shells.

`Depends on: QT–02, S3–24`

PRODUCES

By-time-point and positive-control tables and figures; categorical tables; inputs to the cardiodynamic report.

`Feeds: S4–20, QT–05`

QUALITY CONTROL

Independent double programming of time-matched estimates and the positive-control analysis.

COMMON PITFALLS

- Showing positive-control significance without showing that the study could detect an effect near 5 ms.

STANDARDS AND GUIDANCE

- ICH E14: a negative study has a one-sided 95% upper bound for the largest time-matched mean effect that excludes 10 ms (2.2.4); categorical analyses (3.2.2)
- ICH E14/S7B Q&As (R1): positive control with a lower one-sided 95% bound above 0 ms and able to detect an effect of about 5 ms (Q&A 3.1)

QT–05 Cardiodynamic analysis report

Document · PK & pharmacometrics · Pharmacometrician with Lead biostatistician, ECG core laboratory, Medical writer · Unblinding → CSR draft · Concentration-QTc or thorough QT, Single ascending dose, Multiple ascending dose

CONDITIONAL If a thorough QT study or a concentration-QTc analysis is planned

Report of the QT assessment: ECG methods and quality, model development and diagnostics (including hysteresis and goodness of fit), the predicted placebo-adjusted QTc effect with 90% CI at therapeutic and high clinical exposures, categorical findings and, for a thorough QT study, assay sensitivity, with the conclusion stated against the E14 criteria.

Why it matters. Without a complete report the conclusion on QT risk, and the need for later ECG monitoring, cannot be reviewed.

NEEDS

Concentration-QTc analysis outputs and assumption-check records; by-time-point and positive-control outputs for a thorough QT study; reader-variability results.

`Depends on: QT–04, S4–30`

PRODUCES

Cardiodynamic report appended to the CSR; summary text for the CSR and later submissions.

`Feeds: S5–03`

QUALITY CONTROL

Independent pharmacometrics review; key estimates reproduced from archived run records.

COMMON PITFALLS

- Reporting the effect at therapeutic exposure only and omitting the high clinical exposure.

STANDARDS AND GUIDANCE

- ICH E14/S7B Q&As (R1): documentation of model assumptions, hysteresis and goodness of fit, and a decision based on the upper bound of the two-sided 90% CI at the highest clinically relevant exposure (Q&A 5.1)
- ICH E14: central tendency and categorical analyses (3.2)

IN THIS TOOLKIT

R template: exploratory concentration-QTc delay diagnostic (MAD) (r_tlf_pseudocode/mad/t_cqtc_hysteresis.R)

RI–04 Renal impairment results: PK versus renal function and dosing support

Output · Biostatistics · Statistical programmer with Lead biostatistician, Pharmacometrician, Clinical pharmacologist · Unblinding → Outputs QC-passed · Renal impairment

Regression estimates and plots of individual PK parameters against renal function, descriptive statistics and GMRs versus control by group (total and unbound), dialysis results, and simulations or no-effect boundary comparisons that support any dose adjustment.

Why it matters. Without these outputs the renal dosing statement in labeling lacks a quantitative basis.

NEEDS

ADPP and ADSL with renal variables; renal analysis section; exposure-response information.

`Depends on: RI–03, S2–04`

PRODUCES

Renal TLFs and dosing-support analyses for the CSR and labeling.

`Feeds: S4–20`

QUALITY CONTROL

Independent double programming of regression and GMR tables; simulations reproducible from archived code.

COMMON PITFALLS

- Reporting group means only, hiding the continuous relationship that sets the dose-adjustment threshold.

STANDARDS AND GUIDANCE

- Guideline on the evaluation of the pharmacokinetics of…: individual PK parameters plotted against renal function as a continuous variable; descriptive statistics by group and GMRs versus control (6.2.1)
- Pharmacokinetics in Patients with Impaired Renal…: dosing by simulations within the 5th to 95th percentiles of reference-group exposure, or by no-effect boundaries (V.C)

STAGE 5 · CLOSES AT ARCHIVED

Reporting, disclosure & archiving

Results become the clinical study report and its appendices, the submission data package, public results postings and the archived study record. 23 deliverables.

ID	DELIVERABLE	OWNER	FROM → TO	DESIGNS
S5–01	CSR statistical methods sections (ICH E3 Sections 9.7 and 11.4.2)	Lead biostatistician	G3 → M5	All
S5–02	CSR in-text tables and figures	Statistical programmer	G4 → G5	All
S5–03	Biostatistics contribution to and review of the CSR draft (results text, consistency review)	Lead biostatistician	G5 → M5	All
S5–04	CSR Appendix 16.1.9: documentation of statistical methods, including the final SAP	Lead biostatistician	G5 → M5	All
S5–05	CSR Appendix 16.2: participant data listings	Statistical programmer	G4 → G5	All
S5–06	CSR and submission appendices: Define-XML, reviewer guides (cSDRG, ADRG) and annotated CRF CONDITIONAL	Statistical programmer	G5 → M6	All
S5–07	Safety narrative data support CONDITIONAL	Statistical programmer	G3 → G5	All
S5–08	Participant profiles	Statistical programmer	G3 → G5	All
S5–09	Submission datasets and analysis programs (with the reviewer guides) CONDITIONAL	Statistical programmer	G4 → M6	All
S5–10	Traceability matrix (raw data to SDTM, ADaM, TLF and CSR)	Lead biostatistician	G4 → M5	All
S5–11	Numerical consistency QC: CSR versus TLFs versus datasets	Lead biostatistician	G5 → M5	All
S5–12	Statistical input to the CSR synopsis and briefing documents	Lead biostatistician	G5 → M6	All
S5–13	Biostatistics sign-off and CSR release for submission	Lead biostatistician	M5 → M6	All
S5–14	eCTD Module 5 study-data package assembly and conformance (reviewer guide bundle, study tagging) CONDITIONAL	Regulatory affairs	M5 → M6	All
S5–15	ADRG finalization and cross-document consistency (SAP, Define-XML, ADRG, CSR)	Lead biostatistician	G5 → M5	All
S5–16	CSR Appendix 16.1.7: randomization scheme and codes CONDITIONAL	Lead biostatistician	G4 → M5	All
S5–17	ClinicalTrials.gov results data package CONDITIONAL	Lead biostatistician	M5 → M6	All
S5–18	EU CTR summary of results CONDITIONAL	Lead biostatistician	M5 → M6	All
S5–19	Lay summary of results: statistics review CONDITIONAL	Lead biostatistician	M5 → M6	All
S5–20	TMF filing of biostatistics and programming records	Lead biostatistician	G1 → G6	All
S5–21	Archiving of the analysis environment, programs, datasets and outputs	Lead statistical programmer	M5 → G6	All
S5–22	Statistics support for publications and data-sharing requests CONDITIONAL	Lead biostatistician	M5 → G6	All
ESC–06	Preliminary-versus-final data reconciliation for escalation decisions	Lead biostatistician	G4 → G5	SAD, MAD

S5–01 CSR statistical methods sections (ICH E3 Sections 9.7 and 11.4.2)

Document · Reporting & disclosure · Lead biostatistician with PK scientist, Pharmacometrician, Medical writer · Database lock → CSR final · All designs

Description of the statistical and PK methods as actually executed: analysis sets, estimands and intercurrent-event strategies, handling of missing and BLQ data, nominal versus actual times, summary conventions, NCA methods and software, any modeling, comparative PK and concentration-QTc methods, interim or committee rules, and every change from the SAP with its timing relative to unblinding.

Why it matters. Methods text that drifts from the SAP or from what the programs did undermines the credibility of every reported result.

NEEDS

Final SAP and amendments; SAP change log; protocol and estimands; final TLFs; NCA and modeling run records; reviewer guides.

Depends on: S2–02, S4–21, S4–25, S4–31

PRODUCES

Methods text for the CSR body; basis for Appendix 16.1.9; anchor for the results text and synopsis.

Feeds: S5–03, S5–04, S5–15

QUALITY CONTROL

Second-biostatistician review against the final SAP; check that every method maps to a reported output; medical-writer editorial QC; SAP-to-CSR methods reconciliation checklist.

COMMON PITFALLS

- Methods paraphrased from the SAP although the programs implemented a later clarification.
- Changes made after unblinding described without their timing, contrary to ICH E3 Section 9.8.

STANDARDS AND GUIDANCE

- ICH E3: Sections 9.7, 9.8 and 11.4.2: planned methods, changes to planned analyses and statistical issues
- ICH E9(R1): estimands and sensitivity analyses
- ICH M13A: report detailed enough for the PK and statistical analyses to be repeated

IN THIS TOOLKIT

Worked study: CSR Section 11 statistical text, including 11.4.2 (study/06_csr/csr_section11.html) WORKED STUDY

S5–02 CSR in-text tables and figures

Output · Reporting & disclosure · Statistical programmer with Lead biostatistician, Medical writer · Outputs QC-passed → CSR draft · All designs

Condensed tables and figures placed in the CSR body: disposition, demographics, exposure, key PK concentration and parameter summaries, mean concentration-time figures, dose proportionality, comparative PK ratios with confidence intervals, concentration-QTc and key safety summaries. They are abridged from the full Section 14 outputs and produced by the same validated programs from the locked datasets.

Why it matters. In-text values typed or re-derived by hand drift from the source outputs and fail numerical QC.

NEEDS

Locked ADaM datasets; QC-passed full TLFs; NCA and modeling results; in-text shell specifications.

Depends on: S4–21

PRODUCES

In-text tables and figures in the CSR results sections; values cited in the results text and synopsis.

Feeds: S5–03, S5–10

QUALITY CONTROL

Independent double programming or one-to-one reconciliation of every in-text value against the full TLF and source dataset; QC log signed.

COMMON PITFALLS

- In-text tables built by copying from the full TLFs instead of being generated by program.
- Different rounding in in-text and appendix tables for the same statistic.

STANDARDS AND GUIDANCE

- ICH E3: summary tables and figures in the report text, with the full set in Section 14

IN THIS TOOLKIT

TLF program library by design (R) (r_tlf_pseudocode/index.html)

S5–03 Biostatistics contribution to and review of the CSR draft (results text, consistency review)

Review record · Reporting & disclosure · Lead biostatistician with Medical writer, PK scientist, Medical monitor · CSR draft → CSR final · All designs

Biostatistics-authored results text for disposition, PK, PD and safety, plus a structured review of the integrated CSR draft: statistical and PK statements are correct and not over-interpreted, estimand and analysis-set language is consistent, and every number in the text ties to a table. It includes resolution of medical-writing and clinical comments on statistical content.

Why it matters. Statistical misstatements or over-interpretation are hard to correct once the CSR is final and submitted.

NEEDS

CSR draft from medical writing; in-text and full TLFs; ADaM datasets; SAP and SAP change log; review comments.

Depends on: BA–04, ESC–05, ESC–06, QT–05, S4–19, S5–01, S5–02

PRODUCES

Marked-up CSR draft; statistical review sign-off; findings log; clearance for finalization.

Feeds: S5–12

QUALITY CONTROL

Documented number-by-number consistency check (CSR, TLF, dataset); second-reviewer spot check; comment-resolution log kept as a controlled record.

COMMON PITFALLS

- Absence of an effect claimed from a non-significant test instead of from the confidence interval and prespecified limits.
- Descriptive PK differences between small cohorts interpreted as dose-related trends.

STANDARDS AND GUIDANCE

- ICH E3: content of the results, discussion and conclusions sections
- ICH E9: reporting and evaluation of trial results

IN THIS TOOLKIT

Worked study: CSR Section 11 text with per-paragraph number checks (study/06_csr/csr_section11.html) WORKED STUDY

S5–04 CSR Appendix 16.1.9: documentation of statistical methods, including the final SAP

Document · Reporting & disclosure · Lead biostatistician with Medical writer, Quality assurance · CSR draft → CSR final · All designs

Compiled Appendix 16.1.9: the final approved SAP with version history and amendments, the PK analysis plan when separate, output shells, further statistical documentation, and the record of changes from the SAP. It is the auditable record of how the analyses were specified and executed.

Why it matters. An incomplete or wrong-version appendix prevents reviewers from confirming that the analyses were prespecified.

NEEDS

Final SAP and amendments; PK analysis plan; shells; SAP change log; version-control records; CSR methods text.

Depends on: S5–01

PRODUCES

Appendix 16.1.9 of the CSR; reviewer-facing methods documentation in the submission.

Feeds: S5–13

QUALITY CONTROL

Completeness and version reconciliation (approved versions only); cross-check of SAP content against the CSR methods; QA review of the appendix assembly.

COMMON PITFALLS

- A draft SAP version included instead of the signed final.
- SAP clarifications agreed by email never reach the appendix.

STANDARDS AND GUIDANCE

- ICH E3: Appendix 16.1.9 documentation of statistical methods
- 21 CFR Part 11: audit trails for versioned electronic records

S5–05 CSR Appendix 16.2: participant data listings

Output · Reporting & disclosure · Statistical programmer with Lead statistical programmer, Lead biostatistician · Outputs QC-passed → CSR draft · All designs

By-participant data listings under ICH E3 Section 16.2: discontinued participants, protocol deviations, participants excluded from analysis, demographics, compliance and drug concentration data, individual efficacy or PK response data, adverse events, and individual laboratory measurements, generated from locked data by validated programs.

Why it matters. Listings are the reviewer's route to individual data, and gaps or mismatches with the summary tables trigger questions and rework.

NEEDS

Locked SDTM and ADaM datasets; Section 16.2 listing specifications and shells.

Depends on: S4–21

PRODUCES

Appendix 16.2 listings; source for safety narratives and participant profiles.

Feeds: S5–08

QUALITY CONTROL

Independent double programming or QC review of each listing; reconciliation against the summary TLFs; QC log.

COMMON PITFALLS

- Listings filtered to the analysis set, hiding data from excluded participants.
- Sampling-time deviations not flagged in concentration listings.

STANDARDS AND GUIDANCE

- ICH E3: Appendix 16.2 data listings by participant
- ICH M13A: individual concentrations tabulated, with missed or time-deviant samples identified

IN THIS TOOLKIT

TLF program library by design (R), including listings (r_tlf_pseudocode/index.html)

S5–06 CSR and submission appendices: Define-XML, reviewer guides (cSDRG, ADRG) and annotated CRF

Submission component · Programming & CDISC · Statistical programmer with Lead biostatistician, Regulatory affairs · CSR draft → Results posted, submission package · All designs

CONDITIONAL If study data will be submitted to a regulator

Packaging of the SDTM and ADaM Define-XML, the clinical study data reviewer's guide (cSDRG), the analysis data reviewer's guide (ADRG) and the annotated CRF as CSR appendices and Module 5 components, so the datasets are reviewer-ready and traceable.

Why it matters. Missing or mismatched metadata and reviewer guides slow regulatory review and generate information requests.

NEEDS

Final SDTM and ADaM Define-XML; cSDRG and ADRG; annotated CRF; final conformance report.

Depends on: S5–15

PRODUCES

Define-XML and reviewer guides packaged with the datasets in Module 5 and referenced by the CSR.

Feeds: S5–13

QUALITY CONTROL

Conformance re-run on the final datasets; reviewer-guide completeness check; Define-XML versus dataset consistency check.

COMMON PITFALLS

- Reviewer guide explains conformance findings from an earlier run than the submitted datasets.
- Define-XML stylesheet missing, so the file does not display properly for reviewers.

STANDARDS AND GUIDANCE

- Define-XML v2.1: dataset metadata delivered with the data
- PHUSE cSDRG Package: SDTM reviewer guide template and completion guidelines
- PHUSE ADRG Package: ADaM reviewer guide template and completion guidelines
- Study Data Technical Conformance Guide: Technical…: submission of define.xml, reviewer guides and the annotated CRF

S5–07 Safety narrative data support

Document · Reporting & disclosure · Statistical programmer with Medical writer, Medical monitor · Database lock → CSR draft · All designs

CONDITIONAL If deaths, serious adverse events or other significant adverse events occurred

Programming support for ICH E3 narratives of deaths, serious adverse events and other significant adverse events: structured per-participant extracts (demographics, dosing, event timing, severity and relationship, relevant laboratory, vital sign and ECG data, concomitant medications) for medical writers, and QC that narrative facts match the data.

Why it matters. Narratives written from memory or correspondence contradict the datasets and listings that reviewers check against.

NEEDS

Locked SDTM and ADaM data; Section 16.2 listings; participant profiles; narrative trigger list.

Depends on: S5–08

PRODUCES

Per-participant narrative data packages; QC anchor for narrative text.

Feeds: S5–13

QUALITY CONTROL

Narrative-to-source reconciliation per participant; independent check that the trigger list captures all deaths and serious adverse events; QC log.

COMMON PITFALLS

- Trigger list built from the CRF before final coding, so late-coded significant events are missed.
- Narrative dates use a study-day convention different from the listings.

STANDARDS AND GUIDANCE

- ICH E3: Section 12.3.2 narratives of deaths, other serious and certain other significant adverse events

S5–08 Participant profiles

Output · Reporting & disclosure · Statistical programmer with Medical monitor, Medical writer · Database lock → CSR draft · All designs

Per-participant integrated profiles, tabular or graphical, that bring dosing, PK sampling and concentrations, adverse events, laboratory, vital sign, ECG and PD data onto one timeline for medical review, narrative support and data-cleaning confirmation.

Why it matters. Reviewing each domain separately hides the timing links between dosing, exposure and events.

NEEDS

Locked SDTM and ADaM data across domains; profile specification or template; Section 16.2 listings.

Depends on: S5–05

PRODUCES

Participant profiles for medical review and narrative authoring.

Feeds: S5–07

QUALITY CONTROL

Spot check of profiles against source listings; independent review of the profile program; reconciliation against ADaM.

COMMON PITFALLS

- Profiles produced on a pre-lock snapshot and reused after lock without regeneration.
- Time axis mixes study day and clock time across domains.

STANDARDS AND GUIDANCE

- ICH E3: Section 12.3.2 narratives and Section 16.2 listings that profiles support

S5–09 Submission datasets and analysis programs (with the reviewer guides)

Submission component · Programming & CDISC · Statistical programmer with Lead statistical programmer, Lead biostatistician · Outputs QC-passed → Results posted, submission package · All designs

CONDITIONAL If study data will be submitted to a regulator

Final, submission-ready SDTM and ADaM datasets as XPT version 5 transport files, with the programs that created the ADaM datasets and key outputs, packaged for Module 5 with Define-XML and the reviewer guides. Programs are provided as text files with software versions stated in the ADRG, following the FDA Study Data Technical Conformance Guide.

Why it matters. Datasets or programs that do not conform or cannot be followed delay regulatory review and acceptance of the submission.

NEEDS

Locked, conformance-checked SDTM and ADaM datasets; validated analysis and TLF programs; Define-XML; reviewer guides.

Depends on: S3–17, S3–27, S4–21

PRODUCES

Module 5 dataset and program package supporting the CSR; analyses a reviewer can follow and reproduce.

Feeds: S5–13

QUALITY CONTROL

Final conformance check; independent double-programming evidence retained; program readability and executability review; traceability from collected data to outputs confirmed.

COMMON PITFALLS

- Programs call macros or libraries that are not included, so reviewers cannot follow them.
- Datasets regenerated after QC sign-off without rerunning conformance checks.

STANDARDS AND GUIDANCE

- Study Data Technical Conformance Guide: Technical…: XPT version 5 transport files, dataset organization and submission of programs as text
- FDA Validator Rules v1.6: conformance rules applied to submitted study data
- SDTMIG v3.4: tabulation dataset standard
- ADaMIG v1.3: analysis dataset standard

IN THIS TOOLKIT

Worked study: ADaM datasets, XPT version 5 files and metadata checks (study/03_adam/index.html) · Worked study: run-all script that reproduces every output (study/run_all.R)

WORKED STUDY

S5–10 Traceability matrix (raw data to SDTM, ADaM, TLF and CSR)

Document · Quality & compliance · Lead biostatistician with Lead statistical programmer, Medical writer · Outputs QC-passed → CSR final · All designs

End-to-end traceability linking collected data and the annotated CRF to SDTM variables, SDTM to ADaM derivations, ADaM to each TLF, and TLFs to the numbers reported in the CSR, showing analysis-set and value-level provenance for reviewers and inspectors. It extends the Stage 3 raw-to-ADaM matrix to outputs and report text.

Why it matters. Without demonstrable traceability, a reviewer cannot confirm how a reported number was produced.

NEEDS

Annotated CRF; SDTM and ADaM specifications and Define-XML (including value-level metadata); TLF specifications; raw-to-ADaM traceability matrix; CSR in-text mapping.

Depends on: S3–34, S5–02

PRODUCES

Traceability matrix supporting the reviewer guides and CSR consistency; inspection-readiness record.

Feeds: S5–13

QUALITY CONTROL

Independent verification that each CSR and TLF number traces to a derivation and source; gap review; QA audit.

COMMON PITFALLS

- Matrix built once and not updated after late derivation changes.
- Traceability stops at ADaM, so numbers in the CSR text have no documented source.

STANDARDS AND GUIDANCE

- ADaMIG v1.3: traceability between analysis results, ADaM datasets and SDTM
- Define-XML v2.1: value-level metadata and derivation methods

S5–11 Numerical consistency QC: CSR versus TLFs versus datasets

Review record · Quality & compliance · Lead biostatistician with Statistical programmer, Medical writer, Quality assurance · CSR draft → CSR final · All designs

Formal, documented check that every number in the CSR body, synopsis and in-text tables matches the corresponding full TLF and source dataset, and the validated NCA or modeling output for PK. It produces a discrepancy log closed to zero open critical findings before CSR finalization.

Why it matters. Transcription and version errors between outputs and report text are visible to any reviewer and undermine trust in the whole report.

NEEDS

CSR draft and synopsis; in-text and full TLFs; ADaM datasets; NCA and modeling outputs.

Depends on: S5–12

PRODUCES

Signed numerical-consistency QC report and discrepancy log; release gate for the CSR.

Feeds: S5–13, S5–17, S5–18

QUALITY CONTROL

An independent reviewer performs the check; discrepancy log tracked to closure; second-level QA sign-off.

COMMON PITFALLS

- QC done on a draft and not repeated after late TLF reruns.
- Only in-text tables checked, while numbers in the prose and synopsis are skipped.

STANDARDS AND GUIDANCE

- ICH E3: CSR text, in-text tables, Section 14 tables and synopsis content

IN THIS TOOLKIT

Worked study: CSR text with per-paragraph number checks (study/06_csr/csr_section11.html) WORKED STUDY

S5–12 Statistical input to the CSR synopsis and briefing documents

Document · Reporting & disclosure · Lead biostatistician with Medical writer, Regulatory affairs, Clinical pharmacologist · CSR draft → Results posted, submission package · All designs

Statistical content for the CSR synopsis and for regulatory briefing documents: a concise account of design, estimands and analysis sets with the key PK, PD and safety results, consistent word for word with the CSR and TLFs.

Why it matters. Synopsis and briefing numbers are quoted onward, so any inconsistency with the CSR spreads into regulatory correspondence.

NEEDS

Final CSR results and TLFs; SAP and methods text; NCA and modeling results; prior briefing materials.

Depends on: S5–03

PRODUCES

Synopsis statistical text; briefing-document statistical sections; input to the design of the next study.

Feeds: S5–11

QUALITY CONTROL

Consistency check against the CSR body and TLFs (identical numbers); peer and medical review; sign-off.

COMMON PITFALLS

- Synopsis rounded differently from the body tables.
- Briefing document quotes interim or draft results superseded by the final CSR.

STANDARDS AND GUIDANCE

- ICH E3: synopsis content, including numerical results

S5–13 Biostatistics sign-off and CSR release for submission

Review record · Quality & compliance · Lead biostatistician with Lead statistical programmer, Quality assurance · CSR final → Results posted, submission package · All designs

Final documented biostatistics approval that the CSR statistical content, TLFs, datasets, reviewer guides and traceability are complete, consistent and inspection-ready, releasing the package for eCTD assembly. It records the software and version used for each deliverable.

Why it matters. Without a formal release, late changes can reach the submission without QC and accountability for the statistical package is unclear.

NEEDS

Final CSR; TLFs; submission datasets and programs; Define-XML; reviewer guides; QC and consistency logs; traceability matrix.

Depends on: S4–26, S5–04, S5–06, S5–07, S5–09, S5–10, S5–11, S5–16

PRODUCES

Approved, locked CSR statistical package for Module 5.

Feeds: S5–14, S5–20, S5–22

QUALITY CONTROL

Final QA review; confirmation that all discrepancy logs are closed; electronic signature with audit trail; software and version records attached.

COMMON PITFALLS

- Sign-off given while a known discrepancy is still open in another team's log.
- Outputs rerun after sign-off without a new release.

STANDARDS AND GUIDANCE

- 21 CFR Part 11: electronic signatures and audit trails

S5–14 eCTD Module 5 study-data package assembly and conformance (reviewer guide bundle, study tagging)

Submission component · Reporting & disclosure · Regulatory affairs with Lead statistical programmer, Lead biostatistician · CSR final → Results posted, submission package · All designs

CONDITIONAL If study data will be submitted to a regulator

Packaging of the study-data artifacts into eCTD Module 5: folder structure per the FDA Study Data Technical Conformance Guide, placement of SDTM, ADaM, Define-XML, reviewer guides and programs, package-level conformance validation, and consistency with the Study Data Standardization Plan.

Why it matters. A correct set of files placed or tagged incorrectly can still fail technical checks or confuse reviewers.

NEEDS

Released SDTM and ADaM datasets, Define-XML, reviewer guides and programs; Study Data Standardization Plan.

Depends on: S5–13

PRODUCES

Assembled Module 5 study-data package; package-level conformance report.

QUALITY CONTROL

Submission-readiness review; package-level validation.

COMMON PITFALLS

- Study data folder names or study tagging inconsistent with the study identifier used in the CSR.
- Large datasets not split as the conformance guide describes.

STANDARDS AND GUIDANCE

- Study Data Technical Conformance Guide: Technical…: study data folder structure, Study Data Standardization Plan and file conventions
- ICH M4(R4): Module 5 location of clinical study reports
- ICH eCTD v3.2.2: eCTD electronic submission specification
- FDA Validator Rules v1.6: validation rules applied to submitted study data

S5–15 ADRG finalization and cross-document consistency (SAP, Define-XML, ADRG, CSR)

Review record · Quality & compliance · Lead biostatistician with Lead statistical programmer, Medical writer · CSR draft → CSR final · All designs

One documented reconciliation that the SAP, Define-XML, ADRG and cSDRG, and CSR methods describe the same derivations, analysis sets and terminology versions, with the ADRG of record finalized after the check. It complements the numerical CSR-versus-TLF QC by checking methods-level wording.

Why it matters. Methods described differently across documents lead reviewers to question which version produced the results.

NEEDS

Final SAP; Define-XML; ADRG and cSDRG; CSR methods text.

Depends on: S3–17, S3–27, S5–01

PRODUCES

Cross-document consistency checklist; discrepancy resolutions; final ADRG.

Feeds: S5–06

QUALITY CONTROL

Biostatistician sign-off; documented discrepancy closure.

COMMON PITFALLS

- ADRG finalized before late SAP clarifications, so it describes a superseded derivation.
- Analysis-set names differ between the SAP, Define-XML and CSR.

STANDARDS AND GUIDANCE

- PHUSE ADRG Package: ADRG template and completion guidelines
- Define-XML v2.1: derivation methods and origins
- ICH E3: CSR statistical methods sections

S5–16 CSR Appendix 16.1.7: randomization scheme and codes

Document · Reporting & disclosure · Lead biostatistician with Independent statistician, Statistical programmer, Medical writer · Outputs QC-passed → CSR final · All designs

CONDITIONAL If the study is randomized

Appendix describing the randomization method and how it was carried out, including how random numbers were generated, with a table of randomization codes, participant identifiers and treatment assigned (by site for multicenter studies), released only after unblinding.

Why it matters. Without it, reviewers cannot verify that allocation followed the specification, and the CSR is incomplete.

NEEDS

Randomization specification and schedule; randomization QC report; unblinding record; actual treatment received.

Depends on: S4–24

PRODUCES

Appendix 16.1.7 text and table; cross-check of planned versus actual assignment.

Feeds: S5–13

QUALITY CONTROL

Independent check of the appendix table against the master schedule and IRT assignments.

COMMON PITFALLS

- Publishing the planned list without reconciling actual assignments and replacements.

STANDARDS AND GUIDANCE

- ICH E3: randomization method, its execution and the method of generating random numbers described in appendix 16.1.7, with a table of codes, participant identifiers and treatment assigned (9.4.3, 16.1.7)
- ICH E6(R3): master randomization list and treatment allocation and decoding documentation are essential records (Appendix C)

IN THIS TOOLKIT

Worked study: randomization schedule (source for the codes table) (study/01_design/randomization_schedule.csv) WORKED STUDY

S5–17 ClinicalTrials.gov results data package

Submission component · Reporting & disclosure · Lead biostatistician with Regulatory affairs, Statistical programmer, Medical writer · CSR final → Results posted, submission package · All designs

CONDITIONAL If the trial is an applicable clinical trial or is registered voluntarily on ClinicalTrials.gov

Statistics content for ClinicalTrials.gov results: participant flow, baseline characteristics, outcome measures matching the registered entries and adverse event tables, checked against the CSR. Phase 1 drug trials fall outside the definition of an applicable clinical trial in 42 CFR Part 11, so for them a results submission is voluntary and must then meet the conditions of 42 CFR 11.60.

Why it matters. Results that differ from the registered outcome measures or the CSR create public inconsistencies that are hard to correct.

NEEDS

Registered protocol information and outcome measures; QC-passed outputs or the CSR; sponsor disclosure policy.

PRODUCES

Results data in the registry format; QC record of agreement with the CSR.

Depends on: S5–11

QUALITY CONTROL

Independent check of every posted number against the CSR or QC-passed outputs; outcome measures checked against the registration.

COMMON PITFALLS

- Posting outcome measures that differ from the registered ones without explanation.
- Treating Phase 1 status as settling disclosure when voluntary registration or sponsor and journal policies call for results.

STANDARDS AND GUIDANCE

- 42 CFR Part 11: applicable drug clinical trials exclude phase 1 clinical investigations (11.10(a), 11.22); results due no later than 1 year after the primary completion date (11.44(a)); voluntary submissions must meet the conditions in 11.60
- Food and Drug Administration Amendments Act of 2007,…: statutory definition of an applicable drug clinical trial excludes phase I clinical investigations (PHS Act 402(j)(1)(A)(iii))

S5–18 EU CTR summary of results

Submission component · Reporting & disclosure · Lead biostatistician with Regulatory affairs, Statistical programmer, Medical writer · CSR final → Results posted, submission package · All designs

CONDITIONAL If the trial was conducted in an EU Member State under Regulation (EU) No 536/2014

Statistics content for the summary of results submitted to the EU database under Regulation (EU) No 536/2014: trial information including statistical methods, participant disposition, baseline characteristics, each protocol endpoint with its statistical analyses, adverse events and limitations. Article 37(4) applies irrespective of outcome and makes no exception for Phase 1 trials.

Why it matters. A late or incomplete summary breaches the Regulation and leaves the public record inconsistent with the CSR.

NEEDS

Protocol endpoints; QC-passed outputs or the CSR; date of the end of the trial in all Member States concerned.

PRODUCES

Summary of results with Annex IV content submitted through the EU portal; QC record of agreement with the CSR.

Depends on: S5–11

Feeds: S5–19

QUALITY CONTROL

Independent check of each value against the CSR or QC-passed outputs; endpoint list checked against the protocol.

COMMON PITFALLS

- Planning the summary around CSR timing and missing the one-year deadline from the end of the trial.
- Reporting fewer endpoints than the protocol defines.

STANDARDS AND GUIDANCE

- Regulation (EU) No 536/2014 of the European Parliament…: summary of results within one year of the end of the trial in all Member States concerned, irrespective of outcome, with Annex IV content and a lay summary per Annex V; later submission only for scientific reasons detailed in the protocol (Article 37(4)); summary of an intermediate analysis within one year of its date (Article 37(8)); Annex IV covers as many endpoints as the protocol defines
- EMA/263067/2023: Phase I, BE and BA trials are category 1; for adult-only category 1 trials the summary of results with lay summary is submitted (expected within 12 months of the end of the trial) and published 30 months after the end of the trial in the EU/EEA; trials in children or under a pediatric investigation plan are published when submitted (Section 2, Annex I)

S5–19 Lay summary of results: statistics review

Review record · Reporting & disclosure · Lead biostatistician with Medical writer, Medical monitor, Regulatory affairs · CSR final → Results posted, submission package · All designs

CONDITIONAL If the trial was conducted in an EU Member State under Regulation (EU) No 536/2014

Biostatistics review of the summary for laypersons that accompanies the EU summary of results, checking that numbers, adverse reaction frequencies and descriptions of overall results are correct, consistent with the technical summary and not overstated.

Why it matters. Plain-language simplification can misstate results or frequencies in a public document that is hard to correct.

NEEDS

Draft lay summary; technical summary of results; CSR.

PRODUCES

Statistics review comments and sign-off of the lay summary.

Depends on: S5–18

QUALITY CONTROL

Every number traced to the technical summary; second reviewer for frequencies.

COMMON PITFALLS

- Describing a descriptive Phase 1 comparison as proof that a drug works or is safe.

STANDARDS AND GUIDANCE

- Regulation (EU) No 536/2014 of the European Parliament…: lay summary accompanies the summary of results, with Annex V content including adverse reactions and their frequency and the overall results (Article 37(4), Annex V)

S5–20 TMF filing of biostatistics and programming records

Log or record · Quality & compliance · Lead biostatistician with Lead statistical programmer, Quality assurance, Clinical operations · Protocol final → Archived · All designs

Index and timely filing of the statistics records the TMF needs: sample-size calculations, randomization documents, SAP versions, analysis-set decisions, analysis datasets and programs, QC records and outputs, committee data and minutes, and system validation records, filed as each is finalized rather than at study end.

Why it matters. Records missing from the TMF cannot show inspectors how the analyses were specified and carried out.

NEEDS

TMF plan and index; finalized statistics and programming records from every stage.

`Depends on: S5–13`

PRODUCES

Statistics section of the TMF index with filing status; filed records with version, author, reviewer and approval details.

`Feeds: S5–21`

QUALITY CONTROL

Periodic completeness check against the index; QA spot-check of filed statistics records.

COMMON PITFALLS

- Filing everything at database lock, leaving gaps in contemporaneous evidence.
- Filing unblinded interim material where blinded staff can see it.

STANDARDS AND GUIDANCE

- ICH E6(R3): essential records include statistical documentation (sample-size calculations, analysis-set decisions, analysis datasets, programs, QC records and outputs), the master randomization list and committee minutes; records identifiable, version controlled and filed in a timely manner (Appendix C)
- Regulation (EU) No 536/2014 of the European Parliament…: the TMF holds the essential documents and is readily available and directly accessible to Member States on request (Article 57)

S5–21 Archiving of the analysis environment, programs, datasets and outputs

Log or record · Quality & compliance · Lead statistical programmer with Lead biostatistician, Quality assurance · CSR final → Archived · All designs

Final archive of locked datasets, analysis and QC programs, logs, outputs, software versions and environment records, so that every reported number can be regenerated, with access controls, a retention period set by the applicable regulations and documented transfer of ownership.

Why it matters. Without a complete archive, results cannot be reproduced or defended in later submissions or inspections after the team has moved on.

NEEDS

Locked SDTM and ADaM; final programs, logs and outputs; QC records; software and environment records; applicable retention requirements.

`Depends on: S5–20`

PRODUCES

Archive package with manifest and checksums; archive record with location, access list and retention date.

QUALITY CONTROL

Restore test from the archive that regenerates selected outputs; QA review of the archive record.

COMMON PITFALLS

- Archiving programs without software versions, so outputs cannot be regenerated later.
- Choosing a retention period from habit instead of the regulations that apply.

STANDARDS AND GUIDANCE

- ICH E6(R3): statistical programming records for reported outputs retained with QC records, outputs traceable to programs, dated, protected against change and access controlled (3.16.2(f)); retention per applicable regulatory requirements (3.16.3)
- Regulation (EU) No 536/2014 of the European Parliament…: TMF content archived for at least 25 years after the end of the trial, with traceable alterations and documented transfer of ownership (Article 58)

IN THIS TOOLKIT

Worked study: run manifest (software versions, seed, stage order) (study/run_manifest.json) `WORKED STUDY`

S5–22 Statistics support for publications and data-sharing requests

Document · Reporting & disclosure · Lead biostatistician with Medical writer, Statistical programmer · CSR final → Archived · All designs

`CONDITIONAL` If study results are published or data sharing is requested

Statistical review of manuscripts, abstracts and posters based on the study (numbers traced to the CSR, methods described as carried out, reporting-guideline items for crossover trials), and preparation of de-identified datasets and documentation when data sharing is agreed.

Why it matters. Publications that differ from the CSR, or shared data without adequate de-identification and documentation, damage credibility and can breach privacy.

NEEDS

Final CSR and outputs; archived datasets; publication plan; data-sharing request and agreement.

`Depends on: S5–13`

PRODUCES

Signed statistics review of each publication; de-identified dataset package with documentation where sharing is agreed; publications list for appendix 16.1.11 when available.

QUALITY CONTROL

Number-by-number check against the CSR or archived outputs; independent review of de-identification.

COMMON PITFALLS

- Recomputing results for a manuscript with different software or data cuts than the CSR.

STANDARDS AND GUIDANCE

- ICH E3: publications based on the study listed in appendix 16.1.11
- ICH E8(R1): reporting of study results planned, comprehensive, accurate, timely and publicly accessible (7)
- CONSORT crossover extension: reporting guideline for randomized crossover trials

ESC-06 Preliminary-versus-final data reconciliation for escalation decisions

Document · Biostatistics · Lead biostatistician with PK scientist, Medical monitor, Medical writer · Outputs QC-passed → CSR draft · Single ascending dose, Multiple ascending dose

Comparison of the data used at each escalation review with the locked data and QC-passed outputs (participant counts, key adverse events, preliminary versus final PK parameters), stating whether any difference would have changed a decision. It feeds the CSR description of interim reviews.

Why it matters. Without it, differences between escalation-time data and the CSR surface during review or inspection with no documented assessment.

NEEDS

Escalation decision log and archived package versions; final NCA parameters; final safety tables; locked database.

Depends on: ESC-04, S4-07, S4-13

PRODUCES

Reconciliation table and short memo; text for the CSR sections on dose selection and interim reviews.

Feeds: S5-03

QUALITY CONTROL

Second reviewer checks each comparison against the archived package and the final outputs.

COMMON PITFALLS

- Overwriting archived escalation packages, which makes the comparison impossible.
- Reporting only final PK, leaving reviewers to find unexplained differences from the decision minutes.

STANDARDS AND GUIDANCE

- ICH E3: all interim analyses, formal or informal, described in full, with operating procedures (11.4.2.3)

BY DESIGN

Design modules

The deliverables that belong to a particular study design, in addition to the ones every study needs.

Single ascending dose (SAD)

ID	DELIVERABLE	STAGE	OWNER
S1–07	Safety review committee or DMC charter: statistical decision rules and dose-escalation criteria	1	Lead biostatistician
S1–08	Dose-escalation operating-characteristics simulation (design stage)	1	Lead biostatistician
QT–01	Cardiac safety (QT) design and sample-size basis	1	Lead biostatistician
QT–02	Cardiodynamic (concentration-QTc) analysis plan	2	Pharmacometrician
QT–03	ECG core-laboratory data transfer and blinded-reading specification	2	Data manager
ESC–01	Per-cohort dose-escalation data package	3	Lead biostatistician
ESC–02	Preliminary PK analysis and next-dose exposure projection	3	PK scientist
ESC–03	Sentinel-dosing review record	3	Medical monitor
ESC–04	Dose-escalation decision record and escalation log	3	Safety Review Committee
ESC–05	Cumulative safety and PK summary for part transitions	3	Lead biostatistician
S4–09	Dose-proportionality and accumulation analyses	4	Statistical programmer
S4–11	Concentration-QTc (C-QTc) and QT/QTc analysis	4	Pharmacometrician
S4–30	Concentration-QTc (C-QTc) model assumption checks and QC record	4	Pharmacometrician
QT–05	Cardiodynamic analysis report	4	Pharmacometrician
ESC–06	Preliminary-versus-final data reconciliation for escalation decisions	5	Lead biostatistician

Multiple ascending dose (MAD)

ID	DELIVERABLE	STAGE	OWNER
S1–07	Safety review committee or DMC charter: statistical decision rules and dose-escalation criteria	1	Lead biostatistician
S1–08	Dose-escalation operating-characteristics simulation (design stage)	1	Lead biostatistician
QT–01	Cardiac safety (QT) design and sample-size basis	1	Lead biostatistician
QT–02	Cardiodynamic (concentration-QTc) analysis plan	2	Pharmacometrician
QT–03	ECG core-laboratory data transfer and blinded-reading specification	2	Data manager
ESC–01	Per-cohort dose-escalation data package	3	Lead biostatistician
ESC–02	Preliminary PK analysis and next-dose exposure projection	3	PK scientist
ESC–03	Sentinel-dosing review record	3	Medical monitor
ESC–04	Dose-escalation decision record and escalation log	3	Safety Review Committee
ESC–05	Cumulative safety and PK summary for part transitions	3	Lead biostatistician
S4–09	Dose-proportionality and accumulation analyses	4	Statistical programmer
S4–11	Concentration-QTc (C-QTc) and QT/QTc analysis	4	Pharmacometrician
S4–30	Concentration-QTc (C-QTc) model assumption checks and QC record	4	Pharmacometrician
QT–05	Cardiodynamic analysis report	4	Pharmacometrician
ESC–06	Preliminary-versus-final data reconciliation for escalation decisions	5	Lead biostatistician

Bioequivalence (BE)

ID	DELIVERABLE	STAGE	OWNER
BE–01	BE design and sample-size basis (2x2, replicate or parallel)	1	Lead biostatistician
BE–02	BE analysis section: model, acceptance limits and data-removal rules	2	Lead biostatistician
BE–03	BE analysis-set and period-exclusion derivations	3	Statistical programmer
S4–10	Food-effect, DDI and relative BA or BE statistical analysis (ANOVA, 90% CI, geometric mean ratio)	4	Statistical programmer
BE–04	Reference-scaled BE analysis for a highly variable drug (replicate design)	4	Statistical programmer
BE–05	BE supplementary and sensitivity analyses	4	Statistical programmer
BE–06	BE presentation set: ANOVA tables, individual ratios and profiles	4	Statistical programmer

Relative bioavailability (BA)

ID	DELIVERABLE	STAGE	OWNER
BA–01	Relative bioavailability or formulation-bridging design and precision basis	1	Lead biostatistician
BA–02	BA analysis section: comparisons, reference and data-removal rules	2	Lead biostatistician

ID	DELIVERABLE	STAGE	OWNER
S4–10	Food-effect, DDI and relative BA or BE statistical analysis (ANOVA, 90% CI, geometric mean ratio)	4	Statistical programmer
BA–03	Relative or absolute bioavailability results	4	Statistical programmer
BA–04	Clinical relevance assessment when bioavailability similarity is not shown	4	Pharmacometrician

Food effect (FE)

ID	DELIVERABLE	STAGE	OWNER
FE–01	Food-effect design and sample-size basis	1	Lead biostatistician
FE–02	Food-effect analysis section	2	Lead biostatistician
FE–03	Meal-compliance and dosing-condition derivations	3	Statistical programmer
S4–10	Food-effect, DDI and relative BA or BE statistical analysis (ANOVA, 90% CI, geometric mean ratio)	4	Statistical programmer
FE–04	Fed versus fasted results and labeling support	4	Statistical programmer

Drug-drug interaction (DDI)

ID	DELIVERABLE	STAGE	OWNER
DDI–01	Drug interaction study design and sample-size basis	1	Lead biostatistician
DDI–02	Drug interaction analysis section: no-effect boundaries and model	2	Lead biostatistician
DDI–03	Drug interaction treatment-condition derivations	3	Statistical programmer
S4–10	Food-effect, DDI and relative BA or BE statistical analysis (ANOVA, 90% CI, geometric mean ratio)	4	Statistical programmer
DDI–04	Drug interaction results: GMR, forest plot and inhibitor or inducer classification	4	Statistical programmer

Concentration-QTc or thorough QT (QT)

ID	DELIVERABLE	STAGE	OWNER
QT–01	Cardiac safety (QT) design and sample-size basis	1	Lead biostatistician
QT–02	Cardiodynamic (concentration-QTc) analysis plan	2	Pharmacometrician
QT–03	ECG core-laboratory data transfer and blinded-reading specification	2	Data manager
S4–11	Concentration-QTc (C-QTc) and QT/QTc analysis	4	Pharmacometrician
S4–30	Concentration-QTc (C-QTc) model assumption checks and QC record	4	Pharmacometrician
QT–04	Thorough QT by-time-point analysis and assay sensitivity	4	Statistical programmer
QT–05	Cardiodynamic analysis report	4	Pharmacometrician

Renal impairment (RI)

ID	DELIVERABLE	STAGE	OWNER
RI–01	Renal impairment design and sample-size basis	1	Lead biostatistician
RI–02	Renal impairment analysis section	2	Lead biostatistician
RI–03	Renal function classification, unbound PK and dialysis derivations	3	Statistical programmer
RI–04	Renal impairment results: PK versus renal function and dosing support	4	Statistical programmer

Hepatic impairment (HI)

ID	DELIVERABLE	STAGE	OWNER
HI–01	Hepatic impairment design and sample-size basis	1	Lead biostatistician
HI–02	Hepatic impairment analysis section	2	Lead biostatistician
HI–03	Child-Pugh classification and unbound PK derivations	3	Statistical programmer
HI–04	Hepatic impairment results: PK by Child-Pugh group and dosing support	4	Statistical programmer

SOURCES

Standards and guidance index

Every reference cited in this handbook, checked against the issuing body's own page on 2026-11-08. Confirm the version that applies to your study and submission before relying on it.

REFERENCE	ISSUER	VERSION, DATE	STATUS	CITED BY
ADaM Conformance Rules v5.0 https://www.cdisc.org/standards/foundational/adam/adam-conformance-rules-v5-0	CDISC	5.0, 6 October 2023	current	1
ADaM Structure for Occurrence Data (OCCDS) Implementation Guide v1.1 https://www.cdisc.org/standards/foundational/adam/adam-structure-occurrence-data-implementation-guide-v1-1	CDISC	1.1, 29 November 2021	current	3
ADaMIG for Non-compartmental Analysis Input Data v1.0 https://www.cdisc.org/standards/foundational/adam/adamig-non-compartmental-analysis-input-data-v1-0	CDISC	1.0, 29 November 2021	current	3
Analysis Data Model Implementation Guide (ADaMIG) v1.3 https://www.cdisc.org/standards/foundational/adam/adamig-v1-3	CDISC	1.3, 29 November 2021	current	23
Basic Data Structure for ADaM popPK Implementation Guide v1.0 https://www.cdisc.org/standards/foundational/adam/basic-data-structure-adam-poppk-implementation-guide-v1-0	CDISC	1.0, 6 October 2023	current	1
Clinical Data Acquisition Standards Harmonization Implementation Guide (CDASHIG) v2.3 https://www.cdisc.org/standards/foundational/cdash/cdashig-v2-3	CDISC	v2.3 (related CDASH Model v1.3), 2023-09-28	current	1
Define-XML v2.1 https://www.cdisc.org/standards/data-exchange/define-xml/define-xml-v2-1	CDISC	2.1 specification; package v2.1.11 updated 6 April 2026, 15 May 2019 (v2.1 published)	current	15
SDTM and SDTMIG Conformance Rules v2.0 https://www.cdisc.org/standards/foundational/sdtmig/sdtm-and-sdtmig-conformance-rules-v2-0	CDISC	2.0, 29 November 2021	current	1
Study Data Tabulation Model Implementation Guide: Human Clinical Trials (SDTMIG) v3.4 https://www.cdisc.org/standards/foundational/sdtmig/sdtmig-v3-4	CDISC	3.4, 29 November 2021	current	21
CDISC Controlled Terminology https://www.cdisc.org/standards/terminology/controlled-terminology	CDISC, with NCI Enterprise Vocabulary Services (EVS), which maintains and distributes the files	Release dated 2026-09-25 (package P62), the latest on 8 November 2026, 25 September 2026	current	17
CONSORT 2010 statement: extension to randomised crossover trials https://www.equator-network.org/reporting-guidelines/consort-2010-statement-extension-to-randomised-crossover-trials/	CONSORT Group (journal article, doi:10.1136/bmj.l4378; listed by the EQUATOR Network)	2019;366:l4378 (PMID 31366597), 2019-07-31	current	1
Guideline on reporting the results of population pharmacokinetic analyses https://www.ema.europa.eu/en/scientific-guidelines/reporting-results-population-pharmacokinetic-analyses	EMA (CHMP)	Final, Adopted by CHMP 21 June 2007; effective 1 January 2008	current	3
Guideline on strategies to identify and mitigate risks for first-in-human and early clinical trials with investigational medicinal products https://www.ema.europa.eu/en/scientific-guidelines/strategies-identify-mitigate-risks-first-human-early-clinical-trials-investigational-medicinal-products	EMA (CHMP)	Revision 1, Adopted by CHMP 20 July 2017; effective 1 February 2018	current	9
Guideline on the evaluation of the pharmacokinetics of medicinal products in patients with decreased renal function https://www.ema.europa.eu/en/scientific-guidelines/evaluation-pharmacokinetics-medicinal-products-patients-decreased-renal-function	EMA (CHMP)	Final; replaces CHMP/EWP/225702, Adopted by CHMP 17 December 2015; effective 1 July 2016	current	4
Guideline on the evaluation of the pharmacokinetics of medicinal products in patients with impaired hepatic function https://www.ema.europa.eu/en/scientific-guidelines/evaluation-pharmacokinetics-medicinal-products-patients-impaired-hepatic-function	EMA (CHMP)	Final, Adopted by CHMP February 2005 (document dated 17 February 2005); effective 1 August 2005	current	4
Guideline on the investigation of bioequivalence https://www.ema.europa.eu/en/scientific-guidelines/investigation-bioequivalence	EMA (CHMP)	Revision 1 (with corrections), Adopted by CHMP 20 January 2010; effective 1 August 2010	current (see ICH-M13A, in part (EU, from 25 January 2025))	1
Guideline on the investigation of drug interactions https://www.ema.europa.eu/en/scientific-guidelines/investigation-drug-interactions	EMA (CHMP)	Revision 1 (Corr. 2), Adopted by CHMP 21 June 2012; effective 1 January 2013	current (see ICH-M12, in part (EU, from 30	1

REFERENCE	ISSUER	VERSION, DATE	STATUS	CITED BY
			November 2024))	
Guideline on computerised systems and electronic data in clinical trials https://www.ema.europa.eu/en/documents/regulatory-procedural-guideline/guideline-computerised-systems-and-electronic-data-clinical-trials_en.pdf	EMA (GCP Inspectors Working Group)	Final, Adopted 7 March 2023; first published 10 March 2023; comes into effect 6 months after publication	current	5
Revised CTIS Transparency Rules https://www.ema.europa.eu/en/documents/other/revised-ctis-transparency-rules_en.pdf	EMA (adopted by the EMA Management Board)	Revised rules, Adopted 5 October 2023; applicable from 18 June 2024	current	1
Considerations regarding the implementation of ICH M13A on bioequivalence for immediate-release solid oral dosage forms https://www.ema.europa.eu/en/documents/scientific-guideline/considerations-regarding-implementation-ich-m13a-bioequivalence-immediate-release-solid-oral-dosage-forms_en.pdf	European Medicines Agency (CHMP)	EMA/531548/2024, Adopted by CHMP 2025-02-17; first published 2025-02-26	current	1
Regulation (EU) No 536/2014 of the European Parliament and of the Council of 16 April 2014 on clinical trials on medicinal products for human use, and repealing Directive 2001/20/EC https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32014R0536	European Parliament and Council of the European Union (published in EUR-Lex)	In force; current consolidated version dated 5 December 2022, 16 April 2014; in application since 31 January 2022	current	4
Electronic Systems, Electronic Records, and Electronic Signatures in Clinical Investigations: Questions and Answers https://www.fda.gov/regulatory-information/search-fda-guidance-documents/electronic-systems-electronic-records-and-electronic-signatures-clinical-investigations-questions	FDA	Final (Revision 1, procedural), October 2024 (notice of availability 2 October 2024, 89 FR 80255)	current	1
Adaptive Design Clinical Trials for Drugs and Biologics: Guidance for Industry https://www.fda.gov/regulatory-information/search-fda-guidance-documents/adaptive-design-clinical-trials-drugs-and-biologics-guidance-industry	FDA (CBER and CDER)	Final, November 2019 (date on the guidance cover; the FDA web page shows December 2019)	current	4
Guidance for Clinical Trial Sponsors: Establishment and Operation of Clinical Trial Data Monitoring Committees https://www.fda.gov/regulatory-information/search-fda-guidance-documents/establishment-and-operation-clinical-trial-data-monitoring-committees	FDA (CBER, CDRH and CDER)	Final, March 2006	current	2
Drug-Induced Liver Injury: Premarketing Clinical Evaluation https://www.fda.gov/regulatory-information/search-fda-guidance-documents/drug-induced-liver-injury-premarketing-clinical-evaluation	FDA (CDER and CBER)	Final, July 2009	current	2
Exposure-Response Relationships: Study Design, Data Analysis, and Regulatory Applications https://www.fda.gov/regulatory-information/search-fda-guidance-documents/exposure-response-relationships-study-design-data-analysis-and-regulatory-applications	FDA (CDER and CBER)	Final, April 2003 (date on the guidance cover; the FDA web page lists May 2003)	current	2
FDA Validator Rules https://www.fda.gov/industry/fda-data-standards-advisory-board/study-data-standards-resources	FDA (CDER and CBER)	1.6, December 2022 (finalized)	current	3
Population Pharmacokinetics: Guidance for Industry https://www.fda.gov/regulatory-information/search-fda-guidance-documents/population-pharmacokinetics	FDA (CDER and CBER)	Final (Level 1), February 2022	current	5
Study Data Technical Conformance Guide: Technical Specifications Document https://www.fda.gov/regulatory-information/search-fda-guidance-documents/study-data-technical-conformance-guide-technical-specifications-document	FDA (CDER and CBER)	6.3, October 2026	current	11
Assessing the Effects of Food on Drugs in INDs and NDAs: Clinical Pharmacology Considerations https://www.fda.gov/regulatory-information/search-fda-guidance-documents/assessing-effects-food-drugs-inds-and-ndas-clinical-pharmacology-considerations	FDA (CDER)	Revision 1 (Level 2 revision), May 2026 (Revision 1); first final version June 2022 (notice 24 June 2022, 87 FR 37861)	current	6
Bioavailability Studies Submitted in NDAs or INDs: General Considerations https://www.fda.gov/regulatory-information/search-fda-guidance-documents/bioavailability-studies-submitted-ndas-or-inds-general-considerations	FDA (CDER)	Final, April 2022	current	7
Pharmacokinetics in Patients with Impaired Hepatic Function: Study Design, Data Analysis, and Impact on Dosage https://www.fda.gov/regulatory-information/search-fda-guidance-documents/pharmacokinetics-patients-impaired-hepatic-function-study-design-data-analysis-and-impact-dosing-and	FDA (CDER)	Draft (Level 1), September 2026 (notice 2 September 2026, 91 FR 56449; comments requested by 1 December 2026)	draft	4
Pharmacokinetics in Patients with Impaired Renal Function: Study Design, Data Analysis, and Impact on Dosing https://www.fda.gov/regulatory-information/search-fda-guidance-documents/pharmacokinetics-patients-impaired-renal-function-	FDA (CDER)	Final (Level 1), March 2024	current	4

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Statistical Approaches to Establishing Bioequivalence https://www.fda.gov/regulatory-information/search-fda-guidance-documents/statistical-approaches-establishing-bioequivalence	FDA (CDER)	Final (Level 1), May 2026	current	8
Part 11: Electronic Records; Electronic Signatures https://www.ecfr.gov/api/versioner/v1/full/2026-11-05/title-21.xml?part=11	FDA (Code of Federal Regulations, Title 21)	eCFR text current as of 5 November 2026, Original rule 62 FR 13464, 20 March 1997; latest amendment in the eCFR history 2 March 2023 (11.100)	current	17
Part 312: Investigational New Drug Application https://www.ecfr.gov/api/versioner/v1/full/2026-11-05/title-21.xml?part=312	FDA (Code of Federal Regulations, Title 21)	eCFR text current as of 5 November 2026, Source 52 FR 8831, 19 March 1987, as amended; latest substantive amendments in the eCFR history dated 22 September 2026	current	3
Part 11: Clinical Trials Registration and Results Information Submission https://www.ecfr.gov/api/versioner/v1/full/2026-11-03/title-42.xml?part=11	HHS, NIH (Code of Federal Regulations, Title 42)	eCFR text current as of 3 November 2026, Final rule 81 FR 65138, 21 September 2016; in effect from 18 January 2017; last amended 9 December 2024 (web address updates, 89 FR 97558)	current	1
Addendum on Estimands and Sensitivity Analysis in Clinical Trials to the Guideline on Statistical Principles for Clinical Trials https://database.ich.org/sites/default/files/E9-R1_Step4_Guideline_2019_1203.pdf	ICH	Step 4 final version (document dated 17 November 2019), Adopted 20 November 2019 (Step 4)	current	5
Bioanalytical Method Validation and Study Sample Analysis https://database.ich.org/sites/default/files/M10_Guideline_Step4_2022_0524.pdf	ICH	Step 4 final version, Adopted 24 May 2022 (Step 4)	current	2
Bioequivalence for Immediate-Release Solid Oral Dosage Forms https://database.ich.org/sites/default/files/ICH_M13A_Step4_Final_Guideline_2024_0723.pdf	ICH	Step 4 final version, 23 July 2024 (Step 4); M13A Q&As adopted 23 July 2024	current	29
Clinical Electronic Structured Harmonised Protocol (CeSHarP) https://database.ich.org/sites/default/files/ICH_Step4_M11_Final_Guideline_2025_1119.pdf	ICH	Step 4 final versions of all three documents, 19 November 2025 (Step 4); Step 2 was 27 September 2022	current	7
Clinical Safety Data Management: Definitions and Standards for Expedited Reporting https://database.ich.org/sites/default/files/E2A_Guideline.pdf	ICH	Step 4, 27 October 1994 (Step 4)	current	2
Drug Interaction Studies https://database.ich.org/sites/default/files/ICH_M12_Step4_Guideline_2024_0521_0.pdf	ICH	Step 4 final version, 21 May 2024 (Step 4); M12 Q&As adopted 21 May 2024	current	6
General Considerations for Clinical Studies https://database.ich.org/sites/default/files/ICH_E8-R1_Guideline_Step4_2021_1006.pdf	ICH	Step 4 final version (minor editorial correction 4 August 2022), 6 October 2021 (Step 4)	current	4
General Principles for Model-Informed Drug Development https://database.ich.org/sites/default/files/ICH_M15_Step4_Final_Guideline_2026_0129.pdf	ICH	Step 4 final version, 29 January 2026 (Step 4)	current	2
Guideline for Good Clinical Practice E6(R3) https://database.ich.org/sites/default/files/ICH%20E6%28R3%29_Step4_FinalConsolidatedGuideline_2026_0616_.pdf	ICH	Step 4 final version, consolidated (Principles, Annex 1 and Annex 2), Principles and Annex	current	29

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Organisation of the Common Technical Document for the Registration of Pharmaceuticals for Human Use https://database.ich.org/sites/default/files/M4_R4__Guideline.pdf	ICH	R4, current Step 4 version, 15 June 2016	current	1
Revision of M4E Guideline on Enhancing the Format and Structure of Benefit-Risk Information in ICH: Efficacy, M4E(R2) https://database.ich.org/sites/default/files/M4E_R2__Guideline.pdf	ICH	R2, current Step 4 version, 15 June 2016	current	2
Statistical Principles for Clinical Trials https://database.ich.org/sites/default/files/E9_Guideline.pdf	ICH	Step 4, 5 February 1998 (Step 4)	current	35
Structure and Content of Clinical Study Reports https://database.ich.org/sites/default/files/E3_Guideline.pdf	ICH	Step 4, 30 November 1995 (Step 4)	current	38
The Clinical Evaluation of QT/QTc Interval Prolongation and Proarrhythmic Potential for Non-Antiarrhythmic Drugs https://database.ich.org/sites/default/files/E14_Guideline.pdf	ICH	Step 4, 12 May 2005 (Step 4)	current	10
Clinical and Nonclinical Evaluation of QT/QTc Interval Prolongation and Proarrhythmic Potential: Questions and Answers (R1) https://database.ich.org/sites/default/files/ICH%20E14_S7B%20Q%26As%28R1%29%202026_2008.pdf	ICH (E14/S7B Implementation Working Group)	R1, Adopted under Step 4 on 20 August 2026 (first E14/S7B Q&As adopted 21 February 2022)	current	8
ICH E3 Guideline: Structure and Content of Clinical Study Reports, Questions & Answers (R1) https://database.ich.org/sites/default/files/E3_Q%26As_R1_Q%26As.pdf	ICH (E3 Implementation Working Group)	R1, Step 4 on 7 June 2012; R1 (typographical correction in the answer to Question 6) dated 6 July 2012	current	1
Electronic Common Technical Document Specification (ICH eCTD Specification V 3.2.2) https://admin.ich.org/sites/default/files/inline-files/eCTD_Specification_v3_2_2_0.pdf	ICH (developed by the M2 Expert Working Group; maintained by the M8 EWG/IWG)	3.2.2, 16 July 2008	current	1
Medical Dictionary for Regulatory Activities (MedDRA) https://admin.meddra.org/api/articles	ICH (maintained by the MedDRA Maintenance and Support Services Organization, MSSO)	29.1, English release 1 September 2026 (updated English files 17 September 2026); all languages 15 September 2026	current	7
ICH M13A Guideline: Bioequivalence for Immediate-Release Solid Oral Dosage Forms, Questions and Answers (M13A Q&As) https://database.ich.org/sites/default/files/ICH_M13A_Step4_QAs_2024_0723.pdf	ICH, M13 Expert Working Group	Step 4, 2024-07-23	current	2
ISPE GAMP 5: A Risk-Based Approach to Compliant GxP Computerized Systems (Second Edition) https://ispe.org/publications/guidance-documents/gamp-5-guide-2nd-edition	ISPE	Second Edition, July 2022	current	2
Common Terminology Criteria for Adverse Events (CTCAE) v6.0 https://dctd.cancer.gov/research/ctep-trials/for-sites/adverse-events	NCI (Cancer Therapy Evaluation Program)	6.0, Published 22 July 2025	current	3
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Food and Drug Administration Amendments Act of 2007, Title VIII (Clinical Trial Databases), Section 801: Expanded Clinical Trial Registry Data Bank https://www.govinfo.gov/content/pkg/PLAW-110publ85/pdf/PLAW-110publ85.pdf	U.S. Congress (Public Law 110-85)	As enacted, 27 September 2007	current	1
WHODrug Global (UMC's Drug Dictionary) https://who-umc.org/whodrug/whodrug-global/what-is-whodrug-global/	Uppsala Monitoring Centre (UMC)	New releases on 1 March and 1 September each year; label of the latest release not confirmed, UMC page last modified 7 September 2026	current	3